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Exact Identity Linking Entropy Production and Mutual Information

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We establish an exact identity for overdamped Langevin dynamics: the total entropy production rate equals four times the mutual information rate between an infinitesimal displacement and its actual temporal midpoint, plus a mean-flow term. This provides a forward information-theoretic representation of irreversibility and makes the Shannon chain rule act directly on dissipation. For additive bipartitions, the chain rule reveals nonnegative self and interaction channels, with the self channel coinciding with apparent entropy production. This channel structure sharpens the learning-rate bound by showing that learning is constrained solely by the interaction channel. The same structure is also shown to resolve a recently reported coarse-graining reversal in hydrodynamically coupled colloids. Together, these results exhibit that, within this channel decomposition, learning and coarse-graining probe complementary parts of dissipation in overdamped Langevin systems.

Entropy production (EP), a central measure of nonequilibrium irreversibility, admits equivalent representations that expose different structures. Its path-space form is the Kullback–Leibler divergence between forward and backward path measures, revealing arrow-of-time distinguishability and fluctuation relations [1–5]. Related information-theoretic measures of time asymmetry likewise compare forward and reversed trajectory ensembles [6]. For overdamped diffusions, the equivalent current-quadratic form instead exposes instantaneous geometry and projections [5, 7–10]. Mutual information (MI), a central measure of statistical dependence in information theory, asks a different question: how variables depend on one another within a single joint ensemble. MI has long played an important role in information thermodynamics, entering generalized second laws and relations involving feedback or information exchange [11–16]. Direct relations between MI and EP are also known in special settings [17–19]. Yet it is not obvious that the total EP rate itself admits an exact representation in terms of ordinary MI. To our knowledge, a general identity of this kind has been lacking.

Here we establish such an identity for overdamped Langevin dynamics. The total EP rate equals four times the MI rate between an infinitesimal displacement and its actual temporal midpoint, plus a mean-flow term. Both variables belong to the same forward trajectory segment; no time-direction label or separate reversed ensemble enters the final MI. The midpoint is the natural time-reversal-symmetric reference and converts irreversible current into displacement–midpoint dependence (Fig. 1), while the mean-flow term accounts for uniform translation invisible to MI. The result is therefore a forward information-theoretic representation.

In this representation, the physical degrees of freedom appear explicitly as MI arguments, allowing the Shannon chain rule to act directly on EP. For additive bipartitions, this maps onto the marginal-current projection [7, 10] and identifies nonnegative self and interaction channels. We use this structure to sharpen the learning-rate bound [20] and to resolve a reported coarse-graining reversal in hydrodynamically coupled colloids [21, 22].

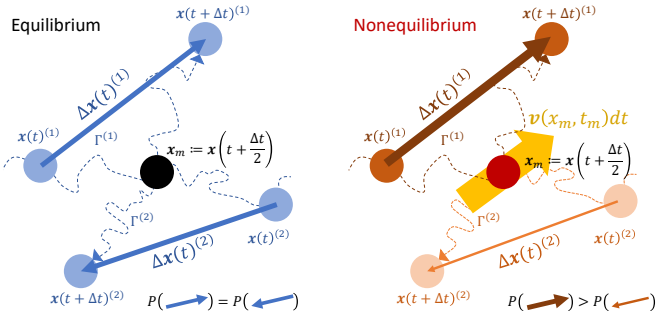

FIG. 1. Schematic illustration of displacement $\Delta\boldsymbol{x}$ conditioned on $\boldsymbol{x}_m$. (Left) In equilibrium, detailed balance implies that $\boldsymbol{x}_m$ contains no information about $\Delta\boldsymbol{x}$. (Right) In nonequilibrium, a finite current velocity $\boldsymbol{v}_t(\boldsymbol{x}_m)$ breaks detailed balance and biases $\Delta\boldsymbol{x}$ conditioned on $\boldsymbol{x}_m$, rendering the midpoint informative. In the infinitesimal-time limit $\mathrm{d}t \to 0$, $\Delta\boldsymbol{x}$ corresponds to $\mathrm{d}\boldsymbol{x}$ in the main text.

Main identity.—We consider standard overdamped Langevin dynamics with arbitrary nonlinear and time-dependent forces

$$\dot{\boldsymbol{x}}_t = \boldsymbol{\mu}_t \boldsymbol{F}_t(\boldsymbol{x}_t) + \sqrt{2\mathbf{D}_t}\,\boldsymbol{\xi}_t, \tag{1}$$

where $\boldsymbol{\mu}_t$ is the mobility, $\boldsymbol{F}_t$ the force, and $\boldsymbol{\xi}_t$ is Gaussian white noise with $\langle\boldsymbol{\xi}_t\rangle = \mathbf{0}$ and $\langle\boldsymbol{\xi}_t\boldsymbol{\xi}_s^{\intercal}\rangle = \mathbf{I}\delta(t-s)$. We assume position-independent noise $\mathbf{D}_t$ in the main text for simplicity; extensions to $\mathbf{D}_t(\boldsymbol{x})$ are given in the Supplemental Material (SM). The total EP is standardly defined by the Kullback–Leibler divergence between the forward and time-reversed path measures, quantifying the distinguishability of the process from its time reverse [3]. For the dynamics in Eq. (1), its instantaneous rate reads $\sigma_t = \langle \boldsymbol{v}_t^{\intercal}\mathbf{D}_t^{-1}\boldsymbol{v}_t\rangle$, where $\langle\cdot\rangle = \int(\cdot)\,p_t(\boldsymbol{x})\,d\boldsymbol{x}$, p_t is the probability density, and $\boldsymbol{v}_t \coloneqq \boldsymbol{\mu}_t\boldsymbol{F}_t - \mathbf{D}_t\nabla\ln p_t$ is the current velocity (also called the local mean velocity) [5].

We show that this path-space distinguishability underlying EP can be reframed as forward within-ensemble dependence between an infinitesimal displacement and its temporal midpoint, both belonging to the same for-

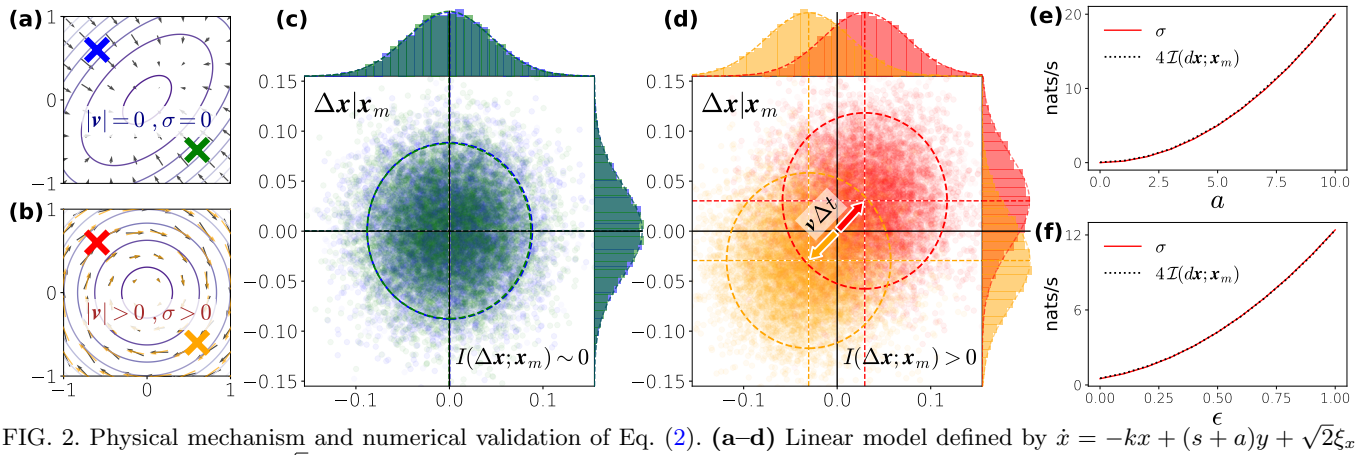

FIG. 2. Physical mechanism and numerical validation of Eq. (2). **(a–d)** Linear model defined by $\dot{x} = -kx + (s+a)y + \sqrt{2}\xi_x$ and $\dot{y} = -ky + (s-a)x + \sqrt{2}\xi_y$ with $k = 10$ and $\langle \xi_i \xi_j \rangle = \delta_{ij}$. Parameters are $(s, a) = (5, 0)$ for equilibrium (a, c), and $(0, 50)$ for nonequilibrium steady state (NESS) (b, d). In (a, b), the force $\boldsymbol{F}(\boldsymbol{x})$ (black) and current velocity $\boldsymbol{v}(\boldsymbol{x})$ (yellow) are overlaid on steady-state density $p(\boldsymbol{x})$. Crosses indicate sampling locations for the time midpoint $\boldsymbol{x}_m$. In (c, d), conditional displacements $\Delta\boldsymbol{x}|\boldsymbol{x}_m$ at the marked locations are shown. Circles are centered at sample means with radius $1.96 \times$ SD. Dashed lines mark the means, and black solid lines denote the origin. In equilibrium (c), the distributions are indistinguishable. In NESS (d), they are separated by the current velocity $\boldsymbol{v}(\boldsymbol{x}_m)\Delta t$, rendering $\boldsymbol{x}_m$ informative about $\Delta\boldsymbol{x}$. In the simulations, $\Delta t = 10^{-3}$. **(e, f)** Coincidence of information rates $4\mathcal{I}(\mathrm{d}\boldsymbol{x}; \boldsymbol{x}_m)$ (dotted) to the total EP rate (solid) for the same linear model with $s = 0$ (e), and for the nonlinear model $\dot{x} = -x + y + \epsilon y^3 + \sqrt{2}\xi_x$, $\dot{y} = -y + \sqrt{2}\xi_y$ (f). Eq. (2) remains exact in both linear and nonlinear models.

ward trajectory segment, as illustrated in Fig. 2a–d. Let $m = t + \frac{\mathrm{d}t}{2}$, and we denote quantities at time m with the subscript m. Considering the infinitesimal displacement $\mathrm{d}\boldsymbol{x}_t = \boldsymbol{x}_{t+\mathrm{d}t} - \boldsymbol{x}_t$ conditioned on the midpoint position $\boldsymbol{x}_m$, we obtain the exact identity at the level of rates,

$$\sigma_t = 4\mathcal{I}(\mathrm{d}\boldsymbol{x}_t; \boldsymbol{x}_m) + \langle \boldsymbol{v}_t \rangle^{\intercal} \mathbf{D}_t^{-1} \langle \boldsymbol{v}_t \rangle, \tag{2}$$

where $\mathcal{I}(A; B) := \lim_{\mathrm{d}t \to 0} I(A; B)/\mathrm{d}t$.

To prove Eq. (2), we use the Markov property at the midpoint m, where conditioning on $\boldsymbol{x}_m$ makes the two half-step increments $(\boldsymbol{x}_{t+\mathrm{d}t} - \boldsymbol{x}_m)$ and $-(\boldsymbol{x}_t - \boldsymbol{x}_m)$ statistically independent. The forward increment follows the Euler–Maruyama step [23, 24] with mean $\boldsymbol{\mu}_m \boldsymbol{F}_m \mathrm{d}t/2$ and covariance $\mathbf{D}_m \mathrm{d}t$. The backward increment $\boldsymbol{x}_t - \boldsymbol{x}_m$ is governed by reverse-time diffusion [25, 26] with score-corrected drift $\boldsymbol{F}^{\mathrm{B}} = -\boldsymbol{F} + 2\boldsymbol{\mu}^{-1}\mathbf{D}\nabla \ln p$ (and the same diffusion matrix), hence $\langle -(\boldsymbol{x}_t - \boldsymbol{x}_m) | \boldsymbol{x}_m \rangle = -\boldsymbol{\mu}_m \boldsymbol{F}_m^{\mathrm{B}} \mathrm{d}t/2$. Consequently, in the infinitesimal-time limit $\mathrm{d}t \to 0$, the sum admits a Gaussian core with conditional mean $\langle \mathrm{d}\boldsymbol{x}_t | \boldsymbol{x}_m \rangle = \boldsymbol{v}_m \mathrm{d}t$ and covariance $\mathrm{Cov}(\mathrm{d}\boldsymbol{x}_t | \boldsymbol{x}_m) = 2\mathbf{D}_m \mathrm{d}t$. Equivalently, to the order relevant for Eq. (2), $p(\mathrm{d}\boldsymbol{x}_t | \boldsymbol{x}_m)$ is captured by the effective Gaussian channel

$$\mathrm{d}\boldsymbol{x}_t = \boldsymbol{v}_m(\boldsymbol{x}_m)\mathrm{d}t + \sqrt{2\mathbf{D}_m \mathrm{d}t}\,\mathbf{N}, \qquad \mathbf{N} \sim \mathcal{N}(0, \mathbf{I}). \tag{3}$$

We emphasize that the midpoint conditioning is essential, as other conditioning does not yield the current velocity. For example, $\langle \mathrm{d}\boldsymbol{x}_t \mid \boldsymbol{x}_t \rangle = \mu_t \boldsymbol{F}_t(\boldsymbol{x}_t)\,\mathrm{d}t$, whereas $\langle \mathrm{d}\boldsymbol{x}_t \mid \boldsymbol{x}_m \rangle = \boldsymbol{v}_m(\boldsymbol{x}_m)\,\mathrm{d}t$. We also note that Eq. (3) is not a stochastic differential equation for the original process, but an effective statistical model for $p(\mathrm{d}\boldsymbol{x}_t | \boldsymbol{x}_m)$.

Since Eq. (3) operates in the low signal-to-noise ratio (SNR) regime as $\mathrm{d}t \to 0$, a standard small-SNR expansion for Gaussian channels [27] yields

$$I(\mathrm{d}\boldsymbol{x}_t; \boldsymbol{x}_m) = \frac{\mathrm{d}t}{4} \langle \delta\boldsymbol{v}_m^{\intercal} \mathbf{D}_m^{-1} \delta\boldsymbol{v}_m \rangle + o(\mathrm{d}t), \tag{4}$$

with $\delta\boldsymbol{v}_m := \boldsymbol{v}_m - \langle \boldsymbol{v}_m \rangle$. Using $\mathrm{Cov}(\boldsymbol{v}_m(\boldsymbol{x}_m)) = \mathrm{Cov}(\boldsymbol{v}_t(\boldsymbol{x}_t)) + O(\mathrm{d}t)$, dividing by $\mathrm{d}t$ and taking the limit $\mathrm{d}t \to 0$ recovers the main identity, Eq. (2). No assumption of steady state, linearity, or time independence is required beyond standard regularity conditions. For position-dependent diffusion, the identity retains the same information-theoretic structure in the local diffusion metric; the SM provides a rigorous derivation of the general result, recovering the present additive-noise case, together with a numerical illustration.

The midpoint $\boldsymbol{x}_m$ plays a special role because it is the sole position left invariant by the infinitesimal time-reversal exchange $t \leftrightarrow t + \mathrm{d}t$. It therefore provides a natural reference for recasting path-space irreversibility in terms of statistics within the forward trajectory ensemble, as conceptualized in Fig. 1.

The mean-flow contribution $\langle \boldsymbol{v}_t \rangle^{\intercal} \mathbf{D}_t^{-1} \langle \boldsymbol{v}_t \rangle$, which can be written as $\langle \boldsymbol{\mu}_t \boldsymbol{F}_t \rangle^{\intercal} \mathbf{D}_t^{-1} \langle \boldsymbol{\mu}_t \boldsymbol{F}_t \rangle$ under usual boundary conditions, has a simple interpretation. The mutual information $I(\mathrm{d}\boldsymbol{x}_t; \boldsymbol{x}_m)$ is invariant under a uniform translation, whereas the path-space EP still detects such a global flow. The mean-flow term thus accounts for the mismatch between the forward/backward reference underlying EP and the midpoint-based forward reference underlying $\mathcal{I}$. When $\langle \boldsymbol{v}_t \rangle = \mathbf{0}$, the mismatch disappears and Eq. (2) reduces to the exact relation $\sigma_t = 4\mathcal{I}(\mathrm{d}\boldsymbol{x}_t; \boldsymbol{x}_m)$. In this case, path-space distinguishability is fully aligned with distinguishability within the forward trajectory ensemble. This includes, for example, confined steady states with vanishing net mean-flow.

Self and interaction EP channels.—Eq. (2) provides a new information-theoretic lens for understanding EP. Consider a bipartite system $\{A, B\}$ with block-diagonal diffusion, and define $\sigma_A^{\rm mf} := \langle \boldsymbol{v}_A \rangle^{\intercal} \mathbf{D}_A^{-1} \langle \boldsymbol{v}_A \rangle$. Applying the same midpoint approach to subsystem A yields an identity analogous to Eq. (2) for the local EP rate of A,

$$\sigma_A = 4\mathcal{I}(\mathrm{d}\boldsymbol{x}_A; \boldsymbol{x}_m) + \sigma_A^{\rm mf}. \tag{5}$$

Now, using the chain rule for $\boldsymbol{x}_m = (\boldsymbol{x}_m^A, \boldsymbol{x}_m^B)$ in the natural A-first ordering appropriate to σ_A gives

$$\sigma_{\rm A} = \underbrace{\sigma_A^{\rm mf} + 4\mathcal{I}(\mathrm{d}\boldsymbol{x}_A; \boldsymbol{x}_m^A)}_{\sigma_A^{\rm self}} + \underbrace{4\mathcal{I}(\mathrm{d}\boldsymbol{x}_A; \boldsymbol{x}_m^B | \boldsymbol{x}_m^A)}_{\sigma_{A|B}^{\rm int}} \tag{6a}$$

$$= \left\langle \tilde{\boldsymbol{v}}_A^{\intercal} \mathbf{D}_A^{-1} \tilde{\boldsymbol{v}}_A \right\rangle + \left\langle (\boldsymbol{v}_A - \tilde{\boldsymbol{v}}_A)^{\intercal} \mathbf{D}_A^{-1} (\boldsymbol{v}_A - \tilde{\boldsymbol{v}}_A) \right\rangle, \tag{6b}$$

where $\tilde{\boldsymbol{v}}_A(\boldsymbol{x}_A) := \mathbb{E}[\boldsymbol{v}_A | \boldsymbol{x}_A]$. The midpoint identity and the Shannon chain rule thus yield an exact information-theoretic decomposition of the local EP rate into two nonnegative channels, Eq. (6a). Equation (6b) is the corresponding current-space representation, obtained by applying the small-SNR expansion [27] to Eq. (6a); the marginal-current projection underlying this representation and the nonnegative remainder it entails have precedents in earlier treatments of apparent EP and incomplete observation [7, 10]. What Eq. (6a) newly shows is that the midpoint identity maps the Shannon chain rule exactly onto this current-space decomposition, identifying its two terms with distinct midpoint information rates. See also [17, 28] for information-theoretic decompositions constructed from different dynamical settings.

For a fixed partition, we refer to the two contributions in Eq. (6a) as the *self* and *interaction* channels. This terminology is resolution-based. The self channel $\sigma_A^{\rm self}$ is the A-sector irreversibility resolved from A's own infinitesimal statistics of $(\mathrm{d}\boldsymbol{x}_A, \boldsymbol{x}_m^A)$ [29]. It may nevertheless depend on A's coupling to B through those marginal statistics. Note that when A is the full system, $\sigma_A^{\rm self}$ coincides with the total EP rate of the system. For a proper subsystem, however, $\sigma_A^{\rm self}$ is an instantaneous marginal-current contribution of the full Markovian dynamics and need not equal the complete path-space irreversibility of the generally non-Markovian A-only process [30]. The interaction channel $\sigma_{A|B}^{\rm int}$ is the additional A-sector irreversibility revealed by resolving $\boldsymbol{x}_m^B$ in addition to $\boldsymbol{x}_m^A$; it need not imply direct dynamical coupling. To illustrate these channels, we use a linear cascade model $z \to y \to x$, partitioned into $A = \{x, y\}$ and $B = \{z\}$ (Fig. 3a–c).

Interaction EP constrains learning.—The interaction channel has an operational meaning beyond being the remainder after subtracting the self channel. To make this statement precise, we begin with the recently derived learning-rate bound [20],

$$\left|\dot{I}^A(\boldsymbol{x}_A; \boldsymbol{x}_B)\right|^2 \mathcal{R}_A \le \sigma_A - \sigma_A^{\rm mf}. \tag{7}$$

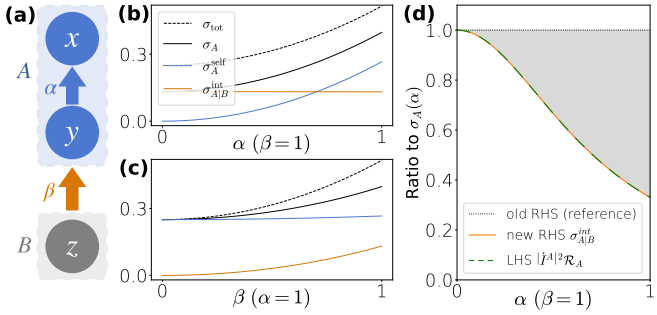

FIG. 3. Demonstration of the self and interaction EP channels and sharpened learning-rate bound. **(a)** Cascade model $\dot{x} = -2x + \alpha y + \sqrt{2}\xi_x$, $\dot{y} = -2y + \beta z + \sqrt{2}\xi_y$, and $\dot{z} = -2z + \sqrt{2}\xi_z$ with $\langle \xi_i \xi_j \rangle = \delta_{ij}$. The system is partitioned into $A = \{x, y\}$ (blue) and $B = \{z\}$ (gray). The coupling constant α controls the internal coupling within A, while β drives A from B. **(b, c)** Local EP rate $\sigma_{\rm A}$ (solid black) and its nonnegative channels, self $\sigma_A^{\rm self}$ (blue) and interaction $\sigma_{A|B}^{\rm int}$ (orange), as functions of coupling strength. The black dashed line denotes the total EP $\sigma_{\rm tot}$. Varying α (with $\beta = 1$) predominantly enhances $\sigma_A^{\rm self}$, while $\sigma_{A|B}^{\rm int}$ remains nearly unchanged (b). Varying β (with $\alpha = 1$) selectively enhances $\sigma_{A|B}^{\rm int}$ with minimal change in $\sigma_A^{\rm self}$ (c). **(d)** Tightening of the learning-rate bound along the same α sweep as in (b), with $\sigma_A^{\rm mf} = 0$ in this model: the previous RHS σ_A (black dotted), sharpened RHS $\sigma_{A|B}^{\rm int}$ (orange solid), and LHS $|\dot{I}^A|^2 \mathcal{R}_A$ (green dashed) are shown, all normalized by σ_A. The gray shading marks the gap between the two bounds. All three coincide at $\alpha = 0$; the sharpened RHS decreases monotonically to 0.330 at $\alpha = 1$, while the LHS nearly saturates throughout, $|\dot{I}^A|^2 \mathcal{R}_A / \sigma_{A|B}^{\rm int} > 0.999$.

Here, $\dot{I}^A$ is the learning rate and $\mathcal{R}_A := \mathrm{Tr}[\mathbf{D}_A \mathbf{F}_A^{|A}]^{-1}$ is the corresponding information-resistance factor, defined from the conditional Fisher information matrix $\mathbf{F}_A^{|A}$ of subsystem A. The channel representation in Eq. (6a) makes explicit that the RHS of Eq. (7) decomposes into two midpoint-information channels,

$$\sigma_A - \sigma_A^{\rm mf} = 4\mathcal{I}(\mathrm{d}\boldsymbol{x}_A; \boldsymbol{x}_m^A) + 4\mathcal{I}(\mathrm{d}\boldsymbol{x}_A; \boldsymbol{x}_m^B | \boldsymbol{x}_m^A).$$

Because $\dot{I}^A$ quantifies the instantaneous change in the statistical dependence between A and B generated by the A-dynamics [31, 32], this decomposition suggests that the corresponding dissipative cost should be controlled not by the A-only channel but by the $B|A$ conditional channel $4\mathcal{I}(\mathrm{d}\boldsymbol{x}_A; \boldsymbol{x}_m^B | \boldsymbol{x}_m^A) = \sigma_{A|B}^{\rm int}$. This structural prediction is rigorously established in current-space (see End Matter), yielding

$$\left|\dot{I}^A(\boldsymbol{x}_A; \boldsymbol{x}_B)\right|^2 \mathcal{R}_A \le \sigma_{A|B}^{\rm int}. \tag{8}$$

Thus, any nonzero A-learning rate necessarily draws on this channel, providing an operational basis for the term *interaction*.

The improvement over Eq. (7) is exactly $4\mathcal{I}(\mathrm{d}\boldsymbol{x}_A; \boldsymbol{x}_m^A)$. Hence, Eq. (8) provides a strictly tighter bound whenever $\mathcal{I}(\mathrm{d}\boldsymbol{x}_A; \boldsymbol{x}_m^A) > 0$. Figure 3d shows that this tightening can be substantial. As $\sigma_A^{\rm self}$ increases from zero,

the previous bound becomes increasingly loose, whereas the interaction-channel bound remains nearly saturated. This behavior illustrates that the self channel contributes to the looseness of the previous bound, while the interaction channel captures the dissipation relevant to learning. A similar substantial tightening is observed in a model fitted to red blood cell data [33]; see the SM.

Midpoint-information resolution of a coarse-graining reversal.—Recently, a counterintuitive coarse-graining reversal was reported in hydrodynamically coupled colloids [21]. Two harmonically trapped particles $X = \{x_1, x_2\}$ are coupled with strength ϵ, while an autonomous Ornstein–Uhlenbeck (OU) drive $\lambda(t)$ modulates the center of the first trap. Increasing ϵ lowers the total EP rate, but raises the particle-subspace EP rate (Fig. 4a,b). A subsequent study traced these trends to redistributed dependencies and information pathways using time-delayed MI and transfer entropy [22]. Here we make this connection exact at the EP level by showing how the physical coarse-graining $(X, \lambda) \mapsto X$ acts directly on the midpoint-information budget.

The linear dynamics can be written as

$$\dot{\boldsymbol{z}} = -\mathbf{F}\boldsymbol{z} + \sqrt{2\mathbf{D}}\,\boldsymbol{\xi}, \quad \boldsymbol{z} = (x_1, x_2, \lambda)^{\mathsf{T}}, \tag{9}$$

where

$$\mathbf{F} = \begin{pmatrix} 1/\tau_1 & \epsilon/\tau_2 & -1/\tau_1 \\ \epsilon/\tau_1 & 1/\tau_2 & -\epsilon/\tau_1 \\ 0 & 0 & 1/\tau_\lambda \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} D_0 & \epsilon D_0 & 0 \\ \epsilon D_0 & D_0 & 0 \\ 0 & 0 & \theta D_0 \end{pmatrix}.$$

Here, $\tau_{1,2}$ are the particle relaxation times, τ_λ is the OU correlation time, D_0 sets the particle diffusion scale, and θ controls the OU drive noise strength relative to that scale. Since $\mathbf{D}_{X\lambda} = 0$, the partition $\{X, \lambda\}$ satisfies the block-diagonal condition of Eq. (5). All mean-flow terms vanish in the centered steady state, so Eq. (5) gives

$$\sigma_{tot} = 4\mathcal{I}(\mathrm{d}X; X_m, \lambda_m) + 4\mathcal{I}(\mathrm{d}\lambda; X_m, \lambda_m).$$

The key point is that physical coarse-graining acts directly on this information budget, because the variables defining the resolution themselves appear as arguments of the MI terms. Under coarse-graining,

$$\{X, \lambda\} \mapsto X \implies \begin{cases} \mathcal{I}(\mathrm{d}X; X_m, \lambda_m) \mapsto \mathcal{I}(\mathrm{d}X; X_m), \\ \mathcal{I}(\mathrm{d}\lambda; X_m, \lambda_m) \mapsto \mathcal{I}(\emptyset; X_m) = 0. \end{cases}$$

The resulting term is precisely the particle-subspace EP σ_X^{sub} considered in the original studies [21, 22],

$$\sigma_X^{\mathrm{sub}} = \sigma_X^{\mathrm{self}} = 4\mathcal{I}(\mathrm{d}X; X_m), \tag{10}$$

as verified from its marginal-current definition given in the End Matter. For the λ-sector, the symmetric increment $\mathrm{d}\lambda$ is uncorrelated with λ_m in the stationary scalar OU marginal. Joint Gaussianity therefore gives $\mathcal{I}(\mathrm{d}\lambda; \lambda_m) = 0$. The Shannon chain rule then identifies

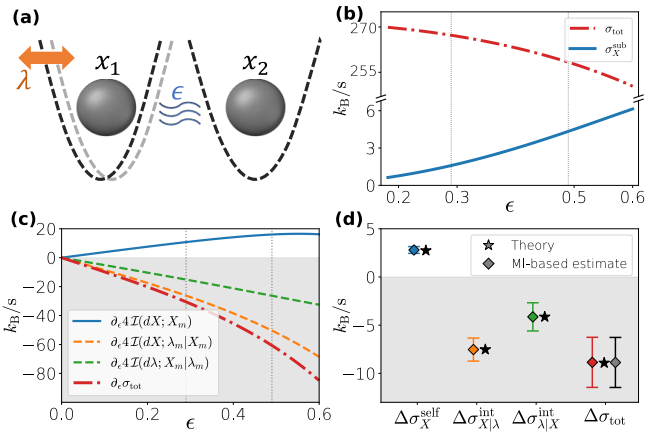

FIG. 4. Midpoint-information resolution of a coarse-graining reversal. Gray shading denotes negative values in panels (c) and (d). **(a)** Schematic of two harmonically trapped particles coupled by hydrodynamic interactions of strength ϵ. The Ornstein–Uhlenbeck drive $\lambda(t)$ displaces the center of the first trap. **(b)** Previously reported coarse-graining reversal. Increasing ϵ decreases the full-system EP rate σ_{tot} but increases the particle-subspace EP rate σ_X^{sub}. **(c)** Responses of the three channels in Eq. (11), $\partial_\epsilon \sigma_X^{\mathrm{self}}$ (blue solid), $\partial_\epsilon \sigma_{X|\lambda}^{\mathrm{int}}$ (orange dashed), and $\partial_\epsilon \sigma_{\lambda|X}^{\mathrm{int}}$ (green dashed), together with $\partial_\epsilon \sigma_{\mathrm{tot}}$ (red dash-dotted). Coarse-graining to X retains only the self channel, which is the only channel with a positive response. Vertical dotted lines mark $\epsilon = 0.29$ and 0.49. **(d)** Finite-resolution recovery of the channel changes $\Delta\sigma_\alpha \equiv \sigma_\alpha(0.49) - \sigma_\alpha(0.29)$. Stars denote continuous-time theoretical values. Diamonds show midpoint-MI rate estimates obtained from 40 independent synthetic trajectories at each value of ϵ, each sampled at 10 kHz for 50 s; error bars are 95% confidence intervals for the differences of ensemble means. For the total channel, the directly estimated change (red) and the sum of the three channel changes (gray) are shown side by side. Their difference is $-0.012\,k_{\mathrm{B}}\,\mathrm{s}^{-1}$, with a 95% confidence interval of $[-0.146, 0.123]\,k_{\mathrm{B}}\,\mathrm{s}^{-1}$. Panels (b–d) use $\theta = 0.56$.

two distinct conditional-information channels as exactly what is removed from each sector. Hence, the total EP has the exact three-channel budget

$$\sigma_{tot} = \underbrace{4\mathcal{I}(\mathrm{d}X; X_m)}_{\sigma_X^{\mathrm{self}} = \sigma_X^{\mathrm{sub}}} + \underbrace{4\mathcal{I}(\mathrm{d}X; \lambda_m|X_m)}_{\sigma_{X|\lambda}^{\mathrm{int}}} + \underbrace{4\mathcal{I}(\mathrm{d}\lambda; X_m|\lambda_m)}_{\sigma_{\lambda|X}^{\mathrm{int}}}. \tag{11}$$

The full-system EP σ_{tot} and the particle-subspace EP σ_X^{sub} thus belong to the same exact information budget. Hydrodynamic coupling redistributes this budget according to

$$\partial_\epsilon \sigma_{X|\lambda}^{\mathrm{int}}, \ \partial_\epsilon \sigma_{\lambda|X}^{\mathrm{int}} < -\partial_\epsilon \sigma_X^{\mathrm{self}} < 0. \tag{12}$$

This means that increasing ϵ suppresses both interaction channels more strongly than it enhances the self channel. As a result, σ_{tot} falls while σ_X^{self} rises. Coarse-graining to X retains only the growing self channel as σ_X^{sub}, producing the reported reversal (Fig. 4c).

These channel responses specify what information is redistributed. Stronger coupling makes X_m more infor-

mative about $\mathrm{d}X$, while λ_m supplies less additional information once X_m is known. The decrease of $\mathcal{I}(\mathrm{d}\lambda; X_m|\lambda_m)$ has a subtler retrodictive origin; a nested chain rule decomposition resolves how this memory is redistributed between the two particles (see End Matter). Using three-point trajectory windows and zero-time extrapolation, finite-resolution synthetic records recover the predicted MI and conditional MI channel changes, whose sum agrees with a direct estimate of $\Delta\sigma_{\mathrm{tot}}$ within uncertainty (Fig. 4d; see SM for estimation details).

Discussion.—Equivalent representations of EP need not be equally revealing for a given physical question. For overdamped Langevin dynamics, the path-space representation naturally exposes time reversal and fluctuation relations [2, 3], whereas the current-quadratic form reveals geometric and projection structures [8, 10]. Their distinct strengths reflect the native operations of the respective representations. The midpoint-information representation adds a third perspective, expressing dissipation itself through ordinary MI and making the Shannon chain rule a native operation on EP.

In additive bipartite Langevin systems, this algebraic access has two concrete consequences. Coarse-graining onto a subsystem removes a precise interaction channel of irreversibility rather than merely underestimating dissipation, which provides an exact resolution of the reported hydrodynamic coarse-graining reversal. The learning-rate bound, by contrast, couples selectively to the interaction channel and not to the self contribution. Taken together, these results show that, within this channel decomposition, coarse-graining and learning probe complementary parts of the same dissipation decomposition. The success of the chain rule in exposing this structure suggests that other information-theoretic operations, such as the data-processing inequality and partial information decomposition, could likewise provide new insights into dissipation. Extending the framework to non-bipartite architectures, underdamped dynamics, and other stochastic systems offers natural directions for future work.

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END MATTER

A. Current-space proof of the sharpened learning-rate bound Eq. (8)

Let $\boldsymbol{s}_A$ denote the conditional score and $\mathbf{F}_A^{|A}$ the corresponding conditional Fisher information matrix for subsystem A,

$$\boldsymbol{s}_A(\boldsymbol{x}) \coloneqq \nabla_{\boldsymbol{x}_A} \ln p(\boldsymbol{x}_B|\boldsymbol{x}_A), \quad \mathbf{F}_A^{|A} \coloneqq \langle \boldsymbol{s}_A \boldsymbol{s}_A^{\intercal} \rangle.$$

The learning rate satisfies [31]

$$\dot{I}_A(\boldsymbol{x}_A;\boldsymbol{x}_B) = \langle \boldsymbol{v}_A \cdot \boldsymbol{s}_A \rangle.$$

Moreover,

$$\mathbb{E}[\boldsymbol{s}_A|\boldsymbol{x}_A] = \int \mathrm{d}\boldsymbol{x}_B\, p(\boldsymbol{x}_B|\boldsymbol{x}_A) \nabla_{\boldsymbol{x}_A} \ln p(\boldsymbol{x}_B|\boldsymbol{x}_A) = 0,$$

so as anticipated by the midpoint-information representation, only the part of $\boldsymbol{v}_A$ not determined by $\boldsymbol{x}_A$ contributes to the learning rate:

$$\dot{I}_A(\boldsymbol{x}_A;\boldsymbol{x}_B) = \big\langle (\boldsymbol{v}_A - \tilde{\boldsymbol{v}}_A) \cdot \boldsymbol{s}_A \big\rangle.$$

Applying Cauchy–Schwarz inequality in the diffusion metric $\mathbf{D}_A$, we obtain

$$\left|\dot{I}_A(\boldsymbol{x}_A;\boldsymbol{x}_B)\right|^2 \le \Big\langle (\boldsymbol{v}_A - \tilde{\boldsymbol{v}}_A)^{\intercal} \mathbf{D}_A^{-1} (\boldsymbol{v}_A - \tilde{\boldsymbol{v}}_A) \Big\rangle \Big\langle \boldsymbol{s}_A^{\intercal} \mathbf{D}_A \boldsymbol{s}_A \Big\rangle. \tag{A.13}$$

The equality condition for the Cauchy–Schwarz inequality gives

$$\boldsymbol{v}_A - \tilde{\boldsymbol{v}}_A \propto \mathbf{D}_A \boldsymbol{s}_A$$

with a state-independent proportionality constant.

Equation (6b) identifies the first factor on the RHS of Eq. (A.13) as the interaction EP channel,

$$\Big\langle (\boldsymbol{v}_A - \tilde{\boldsymbol{v}}_A)^{\intercal} \mathbf{D}_A^{-1} (\boldsymbol{v}_A - \tilde{\boldsymbol{v}}_A) \Big\rangle = \sigma_{A|B}^{\mathrm{int}}.$$

The second factor is $\langle \boldsymbol{s}_A^{\intercal} \mathbf{D}_A \boldsymbol{s}_A \rangle = \mathrm{Tr}[\mathbf{D}_A \mathbf{F}_A^{|A}] = \mathcal{R}_A^{-1}$. Hence we get

$$|\dot{I}_A(\boldsymbol{x}_A;\boldsymbol{x}_B)|^2 \mathcal{R}_A \le \sigma_{A|B}^{\mathrm{int}},$$

which is Eq. (8).

B. Identification of the retained midpoint channel with particle-subspace EP

Denote by σ_X^{sub} the particle-subspace EP considered in [21, 22]. Its marginal current is

$$\boldsymbol{j}_X^{\mathrm{marg}}(X) \coloneqq \int \mathrm{d}\lambda\, \boldsymbol{j}_X(X,\lambda),$$

with $\tilde{\boldsymbol{v}}_X(X) \coloneqq \mathbb{E}[\boldsymbol{v}_X|X] = \boldsymbol{j}_X^{\mathrm{marg}}/p_X(X)$. All mean-flow terms vanish in the centered steady state, giving

$$\begin{aligned}\sigma_X^{\mathrm{sub}} &= \langle \tilde{\boldsymbol{v}}_X^{\intercal}\mathbf{D}_X^{-1}\tilde{\boldsymbol{v}}_X\rangle = \int dX \frac{(\boldsymbol{j}_X^{\mathrm{marg}})^{\intercal}\mathbf{D}_X^{-1}\boldsymbol{j}_X^{\mathrm{marg}}}{p_X} \\ &= \sigma_X^{\mathrm{app}} = \sigma_X^{\mathrm{self}} = 4\mathcal{I}(\mathrm{d}X; X_m).\end{aligned}$$

The equality $\sigma_X^{\mathrm{sub}} = \sigma_X^{\mathrm{app}}$ is the known marginal-current projection [7, 34], whereas the final equality is its midpoint-information representation of Eq. (6a).

Importantly, this identification is not specific to the present linear model. At the level of the instantaneous marginal current, the apparent EP admits the same midpoint-information representation for general nonlinear dynamics and does not rely on the bipartite structure used for the full channel decomposition. More generally, restoring a possible mean-flow gives

$$\sigma_A^{\mathrm{app}} = \sigma_A^{\mathrm{self}} = 4\mathcal{I}(d\boldsymbol{x}_A; \boldsymbol{x}_m^A) + \sigma_A^{\mathrm{mf}}.$$

Thus, the midpoint representation provides an information-theoretic interpretation of the conventional apparent EP. As emphasized in the main text, however, this instantaneous marginal-current quantity generally differs from the full path-space irreversibility of the A-only trajectory, which is generally non-Markovian. For example, in a stationary confined one-dimensional subsystem, $j_A^{\mathrm{marg}} = 0$ and hence $\sigma_A^{\mathrm{self}} = 0$, even though its path-space irreversibility can be positive.

C. Information-theoretic origin of the decrease in $\sigma^{\mathrm{int}}_{\lambda|X} = 4\mathcal{I}(\mathrm{d}\lambda; X_m|\lambda_m)$

Although the causal coupling in the dynamics in Eq. (9) runs from λ to X, $\mathcal{I}(\mathrm{d}\lambda; X_m|\lambda_m)$ is nonzero. The reason is that X_m is the actual temporal midpoint. Writing

$$\mathrm{d}\lambda = [\lambda(t+\mathrm{d}t) - \lambda_m] + [\lambda_m - \lambda(t)],$$

the earlier half-step $\lambda_m - \lambda(t)$ is generally not independent of X_m, even when conditioned on λ_m, because X_m retains information about the recent history of λ through the lagged dynamical response of the particle system. Thus, this channel measures a retrodictive, lagged memory rather than a causal influence from X back to λ. In this sense, the "interaction" captured by $\sigma^{\mathrm{int}}_{\lambda|X}$ reflects memory generated by the dynamical coupling.

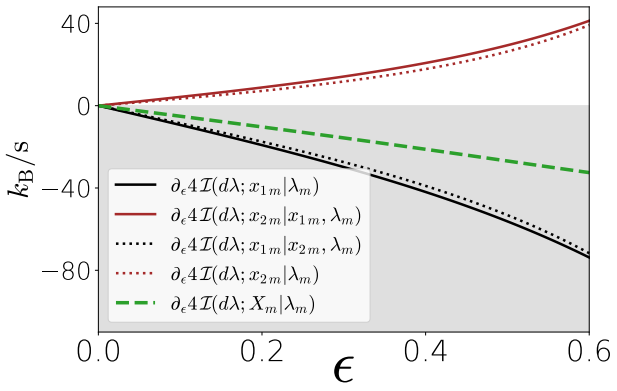

FIG. 5. Nested chain rule decomposition of $\sigma^{\mathrm{int}}_{\lambda|X} = 4\mathcal{I}(\mathrm{d}\lambda; X_m|\lambda_m)$. The curves show the two contributions under the x_1-first (solid) and x_2-first (dotted) chain rule orderings. Regardless of the ordering, the x_1 contribution $\partial_\epsilon 4\mathcal{I}(\mathrm{d}\lambda; x_{1m}|\cdot)$ (black) is negative, whereas the x_2 contribution $\partial_\epsilon 4\mathcal{I}(\mathrm{d}\lambda; x_{2m}|\cdot)$ (brown) is positive. Their sum is independent of the ordering and equals $\partial_\epsilon 4\mathcal{I}(\mathrm{d}\lambda; X_m|\lambda_m)$ (green dashed), which remains negative throughout. All curves are evaluated at $\theta = 0.56$. The horizontal gray line marks zero, and the gray shading denotes negative response. The two vertical gray lines indicate $\epsilon = 0.29$ and 0.49, the coupling strengths analyzed in the experiments.

To resolve how this memory is redistributed between the two particles, consider the x_1-first chain rule decomposition

$$\mathcal{I}(\mathrm{d}\lambda; X_m|\lambda_m) = \mathcal{I}(\mathrm{d}\lambda; x_{1m}|\lambda_m) + \mathcal{I}(\mathrm{d}\lambda; x_{2m}|x_{1m}, \lambda_m).$$

As ϵ increases, the first term decreases whereas the second increases. Stronger hydrodynamic coupling between x_1 and x_2 effectively loads x_1 through its coupling to x_2, thereby reducing the memory of λ retained in x_{1m}. At the same time, x_{2m} acquires part of this memory through the hydrodynamic coupling. This gain, however, is insufficient to compensate for the memory lost by x_{1m}, so the total $\mathcal{I}(\mathrm{d}\lambda; X_m|\lambda_m)$ decreases. Reversing the chain rule ordering of $\{x_{1m}, x_{2m}\}$ preserves these signs and their qualitative magnitude relation, as shown in Fig. 5.

SUPPLEMENTAL MATERIALS

I. DERIVATION OF THE MAIN IDENTITY

The main text focuses on the spatially homogeneous (additive-noise) case for clarity. In this Supplement we establish the general multiplicative-noise result, and recover the additive-noise statement of the main text as a special case. The derivation proceeds as follows. Secs I.A–I.B set notation and assumptions. Sec. I.C recalls the reverse-time diffusion. Sec. I.D derives the midpoint half-step decomposition and the resulting short-time representation of the midpoint-whitened increment. Sec. I.E proves the cancellation of the $O(\sqrt{\mathrm{d}t})$ non-Gaussian terms under midpoint conditioning, and Sec. I.F shows that the remaining deviation from the associated Gaussian proxy contributes only an $o(\mathrm{d}t)$ term and yields the multiplicative-noise identity. Throughout the SM, uppercase $\boldsymbol{X}_t$ denotes the stochastic process, lowercase $\boldsymbol{x}$ a conditioned value or integration variable.

A. Noise convention, model settings, and the midpoint notation

• **Noise convention.** We use a d-dimensional Gaussian white noise $\boldsymbol{\xi}(t)$ with

$$\langle \boldsymbol{\xi}(t)\boldsymbol{\xi}(t')^{\intercal}\rangle = \mathbf{I}_d\,\delta(t-t'). \tag{S1}$$

Equivalently, we introduce a d-dimensional standard Wiener process W_t satisfying $\mathbb{E}[\mathrm{d}W_t] = 0$ and $\mathbb{E}[\mathrm{d}W_t\,\mathrm{d}W_t^{\intercal}] = \mathbf{I}_d\,\mathrm{d}t$. Formally, one may view $\mathrm{d}W_t$ as the time integral of the white noise, i.e. $\mathrm{d}W_t = \boldsymbol{\xi}(t)\,\mathrm{d}t$.

• **Dynamics.** We consider the Itô SDE

$$\mathrm{d}\boldsymbol{X}_t = \boldsymbol{F}_I(\boldsymbol{X}_t,t)\mathrm{d}t + \mathbf{B}(\boldsymbol{X}_t,t)\mathrm{d}W_t, \quad \boldsymbol{X}_t \in \mathbb{R}^d. \tag{S2}$$

We define the diffusion matrix $\mathbf{D}(\boldsymbol{x},t) \coloneqq \frac{1}{2}\mathbf{a}(\boldsymbol{x},t)$, where $\mathbf{a}(\boldsymbol{x},t) \coloneqq \mathbf{B}(\boldsymbol{x},t)\mathbf{B}(\boldsymbol{x},t)^{\intercal}$. Throughout, we fix $\mathbf{B}$ to be the symmetric positive-definite square root of $\mathbf{a}$ (i.e., $\mathbf{B} = \mathbf{a}^{1/2}$), which implies $\mathbf{B} = \mathbf{B}^{\intercal}$. (In a physical overdamped Langevin form $\dot{\boldsymbol{x}} = \boldsymbol{\mu}(\boldsymbol{x},t)\boldsymbol{f}(\boldsymbol{x},t) + \mathbf{B}(\boldsymbol{x},t)\boldsymbol{\xi}$, one may absorb the mobility $\boldsymbol{\mu}$ into the drift and relabel $\boldsymbol{\mu f} \mapsto \boldsymbol{F}_I$; our result depends only on the resulting Itô drift $\boldsymbol{F}_I$ and diffusion matrix $\mathbf{D}$.)

• **Current and current velocity.** Let $p(\boldsymbol{x},t)$ denote the one-time density of $\boldsymbol{X}_t$. It is well known that p evolves according to the Fokker-Planck equation, which can be written in conservation (continuity) form $\partial_t p = -\nabla\cdot\boldsymbol{j}$ with probability current

$$\boldsymbol{j}(\boldsymbol{x},t) \coloneqq \boldsymbol{F}_I(\boldsymbol{x},t)p(\boldsymbol{x},t) - \frac{1}{2}\nabla\cdot(\mathbf{a}(\boldsymbol{x},t)p(\boldsymbol{x},t)) = \boldsymbol{F}_I\,p - \nabla\cdot(\mathbf{D}p), \tag{S3}$$

where $(\nabla\cdot(\mathbf{a}p))_i \coloneqq \sum_j \partial_{x_j}(\mathsf{a}_{ij}p)$. This form of the probability current is standard in the Fokker-Planck description [35, 36]. Following the standard representation [5, 37], we further introduce the current velocity

$$\boldsymbol{v}(\boldsymbol{x},t) \coloneqq \frac{\boldsymbol{j}(\boldsymbol{x},t)}{p(\boldsymbol{x},t)} = \boldsymbol{F}_I - \frac{\nabla\cdot(\mathbf{D}p)}{p}. \tag{S4}$$

• **Entropy production rate.** For overdamped diffusion processes, under suitable smoothness and regularity assumptions (to be specified in the next section), the instantaneous total entropy production rate σ_t can be expressed in terms of the probability current as

$$\sigma_t \;=\; \int \mathrm{d}\boldsymbol{x}\;\frac{\boldsymbol{j}(\boldsymbol{x},t)^{\intercal}\mathbf{D}(\boldsymbol{x},t)^{-1}\boldsymbol{j}(\boldsymbol{x},t)}{p(\boldsymbol{x},t)} \;=\; \big\langle \boldsymbol{v}(\boldsymbol{X}_t,t)^{\intercal}\mathbf{D}(\boldsymbol{X}_t,t)^{-1}\boldsymbol{v}(\boldsymbol{X}_t,t)\big\rangle, \tag{S5}$$

where we used $\boldsymbol{j} = p\boldsymbol{v}$. This is a standard result in stochastic thermodynamics [5].

• **Time Midpoint notation.** Fix a small $\mathrm{d}t > 0$ and define

$$\mathrm{d}\boldsymbol{x}(t) \coloneqq \boldsymbol{X}_{t+\mathrm{d}t} - \boldsymbol{X}_t, \quad t_m \coloneqq t + \frac{\mathrm{d}t}{2}, \quad \boldsymbol{X}_m \coloneqq \boldsymbol{X}_{t_m}. \tag{S6}$$

• **Time midpoint whitened increment.** Let $\mathbf{B}_m \coloneqq \mathbf{B}(\boldsymbol{X}_m, t_m)$. We define

$$\boldsymbol{\eta}_t \coloneqq \frac{1}{\sqrt{\mathrm{d}t}} \mathbf{B}_m^{-1} \mathrm{d}\boldsymbol{x}(t). \tag{S7}$$

Here, we emphasize that the scaling $\mathbf{B}_m^{-1}\mathrm{d}\boldsymbol{x}(t) \mapsto \boldsymbol{\eta}_t$ is deterministic for fixed $\mathrm{d}t$ and therefore preserves mutual information: $I(\mathbf{B}_m^{-1}\mathrm{d}\boldsymbol{x}(t); \boldsymbol{X}_m) = I(\boldsymbol{\eta}_t; \boldsymbol{X}_m)$. In contrast, the whitening map $\mathrm{d}\boldsymbol{x} \mapsto \mathbf{B}_m^{-1}\mathrm{d}\boldsymbol{x}$ depends on the random midpoint $\boldsymbol{X}_m$ and thus does not preserve $I(\mathrm{d}\boldsymbol{x}; \boldsymbol{X}_m)$ in general. We therefore study $I(\mathbf{B}_m^{-1}\mathrm{d}\boldsymbol{x}; \boldsymbol{X}_m)$ as the natural diffusion-metric-normalized quantity, which reduces to $I(\mathrm{d}\boldsymbol{x}; \boldsymbol{X}_m)$ in the additive-noise case (constant $\mathbf{B}$).

B. Assumptions ensuring smooth transition densities and well-defined entropy production rates

We list explicit, checkable sufficient conditions on the coefficients and the law of $\boldsymbol{X}_t$ under which all objects defined in Section I.A are well-posed (in particular the current $\boldsymbol{j}$, current velocity $\boldsymbol{v} = \boldsymbol{j}/p$, and the entropy production rate σ_t), and under which we may invoke standard results in the diffusion literature (time reversal and short-time density expansions) [23–26, 38–43]. Our goal is to establish the main identity rigorously within this broad and physically natural class, rather than to optimize the weakest possible hypotheses.

(A1) Uniform ellipticity. There exist $0 < \lambda \le \Lambda < \infty$ such that $\lambda I \le \mathbf{D}(\boldsymbol{x}, t) \le \Lambda I$ for all $(\boldsymbol{x}, t)$.

(A2) Regularity and smoothness of coefficients. For each t, $\boldsymbol{F}_I(\cdot, t) \in C^\infty(\mathbb{R}^d)$ and $\mathbf{B}(\cdot, t) \in C_b^\infty(\mathbb{R}^d)$ in $\boldsymbol{x}$, where $C_b^\infty(\mathbb{R}^d)$ denotes the class of smooth functions whose spatial derivatives of all orders are globally bounded. Also for each $\boldsymbol{x}$, both $\boldsymbol{F}_I(\boldsymbol{x}, \cdot)$, $\mathbf{B}(\boldsymbol{x}, \cdot)$ and their spatial derivatives are C^1 in time on $[t, t + \mathrm{d}t]$. We denote by $\partial_{\boldsymbol{x}}^a(\cdot)$ and $\partial_t^b(\cdot)$ the corresponding spatial and time derivatives of order a and b.

(A3) Well-posedness and smooth densities. The SDE (S2) admits a unique strong non-explosive solution. For each $t > 0$, the law of $\boldsymbol{X}_t$ admits a strictly positive density $p(\cdot, t)$ such that $p(\cdot, t)$ and $\partial_t p(\cdot, t)$ are $C^\infty(\mathbb{R}^d)$ in $\boldsymbol{x}$. Consequently, the Fokker-Planck equation holds in the classical sense and the current defined in (S3) is well-defined.

(A4) Polynomial growth bounds (drift, score and their derivatives). There exist a constant $C > 0$ and an integer $r \ge 0$ such that, uniformly for $u \in [t, t + \mathrm{d}t]$,

$$\|\partial_{\boldsymbol{x}}^a \boldsymbol{F}_I(\boldsymbol{x}, u)\| + \|\partial_{\boldsymbol{x}}^b \log p(\boldsymbol{x}, u)\| \le C(1 + \|\boldsymbol{x}\|^r), \quad a, b \in \mathbb{N} \cup \{0\}$$

(A5) Non-explosion and moment control. We assume that the process is non-explosive and that sufficiently high moments are finite on the time interval considered. Concretely, we impose either **(A5-A)** or **(A5-B)** below.

• **(A5-A) Bounded drift and current case.** The drift $\boldsymbol{F}_I(\boldsymbol{x}, u)$ and current velocity $\boldsymbol{v}(\boldsymbol{x}, u)$ in (S4) are uniformly bounded in $(\boldsymbol{x}, u) \in \mathbb{R}^d \times [t, t + \mathrm{d}t]$. Also all moments exist, i.e.

$$\sup_{u \in [t, t+\mathrm{d}t]} \mathbb{E}[\|\boldsymbol{X}_u\|^m] < \infty, \quad \forall m \in \mathbb{N}.$$

• **(A5-B) Confined system case.** There exist $\alpha, \theta > 0$ such that

$$\sup_{u \in [t, t+\mathrm{d}t]} \mathbb{E}\big[\exp\big(\theta \|\boldsymbol{X}_u\|^{1+\alpha}\big)\big] < \infty, \quad u \in [t, t + \mathrm{d}t]$$

Assumptions A1–A5 should be viewed as an explicit statement of the regularity conditions that are usually left implicit when one says that the dynamics is a smooth overdamped Langevin diffusion. Physically, A1 is the standard non-degeneracy of the noise, A2–(A3) give the usual smooth well-posed diffusion setting with smooth one-time densities, A4 restricts the drift/score sector to at most polynomial growth, and A5 supplies the corresponding non-explosion and moment control. The two alternatives in A5 cover the most common physical situations: A5-A describes effectively bounded dynamics, including compact spaces such as motion on a ring, while A5-B is tailored to unbounded domains with standard Lyapunov-type dissipative drift. Under these assumptions, the time-reversal and short-time expansions used below are standard, and the entropy production rate in (S5) is finite. For later reference, A1–A3 justify the time-reversal formula and local short-time expansions, A4 controls the polynomial-growth coefficients appearing in the remainder bounds, and A5 is used only for moment/tail control and the localization arguments.

These assumptions are not intended to be minimal. They are a convenient sufficient package without repeated technical bookkeeping. For any fixed truncation order, corresponding finite order hypotheses would suffice. Non-smooth or geometric variants (e.g., periodic identifications, reflecting boundaries, or piecewise-defined drifts) can often be treated by standard localization/truncation and smoothing arguments, but we do not pursue such extensions here.

C. Time reversal diffusion

We briefly recall the reverse-time diffusion needed for the midpoint construction. Under assumptions A1-–A3, the time reversal of an Itô diffusion is again a diffusion with the same diffusion matrix and an explicit reversed drift [25, 26]. Specializing that standard formula at the midpoint yields (S10), the identity used below.

A key ingredient is the drift of the time-reversed diffusion. Consider an Itô diffusion on a finite horizon $[0, T]$

$$\mathrm{d}\boldsymbol{X}_t = \boldsymbol{b}(\boldsymbol{X}_t, t)\mathrm{d}t + \mathbf{B}(\boldsymbol{X}_t, t)\mathrm{d}W_t\,,$$

which satisfies the assumptions A1–A3. Under these assumptions, let $p(\boldsymbol{x}, t)$ denote the one-time density of $\boldsymbol{X}_t$. We then define the time-reversed process by $\bar{\boldsymbol{X}}_t \coloneqq \boldsymbol{X}_{T-t}$ for $t \in [0, T]$. With this definition, $\bar{\boldsymbol{X}}_t$ is again a diffusion process with the same diffusion matrix $\mathbf{a} \coloneqq \mathbf{B}\mathbf{B}^\intercal = 2\mathbf{D}$ and reversed drift $\bar{\boldsymbol{b}}$ given in component form by

$$\bar{b}_i(\boldsymbol{x}, t) = -b_i(\boldsymbol{x}, T-t) + \frac{1}{p(\boldsymbol{x}, T-t)} \sum_{j=1}^{d} \partial_{x_j}\left(\mathsf{a}_{ij}\left(\boldsymbol{x}, T-t\right) p(\boldsymbol{x}, T-t)\right). \tag{S8}$$

Equivalently, in vector notation,

$$\bar{\boldsymbol{b}}(\boldsymbol{x}, t) = -\boldsymbol{b}(\boldsymbol{x}, T-t) + \frac{\nabla \cdot (\mathbf{a}(\boldsymbol{x}, T-t)\, p(\boldsymbol{x}, T-t))}{p(\boldsymbol{x}, T-t)},$$

where $(\nabla \cdot (\mathbf{a}p))_i \coloneqq \sum_j \partial_{x_j}(\mathsf{a}_{ij} p)$ [26].

Specializing this general formula to the local reversal around the midpoint time t_m (i.e. set $T = t_m$ and evaluate at reversed time $t = 0$), and writing the reverse-time drift at time t_m as $\boldsymbol{F}_{rev}(\boldsymbol{x}, t_m) \coloneqq \bar{\boldsymbol{b}}(\boldsymbol{x}, 0)$, we obtain the local identity

$$\boldsymbol{F}_{rev}(\boldsymbol{x}, t_m) = -\boldsymbol{F}_I(\boldsymbol{x}, t_m) + \frac{\nabla \cdot (\mathbf{a}(\boldsymbol{x}, t_m)\, p(\boldsymbol{x}, t_m))}{p(\boldsymbol{x}, t_m)}. \tag{S9}$$

Combining Eq. (S3), (S4) with (S9) at time t_m yields the exact algebraic relation,

$$\frac{\boldsymbol{F}_I(\boldsymbol{x}, t_m) - \boldsymbol{F}_{rev}(\boldsymbol{x}, t_m)}{2} = \boldsymbol{F}_I(\boldsymbol{x}, t_m) - \frac{\nabla \cdot (\mathbf{D}\, p)}{p}(\boldsymbol{x}, t_m) = \frac{\boldsymbol{j}}{p}(\boldsymbol{x}, t_m) = \boldsymbol{v}(\boldsymbol{x}, t_m). \tag{S10}$$

Equation (S10) is the algebraic reason midpoint conditioning is special since it turns the difference between forward and reverse half-step drifts into the current velocity.

D. Time midpoint half-step decomposition

We now focus on the time-midpoint half-step decomposition, which is the key ingredient of the proof. We first introduce the notation and rewrite the normalized time-midpoint whitened increment $\boldsymbol{\eta}_t$ defined in Eq. (S7) from the perspective of time-midpoint conditioning. Hereafter, we use the term "midpoint" to refer to the time midpoint. For convenience, we define

$$h := \mathrm{d}t/2, \quad \boldsymbol{\Delta}^+ := \boldsymbol{X}_{t_m+h} - \boldsymbol{X}_{t_m}, \quad \boldsymbol{\Delta}^- := \boldsymbol{X}_{t_m} - \boldsymbol{X}_{t_m-h}.$$

Then, by the definition, the total increment decomposes as $\mathrm{d}\boldsymbol{x} = \boldsymbol{\Delta}^+ + \boldsymbol{\Delta}^-$. Note that the triplet $(\boldsymbol{X}_t, \boldsymbol{X}_m, \boldsymbol{X}_{t+dt})$ can equivalently be represented as $(\boldsymbol{X}_{t_m-h}, \boldsymbol{X}_{t_m}, \boldsymbol{X}_{t_m+h})$. This representation leads to the following key property:

Lemma S1 (conditional independence). By the Markov property, conditional on $\boldsymbol{X}_m \coloneqq \boldsymbol{X}_{t_m} = \boldsymbol{x}$, the past endpoint $\boldsymbol{X}_{t_m-h}$ and the future endpoint $\boldsymbol{X}_{t_m+h}$ are independent. Consequently, any functionals of the $\boldsymbol{\Delta}^+$ and $\boldsymbol{\Delta}^-$ are conditionally independent given $\boldsymbol{X}_m$.

Proof. For the joint distribution $p(\boldsymbol{X}_{t_m-h} = \boldsymbol{x}^-, \boldsymbol{X}_m = \boldsymbol{x}, \boldsymbol{X}_{t_m+h} = \boldsymbol{x}^+)$, we can write

$$p(\boldsymbol{X}_{t_m-h} = \boldsymbol{x}^-, \boldsymbol{X}_m = \boldsymbol{x}, \boldsymbol{X}_{t_m+h} = \boldsymbol{x}^+) = p(\boldsymbol{X}_{t_m-h} = \boldsymbol{x}^-, \boldsymbol{X}_{t_m+h} = \boldsymbol{x}^+ | \boldsymbol{X}_m = \boldsymbol{x}) p(\boldsymbol{X}_m = \boldsymbol{x}).$$

We also have

$$\begin{aligned} p(\boldsymbol{X}_{t_m-h} = \boldsymbol{x}^-, \boldsymbol{X}_m = \boldsymbol{x}, \boldsymbol{X}_{t_m+h} = \boldsymbol{x}^+) &= p(\boldsymbol{X}_{t_m+h} = \boldsymbol{x}^+ | \boldsymbol{X}_m = \boldsymbol{x}) p(\boldsymbol{X}_m = \boldsymbol{x}, \boldsymbol{X}_{t_m-h} = \boldsymbol{x}^-) \\ &= p(\boldsymbol{X}_{t_m+h} = \boldsymbol{x}^+ | \boldsymbol{X}_m = \boldsymbol{x}) p(\boldsymbol{X}_{t_m-h} = \boldsymbol{x}^- | \boldsymbol{X}_m = \boldsymbol{x}) p(\boldsymbol{X}_m = \boldsymbol{x}), \end{aligned}$$

where the first equality holds because of the Markov property. Hence we get

$$p(\boldsymbol{X}_{t_m-h}=\boldsymbol{x}^-,\boldsymbol{X}_{t_m+h}=\boldsymbol{x}^+|\boldsymbol{X}_m=\boldsymbol{x})=p(\boldsymbol{X}_{t_m+h}=\boldsymbol{x}_+|\boldsymbol{X}_m=\boldsymbol{x})p(\boldsymbol{X}_{t_m-h}=\boldsymbol{x}^-|\boldsymbol{X}_m=\boldsymbol{x}),$$

which means $\boldsymbol{X}_{t_m+h} \perp\!\!\!\perp \boldsymbol{X}_{t_m-h}\,|\boldsymbol{X}_m$. □

Throughout this subsection, we condition on $\boldsymbol{X}_m \coloneqq \boldsymbol{X}_{t_m}=\boldsymbol{x}$.

• **Conditional mean and covariance**

Lemma S1 makes the conditional mean and covariance of $\mathrm{d}\boldsymbol{x}$ easy to compute because $\boldsymbol{\Delta}^+$ and $\boldsymbol{\Delta}^-$ are conditionally independent given $\boldsymbol{X}_m=\boldsymbol{x}$. For the forward half-step, by using standard properties of Itô diffusions, we obtain

$$\mathbb{E}[\boldsymbol{\Delta}^+|\boldsymbol{X}_m=\boldsymbol{x}]=\boldsymbol{F}_I(\boldsymbol{x},t_m)\,h+o(h). \tag{S11}$$

For the backward half-step, define the reversed process $\bar{\boldsymbol{X}}_s \coloneqq \boldsymbol{X}_{t_m-s}$. Then $\bar{\boldsymbol{X}}$ is a diffusion whose drift at $s=0$ equals $\boldsymbol{F}_{rev}(\boldsymbol{x},t_m)$ (Section C). Hence

$$\mathbb{E}\left[\boldsymbol{X}_{t_m-h}-\boldsymbol{X}_{t_m}\,|\,\boldsymbol{X}_m=\boldsymbol{x}\right]=\mathbb{E}\left[\bar{\boldsymbol{X}}_h-\bar{\boldsymbol{X}}_0\,|\,\bar{\boldsymbol{X}}_0=\boldsymbol{x}\right]=\boldsymbol{F}_{rev}(\boldsymbol{x},t_m)\,h+o(h),$$

equivalently

$$\mathbb{E}\left[\boldsymbol{\Delta}^-|\boldsymbol{X}_m=\boldsymbol{x}\right]=-\boldsymbol{F}_{rev}(\boldsymbol{x},t_m)\,h+o(h). \tag{S12}$$

Adding (S11) and (S12) with the relation Eq. (S10) yields

$$\mathbb{E}[\mathrm{d}\boldsymbol{x}|\boldsymbol{X}_m=\boldsymbol{x}]=\boldsymbol{v}(\boldsymbol{x},t_m)\,\mathrm{d}t+o(\mathrm{d}t). \tag{S13}$$

For the covariance, conditional independence (Lemma S1) gives

$$\mathrm{Cov}(\mathrm{d}\boldsymbol{x}\,|\,\boldsymbol{X}_m=\boldsymbol{x})=\mathrm{Cov}(\boldsymbol{\Delta}^+\,|\boldsymbol{X}_m=\boldsymbol{x})+\mathrm{Cov}(\boldsymbol{\Delta}^-\,|\boldsymbol{X}_m=\boldsymbol{x}). \tag{S14}$$

By the standard second-moment expansions for Itô diffusions (or Euler-Maruyama with frozen coefficients [23, 24]),

$$\mathrm{Cov}(\boldsymbol{\Delta}^+\,|\boldsymbol{X}_m=\boldsymbol{x})=\mathbf{a}(\boldsymbol{x},t_m)\,h+o(h). \tag{S15}$$

The reversed diffusion has the same diffusion matrix $\mathbf{a}(\cdot,t_m)$ at $s=0$, hence likewise

$$\mathrm{Cov}(\boldsymbol{\Delta}^-\,|\boldsymbol{X}_m=\boldsymbol{x})=\mathbf{a}(\boldsymbol{x},t_m)\,h+o(h). \tag{S16}$$

Combining (S14)-(S16) gives

$$\mathrm{Cov}(\mathrm{d}\boldsymbol{x}(t)|\boldsymbol{X}_m=\boldsymbol{x})=\mathbf{a}(\boldsymbol{x},t_m)\,\mathrm{d}t+o(\mathrm{d}t). \tag{S17}$$

At this stage the midpoint-conditioned increment already has the correct mean and covariance. The only remaining question is whether multiplicative-noise corrections beyond Gaussian order can affect $I(\boldsymbol{\eta}_t;\boldsymbol{X}_m)$ at order $\mathrm{d}t$.

• **Strong Itô-Taylor expansion and normalized sum**

For short-time approximations of an Itô SDE up to order $O(\mathrm{d}t)$, the Euler–Maruyama (EM) scheme is well known. However, in a strict sense, there exist additional effects of order $O(\mathrm{d}t)$ that influence the distribution beyond the mean drift in multiplicative-noise case. These include deformations of the local shape of the distribution and, in the presence of multidimensional noise, orientation or circulation effects at the pathwise level. Although these contributions have zero mean and therefore do not appear as drift terms, their impact on higher-order statistics beyond the mean and leading-order covariance is of order $O(\mathrm{d}t)$ [24, 44]. The mutual information, however, can be sensitive to these effects. Therefore, to evaluate the mutual information accurately up to order $O(\mathrm{d}t)$, it is necessary to go beyond the EM scheme and consider the strong Itô-Taylor expansion, which incorporates all such $O(\mathrm{d}t)$ contributions.

We use this framework at the midpoint, where both half-steps are expanded around the same space-time point $(\boldsymbol{x},t_m)$ and thus share the same $\mathbf{B}(\boldsymbol{x},t_m)$ and its spatial derivatives (see subsection C). Because of sufficient smoothness and regularity conditions A1-A4, the Itô-Taylor expansion to strong order one is valid locally. Then we can write the normalized and whitened half-steps as

$$\mathbf{B}(\boldsymbol{x},t_m)^{-1}\frac{\boldsymbol{\Delta}^+}{\sqrt{h}}=\boldsymbol{Z}_++\sqrt{h}\,\mathbf{B}_m^{-1}\boldsymbol{F}_I(\boldsymbol{x},t_m)+\sqrt{h}\mathcal{T}_1(\boldsymbol{x}):\left(\boldsymbol{Z}_+\boldsymbol{Z}_+^{\intercal}-\mathbf{I}_d\right)+\sqrt{h}\mathcal{T}_2(\boldsymbol{x}):\mathbf{A}_++O_{L^2}(h), \tag{S18}$$

$$\mathbf{B}(\boldsymbol{x},t_m)^{-1}\frac{\boldsymbol{\Delta}^-}{\sqrt{h}}=-\boldsymbol{Z}_--\sqrt{h}\,\mathbf{B}_m^{-1}\boldsymbol{F}_{rev}(\boldsymbol{x},t_m)-\sqrt{h}\mathcal{T}_1(\boldsymbol{x}):\left(\boldsymbol{Z}_-\boldsymbol{Z}_-^{\intercal}-\mathbf{I}_d\right)-\sqrt{h}\mathcal{T}_2(\boldsymbol{x}):\mathbf{A}_-+O_{L^2}(h), \tag{S19}$$

where the symbol ":" denotes tensor contraction, i.e. $(\mathcal{T} : \mathbf{M})_i := \sum_{j,k=1}^{d} \mathcal{T}_{ijk}\mathbf{M}_{jk}$. Here, $\mathcal{T}_1(\boldsymbol{x})$ and $\mathcal{T}_2(\boldsymbol{x})$ are deterministic tensors depending on $\mathbf{B}(\boldsymbol{x}, t_m)$ and its spatial derivatives evaluated at $(\boldsymbol{x}, t_m)$ (hence they vanish in the additive-noise case). On the other hand, $\boldsymbol{Z}_+, \boldsymbol{Z}_- \sim \mathcal{N}(0, \mathbf{I}_d)$ are normalized Gaussian random variables, and the matrices $\mathbf{A}_\pm$ represent the normalized Lévy-area contribution, defined by

$$(\mathbf{A}_\pm)_{ij} := \frac{1}{2h}\left(\mathbf{J}^{\pm}_{ij} - \mathbf{J}^{\pm}_{ji}\right), \quad \mathbf{J}^{\pm}_{ij} := \int_0^h \left(\int_0^s \mathrm{d}W_u^{\pm,(i)}\right) \mathrm{d}W_s^{\pm,(j)}, \tag{S20}$$

where $\boldsymbol{W}^\pm$ denote d-dimensional standard Wiener processes used in the auxiliary representation below, and $d\boldsymbol{W}^\pm$ denotes their Itô increments. Note that $\|\mathbf{A}^\pm\| = O_{L^2}(1)$ by definition. Roughly, while the first two terms on the RHS of (S18)–(S19) determine the mean and leading order of covariance of the distribution (and this is exactly what the EM scheme captures), the third term $\mathcal{T}_1 : (\boldsymbol{Z}\boldsymbol{Z}^\intercal - \mathrm{I}_d)$ modulates the "shape" of the distribution and the fourth term $\mathcal{T}_2 : \mathbf{A}$ effectively "twists" the individual paths.

We emphasize two key points: First, by Lemma S1, $\boldsymbol{\Delta}^+$ and $\boldsymbol{\Delta}^-$ are independent conditional on $\boldsymbol{X}_m = \boldsymbol{x}$. Therefore, we can represent the two half-step expansions using conditionally independent auxiliary random variables $(\boldsymbol{Z}_+, \mathbf{A}_+)$ and $(\boldsymbol{Z}_-, \mathbf{A}_-)$, which are conditionally independent given $\boldsymbol{X}_m = \boldsymbol{x}$, such that (S18)–(S19) hold. Second, the signs in (S19) flip because $\boldsymbol{\Delta}^- = \boldsymbol{x} - \boldsymbol{X}_{t_m - h}$ is the negative of a forward increment of the reversed diffusion, while the diffusion matrix (hence the coefficients of the third and fourth terms) remain unchanged.

Now, one can rewrite the normalized whitened-increment $\boldsymbol{\eta}_t$ in (S7) as a normalized sum at the midpoint,

$$\boldsymbol{\eta}_t := \frac{\mathbf{B}_m^{-1}\mathrm{d}\boldsymbol{x}}{\sqrt{\mathrm{d}t}} = \frac{\mathbf{B}_m^{-1}(\boldsymbol{\Delta}^+ + \boldsymbol{\Delta}^-)}{\sqrt{2h}} = \frac{1}{\sqrt{2}}\left(\mathbf{B}_m^{-1}\frac{\boldsymbol{\Delta}^+}{\sqrt{h}} + \mathbf{B}_m^{-1}\frac{\boldsymbol{\Delta}^-}{\sqrt{h}}\right), \tag{S21}$$

and adding (S18)-(S19) yields

$$\boldsymbol{\eta}_t = \frac{\boldsymbol{Z}_+ - \boldsymbol{Z}_-}{\sqrt{2}} + \sqrt{\mathrm{d}t}\,\mathbf{B}_m^{-1}\boldsymbol{v}(\boldsymbol{x}, t_m) + \sqrt{\mathrm{d}t}\,\mathcal{T}_1(\boldsymbol{x}) : \frac{\boldsymbol{Z}_+\boldsymbol{Z}_+^\intercal - \boldsymbol{Z}_-\boldsymbol{Z}_-^\intercal}{2} + \sqrt{\mathrm{d}t}\,\mathcal{T}_2(\boldsymbol{x}) : \frac{\mathbf{A}_+ - \mathbf{A}_-}{2} + O_{L^2}(\mathrm{d}t). \tag{S22}$$

Here, the drift term $\mathbf{B}_m^{-1}\boldsymbol{v}(\boldsymbol{x}, t_m)$ comes from (S10). Note that (S22) is a strict extension of Eq. (3) in the main text for the multiplicative-noise case. This equation isolates the entire multiplicative-noise difficulty in two potentially dangerous $O(\sqrt{\mathrm{d}t})$ terms. We will show that such terms are harmless at the order relevant for mutual information in the next section.

The O_{L^2} notation in (S18)–(S19), (S22) means there exist $C > 0$, $r \in \mathbb{N}$ for all $\mathrm{d}t < \mathrm{d}t_0$ (or $h < h_0$) such that

$$\|\boldsymbol{R}_{\boldsymbol{x}}\|_{L^2} \le C\big(1 + \|\boldsymbol{x}\|^r\big)\,\mathrm{d}t \qquad \forall \boldsymbol{x} \in \mathbb{R}^d, \tag{S23}$$

and this is ensured by assumption A1–A4. Especially, the polynomial growth bounds assumption A4 gives the polynomial term $(1 + \|\boldsymbol{x}\|^r)$.

E. Cancellation of the $\sqrt{\mathrm{d}t}$-order correction term in small-time density expansions

Small-time density expansions of multivariate diffusions are naturally organized in half-integer powers of $\mathrm{d}t$ [38–41]. For the present problem, this means that an $O(\sqrt{\mathrm{d}t})$ correction in the conditional distribution of $\boldsymbol{\eta}_t$ could in principle contribute to mutual information at order $O(\mathrm{d}t)$. The purpose of this section is to show that midpoint conditioning removes exactly this contribution. Using (S22), we therefore isolate the two potentially dangerous $O(\sqrt{\mathrm{d}t})$ terms and prove that both are conditionally centered.

Now, for convenience, we denote the dangerous terms of (S22) as follows:

$$\boldsymbol{G}_{\boldsymbol{x}}(\boldsymbol{Z}_+, \boldsymbol{Z}_-) := \mathcal{T}_1(\boldsymbol{x}) : \frac{\boldsymbol{Z}_+\boldsymbol{Z}_+^\intercal - \boldsymbol{Z}_-\boldsymbol{Z}_-^\intercal}{2}, \quad \boldsymbol{H}_{\boldsymbol{x}}(\mathbf{A}_+, \mathbf{A}_-) := \mathcal{T}_2(\boldsymbol{x}) : \frac{\mathbf{A}_+ - \mathbf{A}_-}{2}. \tag{S24}$$

We now analyze these $O(\sqrt{\mathrm{d}t})$ non-Gaussian terms $\boldsymbol{G}_{\boldsymbol{x}}$ and $\boldsymbol{H}_{\boldsymbol{x}}$ in the normalized and whitened increment. Although these terms are present, we show that their conditional expectations vanish under midpoint conditioning, implying that the $O(\sqrt{\mathrm{d}t})$ coefficient in the characteristic function is exactly zero. Consequently, the short-time density expansion contains no non-Gaussian corrections at $O(\sqrt{\mathrm{d}t})$.

• Step 1. Conditional Expectation of $\boldsymbol{G}_{\boldsymbol{x}}$

Under conditioning $\boldsymbol{X}_m = \boldsymbol{x}$, we first reparameterize the independent and normalized Gaussian variables $\boldsymbol{Z}_+, \boldsymbol{Z}_-$ in (S22) using

$$\boldsymbol{Z} := \frac{\boldsymbol{Z}_+ - \boldsymbol{Z}_-}{\sqrt{2}}, \quad \boldsymbol{U} := \frac{\boldsymbol{Z}_+ + \boldsymbol{Z}_-}{\sqrt{2}}, \tag{S25}$$

so $\boldsymbol{Z}, \boldsymbol{U}$ are independent $\mathcal{N}(0, \mathbf{I}_d)$. A direct algebra gives

$$\boldsymbol{Z}_+\boldsymbol{Z}_+^\intercal - \boldsymbol{Z}_-\boldsymbol{Z}_-^\intercal = \boldsymbol{Z}\boldsymbol{U}^\intercal + \boldsymbol{U}\boldsymbol{Z}^\intercal.$$

Hence the $O(\sqrt{\mathrm{d}t})$ non-Gaussian term in (S22) becomes a bilinear form in $(\boldsymbol{Z}, \boldsymbol{U})$:

$$\sqrt{\mathrm{d}t}\,\mathcal{T}_1(\boldsymbol{x}) : \frac{1}{\sqrt{2}}\frac{\boldsymbol{Z}_+\boldsymbol{Z}_+^\intercal - \boldsymbol{Z}_-\boldsymbol{Z}_-^\intercal}{\sqrt{2}} = \sqrt{\mathrm{d}t}\,\boldsymbol{G}_{\boldsymbol{x}}(\boldsymbol{Z}, \boldsymbol{U}),$$

with $\boldsymbol{G}_{\boldsymbol{x}}(\boldsymbol{Z}, \boldsymbol{U})$ linear in $\boldsymbol{U}$ for fixed $\boldsymbol{Z}$. In particular, because $\boldsymbol{U} \perp\!\!\!\perp \boldsymbol{Z}$ and $\mathbb{E}[\boldsymbol{U}] = 0$, we have

$$\mathbb{E}[\boldsymbol{G}_{\boldsymbol{x}}(\boldsymbol{Z}, \boldsymbol{U}) \,|\boldsymbol{X}_m = \boldsymbol{x}, \boldsymbol{Z}\,] = \mathbf{0}. \tag{S26}$$

This cancellation relies crucially on midpoint conditioning. Under other conditionings, such as conditioning on $\boldsymbol{X}_t = \boldsymbol{x}$, the corresponding $O(\sqrt{\mathrm{d}t})$ term involves the quadratic structure $\boldsymbol{Z}_+\boldsymbol{Z}_+^\intercal - \mathbf{I}_d$, which cannot be rewritten into a bilinear form in two independent Gaussians, so the above conditional-centering argument does not apply.

• **Step 2. Conditional Expectation of $\boldsymbol{H}_{\boldsymbol{x}}$**

We work under the auxiliary representation introduced in Section D: conditional on $\boldsymbol{X}_m = \boldsymbol{x}$, the two half-steps can be realized using independent Brownian motions, i.e. the Wiener process $\boldsymbol{W}^\pm$. Under conditioning $\boldsymbol{X}_m = \boldsymbol{x}$, fix one half-step and write $\Delta\boldsymbol{W}^\pm := \boldsymbol{W}_h^\pm$ and $\boldsymbol{Z}_\pm := \Delta\boldsymbol{W}^\pm/\sqrt{h}$. Now consider the time-reversal map on paths on $[0, h]$,

$$\widetilde{\boldsymbol{W}}_s^\pm := \Delta\boldsymbol{W}^\pm - \boldsymbol{W}_{h-s}^\pm, \qquad s \in [0, h].$$

Then $\widetilde{\boldsymbol{W}}_0^\pm = 0$ and $\widetilde{\boldsymbol{W}}_h^\pm = \Delta\boldsymbol{W}^\pm$. Moreover, conditional on the endpoint $\Delta\boldsymbol{W}^\pm$, the Brownian bridge law is invariant under this map: $\boldsymbol{W}^\pm \mid \Delta\boldsymbol{W}^\pm \overset{d}{=} \widetilde{\boldsymbol{W}}^\pm \mid \Delta\boldsymbol{W}^\pm$. (For completeness, this follows since the Brownian bridge is a Gaussian process and the map preserves its conditional mean and covariance.) Hence for any integrable functional f,

$$\mathbb{E}\big[f(\boldsymbol{W}^\pm) \mid \Delta\boldsymbol{W}^\pm\big] = \mathbb{E}\Big[f(\widetilde{\boldsymbol{W}}^\pm) \mid \Delta\boldsymbol{W}^\pm\Big]. \tag{S27}$$

Also, under time-reversal map and by definition of matrix $\mathbf{A}$ (S20), we get the sign-flip property because

$$\begin{aligned}
\left(\mathbf{A}_\pm\left(\widetilde{\boldsymbol{W}}^\pm\right)\right)_{ij} &= \frac{1}{2h}\int_0^h \left(\int_0^s \mathrm{d}\widetilde{W}_u^{\pm,(i)}\right)\mathrm{d}\widetilde{W}_s^{\pm,(j)} - (i \leftrightarrow j) \\
&= \frac{1}{2h}\int_0^h \left(W_h^{\pm,(i)} - W_{h-s}^{\pm,(i)}\right)\left(-\mathrm{d}W_{h-s}^{\pm,(j)}\right) - (i \leftrightarrow j) \\
&= \frac{1}{2h}\int_0^h \left(W_h^{\pm,(i)} - W_\tau^{\pm,(i)}\right)\mathrm{d}W_\tau^{\pm,(j)} - (i \leftrightarrow j) \\
&= \frac{1}{2h}\left(W_h^{\pm,(i)}W_h^{\pm,(j)} - \int_0^h W_\tau^{\pm,(i)}\mathrm{d}W_\tau^{\pm,(j)}\right) - (i \leftrightarrow j) \\
&= \frac{1}{2h}\left(-\int_0^h \left(\int_0^\tau \mathrm{d}W_u^{\pm,(i)}\right)\mathrm{d}W_\tau^{\pm,(j)}\right) - (i \leftrightarrow j) \\
&= -\left(\mathbf{A}_\pm\left(\boldsymbol{W}^\pm\right)\right)_{ij} \quad \forall i, j.
\end{aligned} \tag{S28}$$

In the second equality, we use the definition of the time-reversed Wiener process $\widetilde{W}_s = W_h - W_{h-s}$, which formally yields $\mathrm{d}\widetilde{W}_s = -\,\mathrm{d}W_{h-s}$. This computation is understood in a formal sense and can be rigorously justified by approximating the Itô integrals by Riemann sums and applying the change of variables. In the third equality, we use the change of variables $s \mapsto \tau := h - s$. The $W_h^{\pm,(i)}W_h^{\pm,(j)}$ term cancels out in the fourth line, since $W_h^{\pm,(i)}W_h^{\pm,(j)} - (i \leftrightarrow j) = 0$.

Combining (S27) with (S28) yields

$$\mathbb{E}[\mathbf{A}_\pm(\boldsymbol{W}^\pm) \mid \Delta\boldsymbol{W}^\pm] = \mathbb{E}[\mathbf{A}_\pm(\widetilde{\boldsymbol{W}}^\pm) \mid \Delta\boldsymbol{W}^\pm] = \mathbb{E}[-\mathbf{A}_\pm(\boldsymbol{W}^\pm) \mid \Delta\boldsymbol{W}^\pm] = -\mathbb{E}[\mathbf{A}_\pm(\boldsymbol{W}^\pm) \mid \Delta\boldsymbol{W}^\pm],$$

hence $\mathbb{E}[\mathbf{A}_\pm \mid \Delta\boldsymbol{W}^\pm] = \mathbf{0}$, equivalently $\mathbb{E}[\mathbf{A}_\pm \mid \boldsymbol{Z}_\pm] = \mathbf{0}$. Therefore,

$$\mathbb{E}[\boldsymbol{H}_{\boldsymbol{x}} \,|\, \boldsymbol{Z}_+, \boldsymbol{Z}_-] = \mathcal{T}_2(\boldsymbol{x}) : \frac{\mathbb{E}[\mathbf{A}_+ \mid \boldsymbol{Z}_+] - \mathbb{E}[\mathbf{A}_- \mid \boldsymbol{Z}_-]}{2} = \mathbf{0},$$

and since $\mathcal{T}_2(\boldsymbol{x})$ is deterministic conditional on $\boldsymbol{X}_m = \boldsymbol{x}$, the same conclusion holds under midpoint conditioning:

$$\mathbb{E}[\boldsymbol{H}_{\boldsymbol{x}} \,|\, \boldsymbol{X}_m = \boldsymbol{x},\ \boldsymbol{Z}_+,\ \boldsymbol{Z}_-] = \mathbf{0}. \tag{S29}$$

Unlike $\boldsymbol{G}_{\boldsymbol{x}}$, the centering of the Lévy-area term $\boldsymbol{H}_{\boldsymbol{x}}$ does not rely on midpoint conditioning: it follows from the Brownian-bridge property that the Lévy area has zero conditional mean given the endpoint increment, i.e. $\mathbb{E}[\mathbf{A}\,|\Delta\boldsymbol{W}] = 0$. Consequently, vanishing $\boldsymbol{H}_{\boldsymbol{x}}$ holds regardless of the particular time-point conditioning; it only relies on the endpoint-centered Brownian-bridge property. Intuitively, $\boldsymbol{H}_{\boldsymbol{x}}$ influences how individual trajectories are "twisted" along the path; however, once the endpoints are fixed, these twisting effects average out and do not contribute to one-time marginal distributions.

• **Step 3. Cancellation of the $\sqrt{\mathrm{d}t}$-order non-Gaussian corrections in characteristic function**

Now we show that the coefficient of the $O(\sqrt{\mathrm{d}t})$ term in the non-Gaussian correction to the conditional characteristic function,

$$\varphi_{\boldsymbol{\eta}_t|\boldsymbol{x}}(\boldsymbol{u}) \coloneqq \mathbb{E}[e^{i\langle \boldsymbol{u}, \boldsymbol{\eta}_t\rangle} | \boldsymbol{X}_m = \boldsymbol{x}].$$

Applying the tower property under conditioning on $\boldsymbol{Z}$ in (S25), we have

$$\varphi_{\boldsymbol{\eta}_t|\boldsymbol{x}}(\boldsymbol{u}) = \mathbb{E}[e^{i\langle \boldsymbol{u}, \boldsymbol{\eta}_t\rangle} | \boldsymbol{X}_m = \boldsymbol{x}] = \mathbb{E}\big[\mathbb{E}[e^{i\langle \boldsymbol{u}, \boldsymbol{\eta}_t\rangle} | \boldsymbol{X}_m = \boldsymbol{x}, \boldsymbol{Z}\,] \mid \boldsymbol{X}_m = \boldsymbol{x}\big]. \tag{S30}$$

Also we can write $\boldsymbol{\eta}_t$ by using (S22)–(S25),

$$\boldsymbol{\eta}_t = \Big(\boldsymbol{Z} + \sqrt{\mathrm{d}t}\, \boldsymbol{s}(\boldsymbol{x}, t_m)\Big) + \sqrt{\mathrm{d}t}\, \boldsymbol{G}_{\boldsymbol{x}}(\boldsymbol{Z}, \boldsymbol{U}) + \sqrt{\mathrm{d}t}\, \boldsymbol{H}_{\boldsymbol{x}}(\mathbf{A}_+, \mathbf{A}_-) + \boldsymbol{R}_{\boldsymbol{x}}, \quad \|\boldsymbol{R}_{\boldsymbol{x}}\|_{L^2} = O(\mathrm{d}t),$$

for $\boldsymbol{x}$ in compact where $\boldsymbol{s}(\boldsymbol{x}, t_m)$ denotes $\mathbf{B}_m^{-1}\boldsymbol{v}(\boldsymbol{x}, t_m)$. Expanding (S30) for non-Gaussian term to the second order in $\sqrt{\mathrm{d}t}$, we get the first-order correction terms

$$\begin{aligned}
&\sqrt{\mathrm{d}t}\, i\, \mathbb{E}\left[e^{i\langle \boldsymbol{u}, \boldsymbol{Z} + \sqrt{\mathrm{d}t}\, \boldsymbol{s}\rangle}\, \mathbb{E}\big[\langle \boldsymbol{u}, \boldsymbol{G}_{\boldsymbol{x}}(\boldsymbol{Z}, \boldsymbol{U}) + \boldsymbol{H}_{\boldsymbol{x}}(\mathbf{A}_+, \mathbf{A}_-)\rangle | \boldsymbol{X}_m = \boldsymbol{x}, \boldsymbol{Z}\,\big] \,\middle|\, \boldsymbol{X}_m = \boldsymbol{x}\right] \\
&= \sqrt{\mathrm{d}t}\, i\mathbb{E}\left[e^{i\langle \boldsymbol{u}, \boldsymbol{Z} + \sqrt{\mathrm{d}t}\, \boldsymbol{s}\rangle} \left(\langle \boldsymbol{u}, \mathbb{E}[\boldsymbol{G}_{\boldsymbol{x}}(\boldsymbol{Z}, \boldsymbol{U}) \,|\, \boldsymbol{X}_m = \boldsymbol{x}, \boldsymbol{Z}\,]\rangle + \langle \boldsymbol{u}, \mathbb{E}[\boldsymbol{H}_{\boldsymbol{x}}(\mathbf{A}_+, \mathbf{A}_-) \,|\, \boldsymbol{X}_m = \boldsymbol{x}, \boldsymbol{Z}\,]\rangle\right) \,\middle|\, \boldsymbol{X}_m = \boldsymbol{x}\right].
\end{aligned} \tag{S31}$$

Applying (S26) and (S29) on RHS of (S31), we have

$$\begin{aligned}
\mathbb{E}[\boldsymbol{G}_{\boldsymbol{x}}(\boldsymbol{Z}, \boldsymbol{U}) \,|\boldsymbol{X}_m = \boldsymbol{x}, \boldsymbol{Z}\,] &= 0, \\
\mathbb{E}\big[\boldsymbol{H}_{\boldsymbol{x}}(\mathbf{A}_+, \mathbf{A}_-) | \boldsymbol{X}_m = \boldsymbol{x}, \boldsymbol{Z}\,\big] &= \mathbb{E}\big[\mathbb{E}[\boldsymbol{H}_{\boldsymbol{x}}(\mathbf{A}_+, \mathbf{A}_-) | \boldsymbol{X}_m = \boldsymbol{x}, \boldsymbol{Z}_+, \boldsymbol{Z}_-\,] \,|\, \boldsymbol{X}_m = \boldsymbol{x}, \boldsymbol{Z}\,\big] = 0.
\end{aligned}$$

This means the non-Gaussian corrections of order $\sqrt{\mathrm{d}t}$ in the conditional characteristic function $\varphi_{\boldsymbol{\eta}_t|\boldsymbol{x}}(\boldsymbol{u})$ is perfectly vanishing under midpoint conditioning even though each half-step may have an $O(\sqrt{\mathrm{d}t})$ non-Gaussian correction. Therefore, the conditional characteristic function has the expansion

$$\varphi_{\boldsymbol{\eta}_t|\boldsymbol{x}}(\boldsymbol{u}) = \underbrace{\exp\left(-\frac{\|\boldsymbol{u}\|^2}{2}\right) \exp\left(i\sqrt{\mathrm{d}t}\, \langle \boldsymbol{u}, \boldsymbol{s}(\boldsymbol{x}, t_m)\rangle\right)}_{\text{(normalized) Gaussian part}}\ \underbrace{(1 + O(\mathrm{d}t)(1 + \|\boldsymbol{u}\|^m))}_{\text{non-Gaussian corrections}}, \tag{S32}$$

uniformly for $\boldsymbol{x}$ in compacts. Note that the non-Gaussian corrections start with the $\mathrm{d}t$-order, which means the conditional characteristic function also has no non-Gaussian $O(\sqrt{\mathrm{d}t})$ term.

The $O(\mathrm{d}t)$ notation in (S32) can be expressed in a more rigorous form by using (S22) as follows: There exist constants $\mathrm{d}t_0 > 0$, $C > 0$, and $m, l \in \mathbb{N}$ and a function $r_{\mathrm{d}t}(\boldsymbol{u}; \boldsymbol{x})$ such that,

$$|r_{\mathrm{d}t}(\boldsymbol{u}; \boldsymbol{x})| \le C\big(1 + \|\boldsymbol{x}\|^l\big)\big(1 + \|\boldsymbol{u}\|^m\big) \qquad \forall\, \boldsymbol{x}, \boldsymbol{u} \in \mathbb{R}^d,\ 0 < \mathrm{d}t < \mathrm{d}t_0, \tag{S33}$$

and

$$\varphi_{\boldsymbol{\eta}_t|\boldsymbol{x}}(\boldsymbol{u}) = \exp\left(-\frac{\|\boldsymbol{u}\|^2}{2}\right)\exp\left(i\sqrt{\mathrm{d}t}\,\langle \boldsymbol{u}, \boldsymbol{s}(\boldsymbol{x}, t_m)\rangle\right)\left(1 + r_{\mathrm{d}t}(\boldsymbol{u};\boldsymbol{x})\,\mathrm{d}t\,\right). \tag{S34}$$

Indeed, by (S22) we have $\boldsymbol{\eta}_t = \boldsymbol{Z} + \sqrt{\mathrm{d}t}\,\boldsymbol{s} + \sqrt{\mathrm{d}t}\,\boldsymbol{Q}_{\boldsymbol{x}} + \boldsymbol{R}_{\boldsymbol{x}}$ with $\|\boldsymbol{R}_{\boldsymbol{x}}\|_{L^2(\cdot|\boldsymbol{X}_m=\boldsymbol{x})} = O(\mathrm{d}t)\big(1 + \|\boldsymbol{x}\|^r\big)$ where $\boldsymbol{Q}_{\boldsymbol{x}} = \boldsymbol{G}_{\boldsymbol{x}} + \boldsymbol{H}_{\boldsymbol{x}}$. The $\sqrt{\mathrm{d}t}$-order non-Gaussian correction vanishes, as just shown. The remainder term is controlled by $\|\boldsymbol{R}_{\boldsymbol{x}}\|_{L^2(\cdot|\boldsymbol{X}_m=\boldsymbol{x})} = O(\mathrm{d}t)\big(1 + \|\boldsymbol{x}\|^r\big)$ together with finite-moment bounds for finite Wiener chaoses (and the moment assumption A5): using $|e^{i\langle \boldsymbol{u},\boldsymbol{\delta}\rangle} - 1| \le \min\{2, |\langle \boldsymbol{u},\boldsymbol{\delta}\rangle|\} \le \min\{2, \|\boldsymbol{u}\|\,\|\boldsymbol{\delta}\|\}$ for $\boldsymbol{\delta} = \boldsymbol{R}_{\boldsymbol{x}}$, one obtains $\mathbb{E}[\|\boldsymbol{R}_{\boldsymbol{x}}\|] \le \mathbb{E}\left[\|\boldsymbol{R}_{\boldsymbol{x}}\|^2\right]^{\frac{1}{2}} = O(\mathrm{d}t)\big(1 + \|\boldsymbol{x}\|^r\big)$ therefore $|r_{\mathrm{d}t}(\boldsymbol{u};\boldsymbol{x})| \le C\big(1 + \|\boldsymbol{x}\|^l\big)(1 + \|\boldsymbol{u}\|^m)$ uniformly for $\boldsymbol{x}$. Note that for compact $K \subset \mathbb{R}^d$, we can take the sufficiently large constant $C_K < \infty$ such that $C_K > C\big(1 + \|\boldsymbol{x}\|^l\big)$ for all $\boldsymbol{x} \in K$.

We also remark that the polynomial factor $(1+\|\boldsymbol{u}\|^m)$ in (S32) and (S33) comes from bounding the Taylor remainder of $e^{i\langle \boldsymbol{u},\boldsymbol{\eta}_t\rangle}$ after conditioning on $(\boldsymbol{X}_m = \boldsymbol{x}, \boldsymbol{Z})$. Since the $\sqrt{\mathrm{d}t}$-order term vanishes by (S26)–(S29), the leading non-Gaussian contribution starts at order $\mathrm{d}t$ and is controlled by moments of $|\langle \boldsymbol{u}, \boldsymbol{Q}_{\boldsymbol{x}}\rangle|^j \le \|\boldsymbol{u}\|^j \|\boldsymbol{Q}_{\boldsymbol{x}}\|^j$ for some finite j. Under (A1) the whitening matrix $\mathbf{B}^{-1}$ is uniformly bounded, while (A2)–(A4) imply that the strong Taylor coefficients $\mathcal{T}_1(\boldsymbol{x}), \mathcal{T}_2(\boldsymbol{x})$ in (S22) have at most polynomial growth in $\|\boldsymbol{x}\|$. Together with the moment bound (A5), this yields finite conditional moments of $\boldsymbol{Q}$ up to order m (and uniformity for $\boldsymbol{x}$ on compact sets), leading to a remainder bound of the form $O(\mathrm{d}t)(1 + \|\boldsymbol{u}\|^m)$.

• Step 4. Cancellation of the $\sqrt{\mathrm{d}t}$-order non-Gaussian corrections in probability density function – 4.A Bounded drift case

As stated at the beginning of this subsection, Yang et al. (2019) established that, under Assumptions A1–A4, the transition probability density of the Itô SDE admits an asymptotic expansion in half-integer powers of the time increment $\mathrm{d}t$ [40]. Since Yang et al.'s result is applicable only to the bounded-drift case, we proceed in two steps. We first follow their argument in the case (Assumption A5-A), where the drift is uniformly bounded. We then extend the conclusion to the unbounded-drift regime (Assumption A5-B) by means of a standard truncation-localization argument: after truncating the drift outside a ball $\|\boldsymbol{x}\| \le R$, we apply the result on that compact subset.

For convenience in carrying out this extension, we follow the structure of Yang et al.'s proof but adopt a different representation of the remainder term, based on Kohatsu-Higa et al. (2013) and Bally et al. (2015) [42, 43], in order to make the drift-dependent prefactor explicit because that form is better suited to the localization step. Applying the resulting expansion to each half-steps $\boldsymbol{Y}_\pm := \mathbf{B}_m^{-1}\boldsymbol{\Delta}^\pm/\sqrt{h}$ defined in (S18)–(S19) for bounded-drift case (Assumption A5-A), we obtain the expansion up to order $J \in \mathbb{N}$ for $0 < h \le h_0$ (Note that $h = \mathrm{d}t/2$):

$$f_\pm(\boldsymbol{Y}_\pm = \boldsymbol{z}_\pm|\boldsymbol{X}_m = \boldsymbol{x}) = \phi\Big(\boldsymbol{z}_\pm^{(c)}\Big)\left(1 + \sum_{j=1}^{J} h^{j/2}\, c_{\pm,j}\Big(\boldsymbol{z}_\pm^{(c)}\,;\boldsymbol{x}\Big)\right) + \mathrm{Rem}_\pm\Big(\boldsymbol{z}_\pm^{(c)}, \boldsymbol{x}; J\Big), \quad \boldsymbol{z}_\pm^{(c)} := \boldsymbol{z}_\pm - \sqrt{h}\,\boldsymbol{\mu}_\pm(\boldsymbol{x}) \tag{S35}$$

Here, $\phi(\cdot)$ denotes the probability density function of the d-dimensional standard normal distribution. The vectors $\boldsymbol{\mu}_+$ and $\boldsymbol{\mu}_-$ denote $\mathbf{B}_m^{-1}\boldsymbol{F}_I$ and $-\mathbf{B}_m^{-1}\boldsymbol{F}_{rev}$, respectively, and each $c_{\pm,j}(\cdot;\boldsymbol{x})$ is a finite linear combination of shifted Hermite polynomials (i.e. Hermite polynomials of $\boldsymbol{z}_\pm - \sqrt{h}\,\boldsymbol{\mu}_\pm$) with coefficients depending on $\boldsymbol{x}$. More precisely, under assumptions A1–A5-A, there exist $C_\pm, c_\pm, \lambda_\pm, k_1^\pm, k_2^\pm, h_\pm > 0$ such that

$$\left|\mathrm{Rem}_\pm\Big(\boldsymbol{z}_\pm^{(c)}, \boldsymbol{x}; J\Big)\right| \le C_\pm\, h^{\frac{J+1}{2}}\, e^{c_\pm \|\boldsymbol{\mu}_\pm\|_\infty^2 h}\Big(1 + \|\boldsymbol{x}\|^{k_1^\pm}\Big)\Big(1 + \big\|\boldsymbol{z}_\pm^{(c)}\big\|^{k_2^\pm}\Big)\exp\Big(-\lambda_\pm\big\|\boldsymbol{z}_\pm^{(c)}\big\|^2\Big), \tag{S36}$$

for all $\boldsymbol{x}, \boldsymbol{z} \in \mathbb{R}^d$ and $h < h_\pm$, where $\|\boldsymbol{\mu}_\pm\|_\infty$ denotes the supremum of $\|\boldsymbol{\mu}_\pm\|$. Note that $C_\pm, c_\pm > 0$ are constant. Under A1–A4, the remainder admits a shifted Gaussian-tail bound with polynomial prefactors with some integers $k_1^\pm$, while the drift enters only through the exponential prefactor with $\|\boldsymbol{\mu}_\pm\|_\infty$ [42, 43]. Although the bounded-drift assumption A5-A allows these drift-dependent prefactors to be absorbed into a single constant C_{drift}, we keep them explicit to streamline the extension to the unbounded-drift regime. An integer $k_2^\pm$ and constant $\lambda_\pm > 0$ follow from repeated derivatives of Gaussian kernel structures, as shown in [40, 43]. Accordingly, the remainder can be bounded by a polynomial in the centered increment $\boldsymbol{z}_\pm^{(c)} := \boldsymbol{z}_\pm - \sqrt{h}\,\boldsymbol{\mu}_\pm$ multiplied by a Gaussian tail. Here, $\lambda_\pm$ governs the tail decay and depends only on the diffusion coefficient $\mathbf{B}$ through the uniform ellipticity bounds (A1) and finitely many bounds on spatial derivatives of $\mathbf{B}$ (A2), and is independent of the drift magnitude.

By Lemma S1, $\boldsymbol{Y}_+$ and $\boldsymbol{Y}_-$ are conditionally independent given $\boldsymbol{X}_m = \boldsymbol{x}$. We consider the normalized total increment $\boldsymbol{\eta}_t = (\boldsymbol{Y}_+ + \boldsymbol{Y}_-)/\sqrt{2}$. Its density function can be established via the scaled convolution of f_+ and f_-:

$$p_{\mathrm{d}t}(\boldsymbol{\eta}_t = \boldsymbol{z} \mid \boldsymbol{X}_m = \boldsymbol{x}) = \sqrt{2}^d \int f_+(\boldsymbol{s} \mid \boldsymbol{x})\, f_-\Big(\sqrt{2}\,\boldsymbol{z} - \boldsymbol{s}\,\Big|\,\boldsymbol{x}\Big)\,\mathrm{d}\boldsymbol{s}.$$

Since the class of functions defined by the product of a Gaussian and polynomials (with Gaussian-bounded errors) is closed under convolution, the density of $\boldsymbol{\eta}_t$ retains the same asymptotic structure. More precisely, expanding f_+ and f_- up to order J, convolving term-by-term and collecting powers of $\sqrt{\mathrm{d}t}$ (using standard Gaussian/Hermite identities) and re-express polynomials by combinations of shifted Hermite, we obtain the small-time expansion for $\mathrm{d}t < \mathrm{d}t_0$:

$$p_{\mathrm{d}t}(\boldsymbol{\eta}_t = \boldsymbol{z} \mid \boldsymbol{X}_m = \boldsymbol{x}) = \phi(\boldsymbol{z}_c)\left(1 + \sum_{j=1}^{J} \mathrm{d}t^{\frac{j}{2}}\, c_j(\boldsymbol{z}_c; \boldsymbol{x})\right) + \mathrm{Rem}(\boldsymbol{z}_c, \boldsymbol{x}; J)\,, \quad \boldsymbol{z}_c := \boldsymbol{z} - \sqrt{\mathrm{d}t}\,\boldsymbol{s}(\boldsymbol{x}) \tag{S37}$$

where $\boldsymbol{s} = (\boldsymbol{\mu}_+ + \boldsymbol{\mu}_-)/2 = \mathbf{B}_m^{-1}\boldsymbol{v}$, and each c_j denotes a finite linear combination of shifted Hermite polynomials with coefficients depending on $\boldsymbol{x}$. The remainder term $\mathrm{Rem}(\boldsymbol{z}, \boldsymbol{x}; J)$ satisfies a global bound analogous to (S36):

$$|\mathrm{Rem}(\boldsymbol{z}_c, \boldsymbol{x}; J)| \le C\,\mathrm{d}t^{\frac{J+1}{2}} e^{c\left(\|\boldsymbol{\mu}_+\|_\infty^2 + \|\boldsymbol{\mu}_-\|_\infty^2\right)\mathrm{d}t}\left(1 + \|\boldsymbol{x}\|^{k_1}\right)\left(1 + \|\boldsymbol{z}_c\|^{k_2}\right)\exp\left(-\lambda\,\|\boldsymbol{z}_c\|^2\right), \tag{S38}$$

for all $\boldsymbol{x}, \boldsymbol{z} \in \mathbb{R}^d$ and $\mathrm{d}t < \mathrm{d}t_0$, where C, c, k_1, k_2, and λ are defined similarly to their counterparts in (S36). Note that the remainder term $\mathrm{Rem}(\boldsymbol{z}, \boldsymbol{x}; J)$ is still bounded under the Gaussian tails.

Now applying Fourier transformation $\mathcal{F}$ on (S37), we obtain the conditional characteristic function of $\boldsymbol{\eta}_t$ conditioning $\boldsymbol{X}_m = \boldsymbol{x}$ as

$$\begin{aligned}
\varphi_{\boldsymbol{\eta}_t|\boldsymbol{x}}(\boldsymbol{u}) &= \int e^{i\langle \boldsymbol{u}, \boldsymbol{z}\rangle} p_{\mathrm{d}t}(\boldsymbol{z}|\boldsymbol{x})\mathrm{d}\boldsymbol{z} =: \mathcal{F}[p_{\mathrm{d}t}](\boldsymbol{u}) \\
&= \underbrace{\mathcal{F}\left[\phi(\boldsymbol{z} - \sqrt{\mathrm{d}t}\,\boldsymbol{s})\right](\boldsymbol{u})}_{=:\widetilde{\phi}(\boldsymbol{u},\boldsymbol{s})} + \sum_{j=1}^{J} \sqrt{\mathrm{d}t}^{\,j}\,\mathcal{F}\left[c_j\left(\boldsymbol{z} - \sqrt{\mathrm{d}t}\,\boldsymbol{s}; \boldsymbol{x}\right)\,\phi\left(\boldsymbol{z} - \sqrt{\mathrm{d}t}\,\boldsymbol{s}\right)\right](\boldsymbol{u}) + \underbrace{\mathcal{F}\left[\mathrm{Rem}(\boldsymbol{z}, \boldsymbol{x}; J)\right](\boldsymbol{u})}_{=:\widetilde{\mathrm{Rem}}(\boldsymbol{u},\boldsymbol{x};J)} \\
&= \widetilde{\phi}(\boldsymbol{u}, \boldsymbol{s}) + \sum_{j=1}^{J} \sqrt{\mathrm{d}t}^{\,j}\,\mathcal{F}\left[\left(\sum_{k=0}^{k_j} a_k^{(j)}(\boldsymbol{x}) H_k\left(\boldsymbol{z} - \sqrt{\mathrm{d}t}\,\boldsymbol{s}\right)\right)\phi\left(\boldsymbol{z} - \sqrt{\mathrm{d}t}\,\boldsymbol{s}\right)\right](\boldsymbol{u}) + \widetilde{\mathrm{Rem}}(\boldsymbol{u}, \boldsymbol{x}; J) \\
&= \widetilde{\phi}(\boldsymbol{u}, \boldsymbol{s}) + \sum_{j=1}^{J} \sqrt{\mathrm{d}t}^{\,j} \underbrace{\left(\sum_{k=0}^{k_j} a_k^{(j)}(\boldsymbol{x})\,(i\boldsymbol{u})^k\right)}_{=:\widetilde{c_j}(\boldsymbol{u};\boldsymbol{x})} \widetilde{\phi}(\boldsymbol{u}, \boldsymbol{s}) + \widetilde{\mathrm{Rem}}(\boldsymbol{u}, \boldsymbol{x}; J) \\
&= \underbrace{\exp\left(-\frac{\|\boldsymbol{u}\|^2}{2}\right)\exp\left(i\sqrt{\mathrm{d}t}\,\langle \boldsymbol{u}, \boldsymbol{s}\rangle\right)}_{=\widetilde{\phi}(\boldsymbol{u},\boldsymbol{s})}\left(1 + \sum_{j=1}^{J} \sqrt{\mathrm{d}t}^{\,j}\,\widetilde{c_j}(\boldsymbol{u}, \boldsymbol{x})\right) + \widetilde{\mathrm{Rem}}(\boldsymbol{u}, \boldsymbol{x}; J).
\end{aligned} \tag{S39}$$

In the fourth equality, we use $\mathcal{F}[H_k(\boldsymbol{z} - \boldsymbol{\mu})\phi(\boldsymbol{z} - \boldsymbol{\mu})](\boldsymbol{u}) = \mathcal{F}[(-1)^k \frac{\mathrm{d}^k}{\mathrm{d}\boldsymbol{z}^k}\phi(\boldsymbol{z} - \boldsymbol{\mu})](\boldsymbol{u}) = (i\boldsymbol{u})^k \mathcal{F}[\phi(\boldsymbol{z} - \boldsymbol{\mu})](\boldsymbol{u})$ where H_k is Hermite polynomial of order k. We emphasize that the remainder $\widetilde{\mathrm{Rem}}(\boldsymbol{u}, \boldsymbol{x}; J)$ cannot contribute to the terms of order $\mathrm{d}t^{(j/2)}$ when $j \le J$ because

$$\begin{aligned}
\left|\widetilde{\mathrm{Rem}}(\boldsymbol{u}, \boldsymbol{x}; J)\right| &\le \int \left|e^{i\langle \boldsymbol{u}, \boldsymbol{z}\rangle}\right| |\mathrm{Rem}(\boldsymbol{z}_c, \boldsymbol{x}; J)|\,\mathrm{d}\boldsymbol{z} \\
&= \int C\,\mathrm{d}t^{\frac{J+1}{2}} e^{c(\|\boldsymbol{\mu}_+\|_\infty + \|\boldsymbol{\mu}_-\|_\infty)\mathbf{d}t}\left(1 + \|\boldsymbol{x}\|^{k_1}\right)\left(1 + \|\boldsymbol{z}_c\|^{k_2}\right)\exp\left(-\lambda\|\boldsymbol{z}_c\|^2\right)\mathrm{d}\boldsymbol{z} \quad \text{by (S38)} \\
&= C\,\mathrm{d}t^{\frac{J+1}{2}} e^{c(\|\boldsymbol{\mu}_+\|_\infty + \|\boldsymbol{\mu}_-\|_\infty)\mathbf{d}t}\left(1 + \|\boldsymbol{x}\|^{k_1}\right)\int \left(1 + \|\boldsymbol{z}_c\|^{k_2}\right)\exp\left(-\lambda\|\boldsymbol{z}_c\|^2\right)\mathrm{d}\boldsymbol{z}_c \\
&\le C_Z\,\mathrm{d}t^{\frac{J+1}{2}} e^{c(\|\boldsymbol{\mu}_+\|_\infty + \|\boldsymbol{\mu}_-\|_\infty)\mathbf{d}t}\left(1 + \|\boldsymbol{x}\|^{k_1}\right).
\end{aligned}$$

In third line, the integral variable is changed: $\boldsymbol{z} \mapsto \boldsymbol{z}_c$. The last inequality holds because all moments of the Gaussian distribution are finite.

This means that since there is no non-Gaussian $\sqrt{\mathrm{d}t}$-order term in $\varphi_{\boldsymbol{\eta}_t|\boldsymbol{x}}$ by (S32), $\widetilde{c_1}(\boldsymbol{u}; \boldsymbol{x})$ in (S39) must vanish. The injectivity of Fourier transform ensures that $c_1\left(\boldsymbol{z} - \sqrt{\mathrm{d}t}\,\boldsymbol{s}\right)$ in (S37) must also be zero. Moreover, $\mathrm{d}t^{(j/2)}c_j$ cannot contribute to terms of order $\mathrm{d}t^{k/2}$ when $k < j$ because c_j is polynomial in $\boldsymbol{z} - \sqrt{\mathrm{d}t}\,\boldsymbol{s}$. Hence, one can conclude that there is no non-Gaussian $\sqrt{\mathrm{d}t}$-order term in the conditional density function $p_{\mathrm{d}t}$.

– 4.B Truncation and localization for unbounded drift case

We emphasize that the arguments from (S35) to (S38) hold rigorously only for the bounded-drift case. Therefore, while they apply seamlessly under Assumption A5-A, accommodating the unbounded drift allowed by Assumption A5-B requires an additional truncation and localization procedure. Specifically, let $B(0,R)$ denote a ball of fixed radius R centered at the origin. We introduce an auxiliary process with a variable $\hat{\boldsymbol{X}}_m$, whose drift and diffusion coefficients perfectly match the original system within $B(0,2R)$, but transition smoothly to bounded boundary values outside $B(0,2R+\delta)$ with $\delta>0$. If we restrict our focus to the dynamics within $B(0,R)$, this auxiliary process is nearly identical to the original one; the discrepancy is bounded exactly by the probability that the process exits $B(0,2R)$ during the short-time interval $\mathrm{d}t$.

Quantitatively, let $\hat{p}_{dt}(\cdot|\boldsymbol{x})$ denote the conditional density of the localized auxiliary system with bounded coefficients and $p_{dt}(\cdot|\boldsymbol{x})$ that of the original system. Since the two SDEs have identical coefficients on $B(0,2R)$, the two conditional laws coincide on the event that the trajectory does not exit $B(0,2R)$ during the time window of length $\mathrm{d}t$. Therefore, for $\|\boldsymbol{x}\|\le R$,

$$\begin{aligned}|p_{dt}(\boldsymbol{z}|\boldsymbol{x})-\hat{p}_{dt}(\boldsymbol{z}|\boldsymbol{x})| &\le \mathbb{P}_{\boldsymbol{x}}\big(\tau_{2R}\le \mathrm{d}t\big)\sup_{\boldsymbol{y}\in\mathbb{R}^d}\hat{p}_{dt}(\boldsymbol{z}|\boldsymbol{y})\\ &\le \mathbb{P}_{\boldsymbol{x}}\big(\tau_{2R}\le \mathrm{d}t\big)\Big(C\,e^{\bar{c}\left(1+R^{k_1}\right)\mathrm{d}t}(1+R^{k_2})(1+\|\boldsymbol{z}\|^m)e^{-\gamma_{\mathrm{tr}}\|\boldsymbol{z}\|^2}\Big),\end{aligned} \tag{S40}$$

for certain constants $C,\bar{c},k_1,k_2,m,\gamma_{tr}>0$ where τ_{2R} is the first exit time from $B(0,2R)$. Intuitively, this is because the deviation in probability density is bounded by the escape probability weighted by the maximal density of the auxiliary process within $\boldsymbol{x}\in B(0,R)$. That maximal density is given by bounded-drift expansion (S37)–(S38) with an envelope whose R-dependence is explicit in the prefactors (with numerical constants independent of R), hence one can get the second inequality of (S40). Moreover, since midpoint is the starting point of the forward and backward half-step, $\mathbb{P}_{\boldsymbol{x}}(\tau_{2R}\le \mathrm{d}t)\le C_{\mathrm{tr}}\exp\big(-c_{\mathrm{tr}}(R-C_F(1+R^m)\mathrm{d}t)_+^2/\mathrm{d}t\big)$ for $\|\boldsymbol{x}\|\le R$ by exponential martingale and polynomial growth bounds of drift (Assumption A4), where $(\cdot)_+:=\max\{\cdot,0\}$. Note that the R and $C_F(1+R^m)\mathrm{d}t$ term are coming from the minimum distance to the truncation boundary and the maximum magnitude of the drift, respectively. Combining these yields the truncation error term and we finally obtain

$$p_{\mathrm{d}t}(\boldsymbol{\eta}_t=\boldsymbol{z}\mid \boldsymbol{X}_m=\boldsymbol{x})=\phi(\boldsymbol{z}_c)\big(1+\mathrm{d}t\,c_{\mathrm{d}t}(\boldsymbol{z}_c;\boldsymbol{x})\big)+\mathrm{Rem}_{\mathrm{d}t}(\boldsymbol{z}_c,\boldsymbol{x};R)\,,\quad \forall\|\boldsymbol{x}\|\le R,\ \boldsymbol{z}\in\mathbb{R}^d, \tag{S41}$$

where $\boldsymbol{z}_c=\boldsymbol{z}-\sqrt{\mathrm{d}t}\,\boldsymbol{s}(\boldsymbol{x})$ and $c_{\mathrm{d}t}(\boldsymbol{z}_c;\boldsymbol{x})$ is a finite linear combination of shifted Hermite polynomials in $\boldsymbol{z}_c$ whose coefficients are deterministic functions of $\boldsymbol{x}$ of at most polynomial growth. Equation (S41) is the key output of this subsection since on the localization region $\|\boldsymbol{x}\|\le R(\mathrm{d}t)$, the conditional distribution is a shifted Gaussian plus $O(\mathrm{d}t)$ corrections, with no $O(\sqrt{\mathrm{d}t})$ contribution.

The remainder $\mathrm{Rem}_{\mathrm{d}t}$ satisfies the bounds

$$\mathrm{Rem}_{\mathrm{d}t}\le C_A\,\mathrm{d}t^{3/2}\big(1+\|\boldsymbol{x}\|^{a_1}\big)\big(1+\|\boldsymbol{z}_c\|^{a_2}\big)e^{-\gamma_A\|\boldsymbol{z}_c\|^2}\quad\text{under Assumption A5-A,} \tag{S42-A}$$

$$\mathrm{Rem}_{\mathrm{d}t}\le C_B\Bigg(\mathrm{d}t^{3/2}+\underbrace{e^{-c_B^{(1)}\frac{\left(R-C_F(1+R^{b_1})\mathrm{d}t\right)_+^2}{\mathrm{d}t}}(1+R^{b_2})}_{\text{truncation error}}\Bigg)\times e^{c_B^{(2)}(1+R^{b_3})\mathrm{d}t}\,(1+\|\boldsymbol{x}\|^{b_4})(1+\|\boldsymbol{z}_c\|^{b_5})\,e^{-\gamma_B\|\boldsymbol{z}_c\|^2}\quad\text{under Assumption A5-B,} \tag{S42-B}$$

for certain constants $C_A,C_B,C_F,c_B^{(1)},c_B^{(2)},a_1,a_2,b_1,b_2,b_3,b_4,b_5,\gamma_A,\gamma_B>0$, which are independent of R. We emphasize that there are no $\sqrt{\mathrm{d}t}$-order non-Gaussian correction terms even for the unbounded-drift case.

Note that in (S42), all numerical constants are independent of R, and the R-dependence appears explicitly only in the prefactors. Hence the bound (S42) remains valid for any R. Nevertheless, sending $R\to\infty$ allows the prefactors (e.g., $(1+\|\boldsymbol{x}\|^b)$) to become arbitrarily large. This implies that the exact finite expansion formula (S41) may cease to be asymptotically informative in the joint regime $\mathrm{d}t\to 0$ and $R\to\infty$—specifically, the bound does not guarantee $\mathrm{Rem}_{\mathrm{d}t}=o(\mathrm{d}t)$ uniformly over $\|\boldsymbol{x}\|\le R$, even though the expansion identity (S41) and the estimate (S42) remain formally valid. To prevent the estimate from becoming vacuous, one must choose an appropriate $R=R(\mathrm{d}t)$ to ensure (S41) is meaningful. Therefore, we set $R(\mathrm{d}t)$ hereafter as

$$R(\mathrm{d}t)=\mathrm{d}t^{-\kappa},\quad 0<\kappa<\frac{\epsilon}{a_1}\ \text{with}\ \epsilon>0\ \text{and}\ a_1\ \text{in (S42-A) under Assumption A5-A,} \tag{S43-A}$$

$$R(\mathrm{d}t)=\left(\ln\frac{1}{\mathrm{d}t}\right)^{\beta},\quad \beta>\frac{1}{1+\alpha}\ \text{with}\ \alpha\ \text{in A5-B under Assumption A5-B.} \tag{S43-B}$$

This choice allows us to control the truncation error term in (S42-B) as $o(\mathrm{d}t^{3/2})$ for sufficiently small $\mathrm{d}t$. Indeed, since $C_F(1+R^{b_1})\mathrm{d}t = O(\mathrm{d}t\,|\ln \mathrm{d}t|^{\beta b_1}) \to 0$ as $\mathrm{d}t \to 0$, we have

$$\exp\left(-c_B^{(1)}\frac{\left(|\ln \mathrm{d}t|^{\beta} - C_F(1+|\ln \mathrm{d}t|^{\beta b_1})\mathrm{d}t\right)_+^2}{\mathrm{d}t}\right) \le \exp\left(-\bar{c}\frac{|\ln \mathrm{d}t|^{2\beta}}{\mathrm{d}t}\right) = o(\mathrm{d}t^m) \quad \forall m > 0,$$

and $\mathrm{d}t^m(1+R^k) = O(\mathrm{d}t^m|\ln \mathrm{d}t|^{\beta k}) = o(\mathrm{d}t^{m-\epsilon})$ for all $\epsilon > 0$. Similarly, we can rewrite (S42) in a much simpler form under (S43): There exist $C_{\beta,\kappa,\epsilon} > 0$ and $\mathrm{d}t_0 > 0$ for every $\epsilon > 0$, $\beta > 1/(1+\alpha)$, and $0 < \kappa < \epsilon/a_1$ such that

$$\mathrm{Rem}_{\mathrm{d}t} \le C_{\beta,\kappa,\epsilon}\,\mathrm{d}t^{\frac{3}{2}-\epsilon}\big(1+\|\boldsymbol{z}_c\|^m\big)e^{-\gamma\|\boldsymbol{z}_c\|^2} \qquad \forall \mathrm{d}t < \mathrm{d}t_0, \|\boldsymbol{x}\| \le R(\mathrm{d}t) \text{ in (S43)}, \tag{S44}$$

for certain constants $m, \gamma > 0$; since for (S42-A), $\mathrm{d}t^{3/2}\sup_{\|\boldsymbol{x}\|\in B(0,R)}(1+\|\boldsymbol{x}\|^{a_1}) = \mathrm{d}t^{3/2}(1+R^{a_1}) = o(\mathrm{d}t^{\frac{3}{2}-\epsilon})$; for (S42-B), $e^{c_B^{(2)}(1+R^{b_3})\mathrm{d}t} = e^{o(1)}$ and $\mathrm{d}t^{3/2}\sup_{\|\boldsymbol{x}\|\in B(0,R)}(1+\|\boldsymbol{x}\|^{b_4}) = \mathrm{d}t^{3/2}(1+R^{b_4}) = o(\mathrm{d}t^{\frac{3}{2}-\epsilon})$ for all $\epsilon > 0$. This result is valid for both A5-A and A5-B.

We also remark that even when the exact decomposition (S41) is not asymptotically informative for large R, the representation and the bound (S42) remain useful; for example, they provide a non-asymptotic control of $\|p(\cdot|\boldsymbol{x})\|_\infty$ on prescribed radii R (e.g., dyadic scales $R = 2^k R_0$).

F. Local Gaussian channel proxy and discrepancy of mutual information

From this point on the proof is bookkeeping. We compare the exact midpoint-conditioned channel with its Gaussian proxy, decompose the mutual information difference and show that each resulting term is $o(\mathrm{d}t)$. To this end, we first introduce an auxiliary random variable $\boldsymbol{\eta}_t^G$ as a surrogate for the whitened increment $\boldsymbol{\eta}_t$. We define $\boldsymbol{\eta}_t^G$ as the output of an idealized Gaussian channel driven by the local signal $\sqrt{\mathrm{d}t}\,\boldsymbol{s}(\boldsymbol{X}_m, t_m)$,

$$\boldsymbol{\eta}_t^G = \sqrt{\mathrm{d}t}\,\boldsymbol{s}(\boldsymbol{X}_m, t_m) + \boldsymbol{N}, \quad \boldsymbol{N} \sim \mathcal{N}(0, \mathbf{I}_d), \quad \boldsymbol{N} \perp\!\!\!\perp \boldsymbol{X}_m, \tag{S45}$$

where $\boldsymbol{s} = \mathbf{B}_m^{-1}\boldsymbol{v}$ represents the diffusion-metric normalized current velocity. For brevity, hereafter we write $p_{\mathrm{d}t}(\boldsymbol{z}|\boldsymbol{x}) = p(\boldsymbol{\eta}_t = \boldsymbol{z}|\boldsymbol{X}_m = \boldsymbol{x})$, $p_{\mathrm{d}t}(\boldsymbol{z},\boldsymbol{x}) = p(\boldsymbol{\eta}_t = \boldsymbol{z}, \boldsymbol{X}_m = \boldsymbol{x})$, and $p_{\mathrm{d}t}(\boldsymbol{z}) = \int p_{\mathrm{d}t}(\boldsymbol{z},\boldsymbol{x})\mathrm{d}\boldsymbol{x}$ and similarly for $q_{\mathrm{d}t}$. The arguments will always indicate which object is meant.

Note that the conditional density of $\boldsymbol{\eta}_t^G$ is exactly given by the shifted Gaussian kernel appearing in the leading order of our small-time expansion (S41). Explicitly,

$$q_{\mathrm{d}t}(\boldsymbol{z}|\boldsymbol{x}) \coloneqq q_{\mathrm{d}t}\big(\boldsymbol{\eta}_t^G = \boldsymbol{z}|\boldsymbol{X}_m = \boldsymbol{x}\big) = \phi\Big(\boldsymbol{z} - \sqrt{\mathrm{d}t}\boldsymbol{s}(\boldsymbol{x})\Big) = \phi\big(\boldsymbol{z}_c\big), \quad \boldsymbol{z}_c \coloneqq \boldsymbol{z} - \sqrt{\mathrm{d}t}\,\boldsymbol{s}(\boldsymbol{x}), \tag{S46}$$

where $\phi(\cdot)$ is the probability density function of the d-dimensional standard normal distribution. This identifies $\boldsymbol{\eta}_t^G$ as the "Gaussian core" of the dynamics, stripped of the non-Gaussian corrections and the remainder terms $\mathrm{Rem}_{\mathrm{d}t}$ established in (S41). In the subsequent steps, $\boldsymbol{\eta}_t^G$ will serve as the reference for quantifying the mutual information discrepancy induced by the non-Gaussianity of the actual dynamics.

We now demonstrate that the discrepancy $\Delta I \coloneqq I\big(\boldsymbol{\eta}_t; \boldsymbol{X}_m\big) - I\big(\boldsymbol{\eta}_t^G; \boldsymbol{X}_m\big)$ is of order $o(\mathrm{d}t)$ under assumptions A1–A5 by the following steps.

• Step 1. Decomposing upper bound of the $|\Delta I|$

As identified in Eq. (S42), the polynomial growth of the coefficients (Assumption A4) implies that the remainder terms scale with $\|\boldsymbol{X}_m\|$, making a global uniform bound difficult to establish. To circumvent this, we consider a compact ball $B(0,R)$ of radius $R = R(\mathrm{d}t)$ defined in (S43) and centered at the origin, then decompose the discrepancy ΔI into contributions from the interior and exterior of this region. We explicitly handle this by introducing an indicator variable S defined as $S = 1$ if $\boldsymbol{X}_m \in B(0,R)$ and $S = 0$ otherwise.

Applying the information chain rule $I(\boldsymbol{\eta}_t; \boldsymbol{X}_m) = I(\boldsymbol{\eta}_t; (\boldsymbol{X}_m, S)) = I(\boldsymbol{\eta}_t; S) + I(\boldsymbol{\eta}_t; \boldsymbol{X}_m|S)$ to both the actual dynamics and the proxy, we can decompose the magnitude of the discrepancy as:

$$|\Delta I| \le \big|I(\boldsymbol{\eta}_t; S) - I(\boldsymbol{\eta}_t^G; S)\big| + \mathbb{P}(S=1)|\Delta I_1| + \mathbb{P}(S=0)|\Delta I_0|, \tag{S47}$$

where $\Delta I_k \coloneqq I(\boldsymbol{\eta}_t; \boldsymbol{X}_m|S=k) - I(\boldsymbol{\eta}_t^G; \boldsymbol{X}_m|S=k)$ denotes the conditional information difference for each region.

The first term represents the information about the region index carried by the increments. Since S is a discrete binary variable, this term is bounded by twice the Shannon entropy of S, i.e., $|I(\boldsymbol{\eta}_t; S) - I(\boldsymbol{\eta}_t^G; S)| \le 2H(S)$.

The core of the analysis thus reduces to the conditional terms. For each region ($S = 0, 1$), let p_k and q_k denote the conditional joint densities of the actual and proxy processes, respectively (i.e., $p_k(\cdot) = p_{\mathrm{d}t}(\cdot|S = k)$ and $q_k(\cdot) = q_{\mathrm{d}t}(\cdot|S = k)$). We utilize the following exact decomposition identity for the mutual information difference:

$$\Delta I_1 = D_{\mathrm{KL}}\big(p_1(\boldsymbol{z}, \boldsymbol{x})\|q_1(\boldsymbol{z}, \boldsymbol{x})\big) - D_{\mathrm{KL}}\big(p_1(\boldsymbol{z})\|q_1(\boldsymbol{z})\big) + \int_{B(0,R)} \mathrm{d}\boldsymbol{x} \int_{\mathbb{R}^d} \mathrm{d}\boldsymbol{z} \left(p_1(\boldsymbol{z}, \boldsymbol{x}) - q_1(\boldsymbol{z}, \boldsymbol{x})\right) \ln \frac{q_1(\boldsymbol{z}|\boldsymbol{x})}{q_1(\boldsymbol{z})},$$

where $D_{\mathrm{KL}}(p||q) := \int p \ln(p/q)$ is the relative entropy. Note that since $p_1(\boldsymbol{z})$ and $q_1(\boldsymbol{z})$ are marginals, the data processing inequality yields $D_{\mathrm{KL}}\big(p_1(\boldsymbol{z})\|q_1(\boldsymbol{z})\big) \le D_{\mathrm{KL}}\big(p_1(\boldsymbol{z}, \boldsymbol{x})\|q_1(\boldsymbol{z}, \boldsymbol{x})\big)$. Also because $p_1(\boldsymbol{x}) = q_1(\boldsymbol{x})$, the equality $D_{\mathrm{KL}}\big(p_1(\boldsymbol{z}, \boldsymbol{x})\|q_1(\boldsymbol{z}, \boldsymbol{x})\big) = \mathbb{E}_{\boldsymbol{x}\in B(0,R)}\big[D_{\mathrm{KL}}\big(p_1(\boldsymbol{z}|\boldsymbol{x})\|q_1(\boldsymbol{z}|\boldsymbol{x})\big)\big]$ holds. Therefore we obtain

$$|\Delta I_1| \le \underbrace{\mathbb{E}_{\boldsymbol{x}\in B(0,R)}\big[D_{\mathrm{KL}}\big(p_1(\boldsymbol{z}|\boldsymbol{x})\|q_1(\boldsymbol{z}|\boldsymbol{x})\big)\big]}_{A} + \underbrace{\left|\mathbb{E}_{\boldsymbol{x}\in B(0,R)}\left[\int_{\mathbb{R}^d} \mathrm{d}\boldsymbol{z} \left(p_1(\boldsymbol{z}|\boldsymbol{x}) - q_1(\boldsymbol{z}|\boldsymbol{x})\right) \ln \frac{q_1(\boldsymbol{z}|\boldsymbol{x})}{q_1(\boldsymbol{z})}\right]\right|}_{B}. \tag{S48}$$

Now for $|\Delta I_0|$, we divide the "outside" $B(0,R)^c$ into infinitely many shells by introducing $R_k := 2^k R(\mathrm{d}t)$. More precisely, let A_k denote the shell $2^{k-1}R < \|\boldsymbol{X}_m\| \le 2^k R$ for $k \in \mathbb{N}$. Then $B(0,R)^c = \cup_{k=1}^{\infty} A_k$. This division is useful to obtain the upper bound of $I(\boldsymbol{\eta}_t; \boldsymbol{X}_m|S = 0)$ because

$$\begin{aligned} I(\boldsymbol{\eta}_t; \boldsymbol{X}_m|S = 0) &= h(\boldsymbol{\eta}_t|S = 0) - \mathbb{E}_{\boldsymbol{x}|S=0}\left[h(\boldsymbol{\eta}_t|\boldsymbol{X}_m = \boldsymbol{x}, S = 0)\right] \\ &= h(\boldsymbol{\eta}_t|S = 0) - \sum_{k=1}^{\infty} \frac{\mathbb{P}(\boldsymbol{X}_m \in A_k)}{\mathbb{P}(S = 0)} \mathbb{E}_{\boldsymbol{x}\in A_k}[h(\boldsymbol{\eta}_t|\boldsymbol{X}_m = \boldsymbol{x}, S = 0)] \\ &\le \underbrace{\frac{1}{2} \ln\left((2\pi e)^d \det \mathrm{Cov}(\boldsymbol{\eta}_t|S = 0)\right)}_{G_0} + \frac{1}{\mathbb{P}(S = 0)} \sum_{k=1}^{\infty} \underbrace{\mathbb{P}(\boldsymbol{X}_m \in A_k) \sup_{\boldsymbol{x}\in A_k} \ln\|p_0(\cdot|\boldsymbol{x})\|_\infty}_{S_k}, \end{aligned} \tag{S49}$$

where $h(\cdot)$ is differential entropy and $\mathrm{Cov}(\boldsymbol{\eta}_t|S = 0)$ denotes the conditional covariance matrix of $\boldsymbol{\eta}_t$ given $S = 0$. We use the fact that Gaussian distribution maximizes the differential entropy for fixed covariance and $-h(\cdot) = \int p \ln p \le \int p \ln\|p\|_\infty = \ln\|p\|_\infty$ for the last inequality in (S49).

Since $|\Delta I_0| \le I(\boldsymbol{\eta}_t; \boldsymbol{X}_m|S = 0) + I(\boldsymbol{\eta}_t^G; \boldsymbol{X}_m|S = 0)$, we obtain

$$\mathbb{P}(S = 0)|\Delta I_0| \;\le\; \underbrace{\mathbb{P}(S = 0)\big(I\big(\boldsymbol{\eta}_t^G; \boldsymbol{X}_m|S = 0\big) + G_0\big)}_{K} + \sum_{k=1}^{\infty} S_k, \tag{S50}$$

and therefore

$$|\Delta I| \le 2H(S) + A + B + K + \sum_{k=1}^{\infty} S_k. \tag{S51}$$

In the remaining steps, we show that $H(S), A, B, K$ and $\sum_{k=1}^{\infty} S_k$ are $o(\mathrm{d}t)$. Here $H(S)$ controls the region label, A and B the interior relative-entropy and information-density errors, K the exterior Gaussian contribution, and $\sum_k S_k$ the far-tail shells.

- **Step 2. Bounding the Indicator Entropy:** $H(S) = o(\mathrm{d}t)$

Let P_0 denote the tail probability $\mathbb{P}(S = 0) = \mathbb{P}(\|\boldsymbol{X}_m\| > R)$. The entropy of the indicator S is given by $H(S) = -P_0 \ln P_0 - (1 - P_0) \ln(1 - P_0)$. Note that if $P_0 = 0$ or $P_0 = 1$, then $H(S) = 0$, so the condition $H(S) = o(\mathrm{d}t)$ holds trivially. For $P_0 \in (0, 1)$, we utilize the inequality

$$H(S) \le -P_0 \ln P_0 + P_0. \tag{S52}$$

Now, applying Markov's inequality $\mathbb{P}(X > c) \le \mathbb{E}[X]/c$ with $R(\mathrm{d}t)$ defined in (S43) yields

$$P_0 = \mathbb{P}(\|\boldsymbol{X}_m\|^n > R^n) \le \frac{\mathbb{E}[\|\boldsymbol{X}_m\|^n]}{R^n} = \mathbb{E}[\|\boldsymbol{X}_m\|^n]\mathrm{d}t^{\kappa n} \quad \forall n \in \mathbb{N} \text{ under Assumption A5-A}, \tag{S53-A}$$

$$P_0 = \mathbb{P}(e^{\theta\|\boldsymbol{X}_m\|^{1+\alpha}} > e^{\theta R^{1+\alpha}}) \le \frac{\mathbb{E}[e^{\theta\|\boldsymbol{X}_m\|^{1+\alpha}}]}{e^{\theta R^{1+\alpha}}} = \mathbb{E}\left[e^{\theta\|\boldsymbol{X}_m\|^{1+\alpha}}\right] e^{-\theta(\ln(1/\mathrm{d}t))^{\beta(1+\alpha)}} \text{ under Assumption A5-B}. \tag{S53-B}$$

Assumption A5 ensures that each RHS of (S53) remains finite.

In (S53-A), for any target order $k \in \mathbb{N}$, we can choose a sufficiently large n such that $\kappa n > k$, which implies $P_0 = o(\mathrm{d}t^k)$. Similarly, in (S53-B), since $\beta(1+\alpha) > 1$ implies that the exponential factor dominates any polynomial power $\mathrm{d}t^k = e^{-k|\ln \mathrm{d}t|}$, we also have $P_0 = o(\mathrm{d}t^k)$ for any $k \in \mathbb{N}$. Therefore, we obtain the uniform bound

$$P_0 = \mathbb{P}(\|\boldsymbol{X}_m\| > R) = o(\mathrm{d}t^k) \quad \forall k \in \mathbb{N}, \tag{S54}$$

which is valid for both A5-A and A5-B.

Using (S54) with $k = 2$, we write $P_0 = \mathrm{d}t^2 \epsilon(\mathrm{d}t)$, where $\epsilon(\mathrm{d}t) \to 0$ as $\mathrm{d}t \to 0$. This yields $P_0 \ln P_0 = o(\mathrm{d}t)$ since

$$\begin{aligned} \lim_{\mathrm{d}t\to 0} \frac{P_0 \ln P_0}{\mathrm{d}t} &= \lim_{\mathrm{d}t\to 0} \mathrm{d}t\, \epsilon(\mathrm{d}t)\,[2\ln \mathrm{d}t + \ln \epsilon(\mathrm{d}t)] \\ &= \lim_{\mathrm{d}t\to 0} 2\epsilon(\mathrm{d}t)\big(\mathrm{d}t \ln \mathrm{d}t\big) + \lim_{\mathrm{d}t\to 0} \big(\epsilon(\mathrm{d}t) \ln \epsilon(\mathrm{d}t)\big)\mathrm{d}t \\ &= 0 + 0 = 0. \end{aligned} \tag{S55}$$

Consequently, we verify that

$$H(S) \le -P_0 \ln P_0 + P_0 = o(\mathrm{d}t). \tag{S56}$$

• **Step 3. Bounding the Interior Relative Entropy:** $A = o(\mathrm{d}t)$

To establish $A := \mathbb{E}_{\boldsymbol{x}\in B(0,R)}\big[D_{\mathrm{KL}}\big(p_1(\boldsymbol{z}|\boldsymbol{x})\|q_1(\boldsymbol{z}|\boldsymbol{x})\big)\big] = o(\mathrm{d}t)$, it suffices to show that $D_{\mathrm{KL}}\big(p_1(\boldsymbol{z}|\boldsymbol{x})\|q_1(\boldsymbol{z}|\boldsymbol{x})\big) = o(\mathrm{d}t)$ holds uniformly for $\boldsymbol{x} \in B(0,R)$. Within the region $\|\boldsymbol{X}_m\| \in B(0,R)$, we have $p_1 = p_{\mathrm{d}t}$ and $q_1 = q_{\mathrm{d}t}$; hence, we may directly invoke the expansion (S41) and the bound (S44). However, a direct application of the Chi-square approximation is technically subtle since the tail behavior of $\mathrm{Rem}_{\mathrm{d}t}$ relative to the Gaussian q_1 does not strictly guarantee the integrability of the ratio p_1/q_1. Therefore, we explicitly compute $D_{KL}(p_1\|q_1)$ by decomposing the domain to demonstrate the $o(\mathrm{d}t)$ bound.

To this end, we introduce a cutoff radius $L(\mathrm{d}t) := \sqrt{(5/2)\ln(1/\mathrm{d}t)}$ for the centered increment variable $\boldsymbol{z}_c$, distinct from the spatial truncation $B(0,R)$ for $\boldsymbol{X}_m$. Thus, there are two cutoffs in the proof: $R(\mathrm{d}t)$ truncates the position variable $\boldsymbol{x}$, whereas $L(\mathrm{d}t)$ truncates the centered increment $\boldsymbol{z}_c$. We further assume that ϵ in (S43) is chosen sufficiently small (< 0.1). For notational brevity, let B_R and B_L denote $B(0,R)$ and $B(0,L)$, respectively. The KL divergence can then be decomposed as:

$$\begin{aligned} D_{\mathrm{KL}}\big(p_1(\boldsymbol{z}|\boldsymbol{x})\|q_1(\boldsymbol{z}|\boldsymbol{x})\big) &= \int_{\mathbb{R}^d} p_1(\boldsymbol{z}|\boldsymbol{x}) \ln \frac{p_1(\boldsymbol{z}|\boldsymbol{x})}{q_1(\boldsymbol{z}|\boldsymbol{x})} \mathrm{d}\boldsymbol{z} = \int_{\mathbb{R}^d} p_1(\boldsymbol{z}|\boldsymbol{x}) \ln \left(1 + \underbrace{c_{\mathrm{d}t}(\boldsymbol{z}_c)\,\mathrm{d}t + \frac{\mathrm{Rem}_{\mathrm{d}t}(\boldsymbol{z}_c)}{q_1(\boldsymbol{z}|\boldsymbol{x})}}_{u(\boldsymbol{z}_c;\boldsymbol{x})}\right) \mathrm{d}\boldsymbol{z}_c \\ &= \underbrace{\int_{B_L} p_1(\boldsymbol{z}|\boldsymbol{x}) \ln(1+u) \mathrm{d}\boldsymbol{z}_c}_{A_L} + \underbrace{\int_{(B_L)^c} p_1(\boldsymbol{z}|\boldsymbol{x}) \ln(1+u) \mathrm{d}\boldsymbol{z}_c}_{A_{L^c}}. \end{aligned} \tag{S57}$$

In the first line, the change of variables $\boldsymbol{z} \mapsto \boldsymbol{z}_c = \boldsymbol{z} - \sqrt{\mathrm{d}t}\,\boldsymbol{s}(\boldsymbol{x})$ was employed.

We first examine the term $A_L := \int_{B_L} q_1(\boldsymbol{z}|\boldsymbol{x})(1+u)\ln(1+u)\mathrm{d}\boldsymbol{z}_c$. In the region $\boldsymbol{z}_c \in B_L$, since $c_{\mathrm{d}t}(\cdot;\boldsymbol{x})$ is a polynomial in $\boldsymbol{z}_c$ with coefficients uniformly bounded for $\boldsymbol{x} \in B_R$, it follows that $|c_{\mathrm{d}t}\,\mathrm{d}t| \to 0$ as $\mathrm{d}t \to 0$ uniformly for $(\boldsymbol{z}_c, \boldsymbol{x}) \in B_L \times B_R$. Furthermore, using the bound (S44) and the definition $\phi(\boldsymbol{z}_c) = (2\pi)^{-d/2} e^{-\|\boldsymbol{z}_c\|^2/2}$, we obtain

$$\begin{aligned} \sup_{(\boldsymbol{z}_c,\boldsymbol{x})\in B_L\times B_R} \left|\frac{\mathrm{Rem}_{\mathrm{d}t}(\boldsymbol{z}_c,\boldsymbol{x};R)}{\phi(\boldsymbol{z}_c)}\right| &\le \sup_{(\boldsymbol{z}_c,\boldsymbol{x})\in B_L\times B_R} C\,\mathrm{d}t^{\frac{3}{2}-\epsilon}(1+\|\boldsymbol{z}_c\|^m)\exp\left(\left(\frac{1}{2}-\gamma\right)\|\boldsymbol{z}_c\|^2\right) \\ &\le C\,\mathrm{d}t^{\frac{3}{2}-\epsilon}\,\mathrm{polylog}(1/\mathrm{d}t)\exp\left(-\frac{5}{2}\left(\frac{1}{2}-\gamma\right)_+ \ln \mathrm{d}t\right) \\ &\le C\,\mathrm{d}t^{\frac{1}{4}-\epsilon}\,\mathrm{polylog}(1/\mathrm{d}t) \to 0 \text{ as } \mathrm{d}t \to 0. \end{aligned} \tag{S58}$$

This implies that $u(\boldsymbol{z}_c) \to 0$ uniformly in A_L as $\mathrm{d}t \to 0$. For notational clarity and to distinguish the conditional density from joint or marginal forms, we hereafter denote $q_1(\boldsymbol{z}|\boldsymbol{x})$ as $q_{1|\boldsymbol{x}}$. By applying the Taylor expansion of $\ln(1+u)$ with respect to u, we obtain the following bound for sufficiently small $\mathrm{d}t$:

$$|A_L| = \left|\int_{B_L} q_{1|\boldsymbol{x}}(1+u)\ln(1+u)\mathrm{d}\boldsymbol{z}_c\right| \le \left|\int_{B_L} q_{1|\boldsymbol{x}} u\,\mathrm{d}\boldsymbol{z}_c\right| + O\left(\int_{B_L} q_{1|\boldsymbol{x}} u^2 \mathrm{d}\boldsymbol{z}_c\right). \tag{S59}$$

Since the probability densities integrate to unity, $\int_{\mathbb{R}^d} p_1(\cdot|\boldsymbol{x}) = \int_{\mathbb{R}^d} q_1(\cdot|\boldsymbol{x}) = 1$, and noting that $p_1(\cdot|\boldsymbol{x}) = q_{1|\boldsymbol{x}} + q_{1|\boldsymbol{x}}u$, we have the identity $\int_{\mathbb{R}^d} q_{1|\boldsymbol{x}}u = 0$. Consequently, the first term on the RHS of (S59) can be rewritten as an integral over the complement set:

$$\left|\int_{B_L} q_{1|\boldsymbol{x}} u \,\mathrm{d}\boldsymbol{z}_c\right| = \left|\int_{(B_L)^c} q_{1|\boldsymbol{x}} u \,\mathrm{d}\boldsymbol{z}_c\right| \le \mathrm{d}t \int_{(B_L)^c} q_{1|\boldsymbol{x}}|c_{\mathrm{d}t}|\mathrm{d}\boldsymbol{z}_c + \int_{(B_L)^c} |\mathrm{Rem}_{\mathrm{d}t}|\mathrm{d}\boldsymbol{z}_c. \tag{S60}$$

For the first term of (S60), since $c_{\mathrm{d}t}$ is a polynomial in $\boldsymbol{z}_c$ with bounded coefficients (for $\boldsymbol{x} \in B_R$), the Gaussian tail $\exp(-\|\boldsymbol{z}_c\|^2/2)$ of $q_1(\boldsymbol{z}|\boldsymbol{x}) = \phi(\boldsymbol{z}_c)$ in the region $\boldsymbol{z}_c \in (B_L)^c$ (i.e., $\|\boldsymbol{z}_c\| > \sqrt{2.5\ln(1/\mathrm{d}t)}$) ensures that the integral is of order $O(\mathrm{d}t^{1+(5/4)}) = o(\mathrm{d}t)$. Similarly, for the second term, incorporating the factor $\mathrm{d}t^{3/2-\epsilon}$ from (S44) and the Gaussian tail decay $\exp(-\gamma\|\boldsymbol{z}_c\|^2)$ of $\mathrm{Rem}_{\mathrm{d}t}$, we find it is of order $o(\mathrm{d}t^{3/2-\epsilon}) = o(\mathrm{d}t)$ (given our choice of $\epsilon < 0.1$). Thus, the first term of (S59) is $o(\mathrm{d}t)$.

Next, we analyze the second term of (S59). Using (S44) and the inequality $(a+b)^2 \le 2a^2 + 2b^2$, we have

$$\int_{B_L} q_{1|\boldsymbol{x}}\, u^2 \mathrm{d}\boldsymbol{z}_c \le 2\mathrm{d}t^2 \int_{B_L} q_{1|\boldsymbol{x}}\, c_{\mathrm{d}t}^2 \,\mathrm{d}\boldsymbol{z}_c + 2\int_{B_L} \frac{\mathrm{Rem}_{\mathrm{d}t}^2}{q_{1|\boldsymbol{x}}}\,\mathrm{d}\boldsymbol{z}_c.$$

The first integral on the RHS is $O(\mathrm{d}t^2) = o(\mathrm{d}t)$, as moments of polynomials under a Gaussian measure are finite. For the second integral, proceeding analogously to (S58) by utilizing the bound $\|\boldsymbol{z}_c\| \le L(\mathrm{d}t)$, we estimate its order as $o(\mathrm{d}t^{3-2\epsilon-5/4})$, which is $o(\mathrm{d}t)$. Thus, all terms on the RHS of (S59) vanish faster than $\mathrm{d}t$, leading to the conclusion $A_L = o(\mathrm{d}t)$.

Next, we consider the contribution from the outer region, $A_{L^c} \coloneqq \int_{(B_L)^c} p_1(\cdot|\boldsymbol{x}) \ln(p_1(\cdot|\boldsymbol{x})/q_1(\cdot|\boldsymbol{x}))$. Recalling that $\ln p \le p$ for all $p > 0$, we have $p_1(\cdot|\boldsymbol{x}) \ln p_1(\cdot|\boldsymbol{x}) \le p_1(\cdot|\boldsymbol{x})^2$. Furthermore, given that $-\ln q_{1|\boldsymbol{x}} = \mathrm{const} + \|\boldsymbol{z}_c\|^2/2$, we can combine these with the expansion (S41) to obtain

$$\begin{aligned} A_{L^c} &\le \int_{(B_L)^c} p_1(\boldsymbol{z}|\boldsymbol{x})^2 \,\mathrm{d}\boldsymbol{z}_c + \int_{(B_L)^c} p_1(\boldsymbol{z}|\boldsymbol{x}) \left(C + \frac{\|\boldsymbol{z}_c\|^2}{2}\right) \mathrm{d}\boldsymbol{z}_c \\ &\le \underbrace{2\int_{\|\boldsymbol{z}_c\|>L} q_{1|\boldsymbol{x}}^2 (1 + c_{\mathrm{d}t}\,\mathrm{d}t)^2 \mathrm{d}\boldsymbol{z}_c}_{T_1} + \underbrace{2\int_{\|\boldsymbol{z}_c\|>L} \mathrm{Rem}_{\mathrm{d}t}^2 \mathrm{d}\boldsymbol{z}_c}_{T_2} \\ &\quad + \underbrace{C\int_{\|\boldsymbol{z}_c\|>L} q_{1|\boldsymbol{x}}(1 + |c_{\mathrm{d}t}|\mathrm{d}t)(1 + \|\boldsymbol{z}_c\|^2)\mathrm{d}\boldsymbol{z}_c}_{T_3} + \underbrace{C\int_{\|\boldsymbol{z}_c\|>L} |\mathrm{Rem}_{\mathrm{d}t}|(1 + \|\boldsymbol{z}_c\|^2)\,\mathrm{d}\boldsymbol{z}_c}_{T_4}, \end{aligned} \tag{S61}$$

where the inequality $(a+b)^2 \le 2a^2 + 2b^2$ is used for the first two terms.

Analyzing T_1, since $c_{\mathrm{d}t}(\boldsymbol{z}_c)$ is a polynomial in $\boldsymbol{z}_c$, the integrand takes the form $\mathrm{poly}(\boldsymbol{z}_c)\exp(-\|\boldsymbol{z}_c\|^2)$. Integrating over the region $\|\boldsymbol{z}_c\| > L$, the integral scales as $\mathrm{poly}(L)\exp(-L^2)$, which is $O(\mathrm{polylog}(1/\mathrm{d}t)\,\mathrm{d}t^{5/2}) = o(\mathrm{d}t)$. Similarly, T_3 is characterized by the form $\mathrm{poly}(L)\exp(-L^2/2)$, yielding $O(\mathrm{polylog}(1/\mathrm{d}t)\,\mathrm{d}t^{5/4}) = o(\mathrm{d}t)$.

For the remainder terms, we utilize (S44), which gives $\mathrm{Rem}_{\mathrm{d}t}^2 \le C\mathrm{d}t^{3-2\epsilon}\mathrm{poly}(\boldsymbol{z}_c)\exp(-2\gamma\|\boldsymbol{z}_c\|^2)$. Integrating this bound leads to $T_2 \le O(\mathrm{polylog}(1/\mathrm{d}t)\,\mathrm{d}t^{3-2\epsilon+5\gamma}) = o(\mathrm{d}t)$. By a similar logic, we obtain $T_4 \le O(\mathrm{polylog}(1/\mathrm{d}t)\,\mathrm{d}t^{3/2-\epsilon+5\gamma/2}) = o(\mathrm{d}t)$ (recalling our choice of $\epsilon < 0.1$). As all terms on the RHS of (S61) are $o(\mathrm{d}t)$, it follows that $A_{L^c} = o(\mathrm{d}t)$. Consequently, we arrive at the final conclusion for Step 3,

$$A = \mathbb{E}_{\boldsymbol{x}\in B(0,R)}[A_L + A_{L^c}] = \mathbb{E}_{\boldsymbol{x}\in B(0,R)}[o(\mathrm{d}t)] = o(\mathrm{d}t). \tag{S62}$$

- **Step 4. Bounding the Interior Information Density Difference:** $B = o(\mathrm{d}t)$

Now, let us examine the term $B \coloneqq |\mathbb{E}_{\boldsymbol{x}\in B_R}[\int (p_{1|\boldsymbol{x}}(\boldsymbol{z}) - q_{1|\boldsymbol{x}}(\boldsymbol{z}))\ln(q_{1|\boldsymbol{x}}(\boldsymbol{z})/q_1(\boldsymbol{z}))\,\mathrm{d}\boldsymbol{z}]|$. We begin by analyzing the information density, which is given by logarithmic ratio $\ln(q_{1|\boldsymbol{x}}/q_1)$. Recalling that $q_{1|\boldsymbol{x}}(\boldsymbol{z}) = \phi(\boldsymbol{z}_c) = \phi(\boldsymbol{z} - \sqrt{\mathrm{d}t}\,\boldsymbol{s}(\boldsymbol{x}))$, the marginal density $q_1(\boldsymbol{z})$ is given by the mixture:

$$q_1(\boldsymbol{z}) = \phi(\boldsymbol{z})\,\mathbb{E}_{\boldsymbol{x}\in B_R}\left[\exp\left(\boldsymbol{z}\cdot\left(\sqrt{\mathrm{d}t}\,\boldsymbol{s}\right) - \frac{\|\boldsymbol{s}\|^2\mathrm{d}t}{2}\right)\right]. \tag{S63}$$

Here, $\boldsymbol{s}(\boldsymbol{x}) = \mathbf{B}_m^{-1}\boldsymbol{v}(\boldsymbol{x})$. The inverse diffusion matrix $\mathbf{B}_m^{-1}$ is globally bounded due to the uniform ellipticity condition in Assumption A1. If we assume bounded $\boldsymbol{v}$ via A5-A, then $\boldsymbol{s}$ is globally bounded. Alternatively, under Assumption A5-B, the relation $\boldsymbol{v} = (\boldsymbol{F}_I - \boldsymbol{F}_{rev})/2$ from (S10), combined with the structural form of $\boldsymbol{F}_{rev}$ in (S9) (involving linear

combinations of $\boldsymbol{F}_I$, $\nabla \ln p$, $\mathbf{B}_m$, and their derivatives), implies that $\boldsymbol{s}(\boldsymbol{x})$ exhibits at most polynomial growth in $\boldsymbol{x}$. This follows because $\mathbf{B}_m$ and its derivatives are globally bounded by A2, while the drift and score satisfy polynomial growth bounds by A4. Thus, strictly speaking, we have the global bounds

$$\|\boldsymbol{s}(\boldsymbol{x})\| \le C_{s,A} \text{ under Assumption A5-A,} \tag{S64-A}$$

$$\|\boldsymbol{s}(\boldsymbol{x})\| \le C_{s,B}(1+\|\boldsymbol{x}\|^m) \text{ under Assumption A5-B.} \tag{S64-B}$$

Under A5-A, $\boldsymbol{s}(\boldsymbol{x})$ is globally bounded. Under A5-B, by choosing $R(\mathrm{d}t)$ as in (S43-B), for $\boldsymbol{x} \in B_R$ (i.e., $\|\boldsymbol{x}\| \le (\ln(1/dt))^\beta$), the term $(1+\|\boldsymbol{x}\|^m)$ scales as polylog$(1/\mathrm{d}t)$. Consequently, regardless of the specific assumption (A5-A or A5-B), we can uniformly bound $\|\boldsymbol{s}(\boldsymbol{x})\| \le C_s \,\mathrm{polylog}(1/\mathrm{d}t)$ within the region B_R. This implies $\|\sqrt{\mathrm{d}t}\,\boldsymbol{s}(\boldsymbol{x})\| \le C_s\sqrt{\mathrm{d}t}\,\mathrm{polylog}(1/\mathrm{d}t)$. Therefore, for any arbitrary $\epsilon > 0$, there exists $C_{s,\epsilon}, \mathrm{d}t_0(\epsilon) > 0$ such that

$$\|\sqrt{\mathrm{d}t}\,\boldsymbol{s}\| \le \mathrm{C}_{s,\epsilon}\, dt^{1/2-\epsilon} \quad \forall \mathrm{d}t < \mathrm{d}t_0(\epsilon). \tag{S65}$$

In the remainder of Step 4, we assume $\mathrm{d}t$ is sufficiently small to satisfy (S65). Using the inequalities $\left|\boldsymbol{z}\cdot(\sqrt{\mathrm{d}t}\,\boldsymbol{s})\right| \le \|\boldsymbol{z}\|\|\sqrt{\mathrm{d}t}\boldsymbol{s}\| \le C_{s,\epsilon}\|\boldsymbol{z}\|\mathrm{d}t^{1/2-\epsilon}$ and $\|\boldsymbol{s}\|^2\mathrm{d}t \le C_{s,\epsilon}^2\,\mathrm{d}t^{1-2\epsilon}$, we can derive upper and lower bounds for the expectation term in (S63):

$$\phi(\boldsymbol{z})\exp\left(-C_{s,\epsilon}\|\boldsymbol{z}\|dt^{1/2-\epsilon} - C_{s,\epsilon}^2\,dt^{1-2\epsilon}/2\right) \le q_1(\boldsymbol{z}) \le \phi(\boldsymbol{z})\exp\left(C_{s,\epsilon}\|\boldsymbol{z}\|dt^{1/2-\epsilon}\right), \tag{S66}$$

which leads to the following bound on the log-ratio $\ln(q_1(\boldsymbol{z})/\phi(\boldsymbol{z}))$,

$$\left|\ln\frac{q_1(\boldsymbol{z})}{\phi(\boldsymbol{z})}\right| \le C_{s,\epsilon}\|\boldsymbol{z}\|dt^{1/2-\epsilon} + \frac{C_{s,\epsilon}^2}{2}dt^{1-2\epsilon}. \tag{S67}$$

Similarly, applying the same logic to $\phi(\boldsymbol{z}_c) = \phi(\boldsymbol{z})\exp\left(\boldsymbol{z}\cdot\boldsymbol{s}\sqrt{\mathrm{d}t} - \|\boldsymbol{s}\|^2\mathrm{d}t/2\right)$ yields an analogous bound

$$\left|\ln\frac{\phi(\boldsymbol{z}_c)}{\phi(\boldsymbol{z})}\right| \le C_{s,\epsilon}\|\boldsymbol{z}\|dt^{1/2-\epsilon} + \frac{C_{s,\epsilon}^2}{2}dt^{1-2\epsilon}. \tag{S68}$$

Combining (S67) and (S68), we obtain

$$\left|\ln\frac{q_1(\boldsymbol{z}|\boldsymbol{x})}{q_1(\boldsymbol{z})}\right| \le \left|\ln\frac{\phi(\boldsymbol{z}_c)}{\phi(\boldsymbol{z})}\right| + \left|\ln\frac{q_1(\boldsymbol{z})}{\phi(\boldsymbol{z})}\right| \le 2C_{s,\epsilon}\|\boldsymbol{z}\|\mathrm{d}t^{\frac{1}{2}-\epsilon} + C_{s,\epsilon}^2\mathrm{d}t^{1-2\epsilon} \le 2C_{s,\epsilon}\|\boldsymbol{z}_c\|\mathrm{d}t^{\frac{1}{2}-\epsilon} + 3C_{s,\epsilon}^2\mathrm{d}t^{1-2\epsilon}, \tag{S69}$$

where the last inequality utilizes $\boldsymbol{z} = \boldsymbol{z}_c + \sqrt{\mathrm{d}t}\,\boldsymbol{s}$ and the bound (S65).

Finally, combining this estimate with the density expansion (S41) and the remainder bound (S44), we obtain the bound of the integral for B:

$$\begin{aligned}
\int |p_1(\boldsymbol{z}|\boldsymbol{x}) - q_1(\boldsymbol{z}|\boldsymbol{x})|\left|\ln\frac{q_1(\boldsymbol{z}|\boldsymbol{x})}{q_1(\boldsymbol{z})}\right|\mathrm{d}\boldsymbol{z}_c &\le \int\left(\mathrm{d}t\,|c_{\mathrm{d}t}|\,\phi(\boldsymbol{z}_c) + C\,\mathrm{d}t^{\frac{3}{2}-\epsilon}(1+\|\boldsymbol{z}_c\|^m)e^{-\gamma\|\boldsymbol{z}_c\|^2}\right)\left|\ln\frac{q_1(\boldsymbol{z}|\boldsymbol{x})}{q_1(\boldsymbol{z})}\right| \\
&\le C_1\,\mathrm{d}t^{\frac{3}{2}-\epsilon}\underbrace{\int|c_{\mathrm{d}t}|\|\boldsymbol{z}_c\|\phi(\boldsymbol{z}_c)\mathrm{d}\boldsymbol{z}_c}_{B_1} + C_2\,\mathrm{d}t^{2-2\epsilon}\underbrace{\int|c_{\mathrm{d}t}|\,\phi(\boldsymbol{z}_c)\mathrm{d}\boldsymbol{z}_c}_{B_2} \\
&\quad + C_3\,\mathrm{d}t^{2-2\epsilon}\underbrace{\int(1+\|\boldsymbol{z}_c\|^{m+1})e^{-\gamma\|\boldsymbol{z}_c\|^2}\mathrm{d}\boldsymbol{z}_c}_{B_3} + C_4\,\mathrm{d}t^{\frac{5}{2}-3\epsilon}\underbrace{\int(1+\|\boldsymbol{z}_c\|^m)e^{-\gamma\|\boldsymbol{z}_c\|^2}\mathrm{d}\boldsymbol{z}_c}_{B_4}.
\end{aligned} \tag{S70}$$

Since $c_{\mathrm{d}t}$ is a polynomial in $\boldsymbol{z}_c$ with coefficients bounded for $\boldsymbol{x} \in B_R$, the integrals B_1 and B_2 are finite due to the finiteness of Gaussian moments. Similarly, B_3 and B_4 are finite as they involve integrals of polynomials against Gaussian tails $\exp(-\gamma\|\boldsymbol{z}_c\|^2)$. Consequently, the RHS of (S70) is dominated by the leading order term $O(\mathrm{d}t^{3/2-\epsilon})$, which is $o(\mathrm{d}t)$ when taking small ϵ. This yields the conclusion for B,

$$B = \left|\mathbb{E}_{\boldsymbol{x}\in B_R}[o(\mathrm{d}t)]\right| = o(\mathrm{d}t). \tag{S71}$$

• **Step 5. Bounding the Exterior Information:** $K = o(\mathrm{d}t)$

We now turn our attention to the term $K := \mathbb{P}(S=0)\left[I(\boldsymbol{\eta}_t^G;\boldsymbol{X}_m|S=0) + \frac{1}{2}\ln\left((2\pi e)^d\det\mathrm{Cov}(\boldsymbol{\eta}_t|S=0)\right)\right]$, where $\mathrm{Cov}(\boldsymbol{\eta}_t|S=0)$ denotes the conditional covariance matrix of $\boldsymbol{\eta}_t$ given $S=0$. Conditioned on $S=0$, $\boldsymbol{\eta}_t^G$ represents the

output of a Gaussian channel with input $\sqrt{\mathrm{d}t}\,\boldsymbol{s}(\boldsymbol{X}_m)|S=0$ (as defined in (S45)). Therefore, invoking the standard capacity bound for Gaussian channels yields:

$$I(\boldsymbol{\eta}_t^G;\boldsymbol{X}_m|S=0)\le\frac{1}{2}\ln\det\big(\mathbf{I}_d+\mathrm{d}t\,\mathrm{Cov}(\boldsymbol{s}(\boldsymbol{X}_m)|S=0)\big)\le\frac{\mathrm{d}t}{2}\mathbb{E}\big[\|\boldsymbol{s}(\boldsymbol{X}_m)\|^2|S=0\big]\,. \tag{S72}$$

Here, the final inequality follows from the relation $\ln\det(\mathbf{I}_d+\mathbf{A})\le\mathrm{Tr}(\mathbf{A})$ for any positive semi-definite matrix $\mathbf{A}$.

Multiplying both sides of (S72) by $\mathbb{P}(S=0)$, we obtain

$$\mathbb{P}(S=0)I(\boldsymbol{\eta}_t^G;\boldsymbol{X}_m|S=0)\le\frac{\mathrm{d}t}{2}\mathbb{P}(S=0)\mathbb{E}[\|\boldsymbol{s}(\boldsymbol{X}_m)\|^2|S=0]=\frac{\mathrm{d}t}{2}\mathbb{E}[\|\boldsymbol{s}(\boldsymbol{X}_m)\|^2\mathbf{1}_{S=0}], \tag{S73}$$

where $\mathbf{1}_{S=0}$ denotes the indicator function. Applying the Cauchy-Schwarz inequality to the RHS of (S73) gives

$$\mathbb{E}[\|\boldsymbol{s}\|^2\mathbf{1}_{S=0}]\le\mathbb{E}[\|\boldsymbol{s}\|^4]^{\frac{1}{2}}\mathbb{E}[\mathbf{1}_{S=0}]^{\frac{1}{2}}=\mathbb{E}[\|\boldsymbol{s}\|^4]^{\frac{1}{2}}\mathbb{P}(S=0)^{\frac{1}{2}}. \tag{S74}$$

As discussed in Step 4, $\boldsymbol{s}(\boldsymbol{x})$ exhibits at most polynomial growth in $\boldsymbol{x}$; combined with the moment assumption A5, this ensures $\mathbb{E}[\|\boldsymbol{s}\|^4]<\infty$. Recalling from (S54) that $\mathbb{P}(S=0)^{1/2}=o(\mathrm{d}t)$, the estimates (S73)–(S74) imply $\mathbb{P}(S=0)I(\boldsymbol{\eta}_t^G;\boldsymbol{X}_m|S=0)=o(\mathrm{d}t)$.

Next, to bound $G_0:=\frac{1}{2}\ln\big((2\pi e)^d\det\mathrm{Cov}(\boldsymbol{\eta}_t|S=0)\big)$, we invoke the inequality $\det\mathbf{A}\le(\mathrm{Tr}\mathbf{A}/d)^d$ for positive semi-definite matrices. This yields:

$$G_0\le C+\frac{d}{2}\ln\mathrm{Tr}[\mathrm{Cov}(\boldsymbol{\eta}_t|S=0)]\le C+\frac{d}{2}\ln\mathbb{E}[\|\boldsymbol{\eta}_t\|^2|S=0]\le C+\frac{d}{2}\ln\frac{\mathbb{E}[\|\boldsymbol{\eta}_t\|^2\mathbf{1}_{S=0}]}{\mathbb{P}(S=0)}\le C_1+C_2\ln\frac{1}{\mathbb{P}(S=0)}. \tag{S75}$$

The last inequality utilizes $\mathbb{E}[\|\boldsymbol{\eta}_t\|^2\mathbf{1}_{S=0}]\le\mathbb{E}[\|\boldsymbol{\eta}_t\|^2]$ and the finiteness of the second moment $\mathbb{E}[\|\boldsymbol{\eta}_t\|^2]<\infty$, which follows from the Itô-Taylor expansion (S22). Multiplying (S75) by $\mathbb{P}(S=0)$ and applying the decay rates established in (S54) and (S55), we find $\mathbb{P}(S=0)G_0\le C_1\mathbb{P}(S=0)+C_2\mathbb{P}(S=0)\ln(1/\mathbb{P}(S=0))=o(\mathrm{d}t)$. Consequently, we arrive at the desired conclusion:

$$K=\mathbb{P}(S=0)(I(\boldsymbol{\eta}_t^G;\boldsymbol{X}_m|S=0)+G_0)=o(\mathrm{d}t). \tag{S76}$$

• **Step 6. Bounding the Infinite Shell sum: $\sum_{k=1}^{\infty}S_k=o(\mathrm{d}t)$ and Concluding $\Delta I=o(\mathrm{d}t)$**

To analyze the total sum $\sum_k S_k$, we first consider the individual terms defined by $S_k:=\mathbb{P}(\boldsymbol{X}_m\in A_k)\times\sup_{\boldsymbol{x}\in A_k}\ln\|p_0(\cdot|\boldsymbol{x})\|_\infty$. The event $\boldsymbol{X}_m\in A_k$ corresponds to the shell $2^{k-1}R<\|\boldsymbol{X}_m\|\le 2^kR$; hence, we have the bound $\mathbb{P}(\boldsymbol{X}_m\in A_k)\le P_k:=\mathbb{P}(\|\boldsymbol{X}_m\|>2^{k-1}R)$. Applying Markov's inequality analogously to (S53) in Step 2, we obtain the following upper bounds:

$$P_k=\mathbb{P}(\|\boldsymbol{X}_m\|^n>2^{n(k-1)}R^n)\le\frac{\mathbb{E}[\|\boldsymbol{X}_m\|^n]}{2^{n(k-1)}R^n}=\mathbb{E}[\|\boldsymbol{X}_m\|^n]\,2^{-n(k-1)}\mathrm{R}^{-n}\quad\forall n\in\mathbb{N}\text{ under Assumption A5-A}, \tag{S77-A}$$

$$P_k=\mathbb{P}(e^{\theta\|\boldsymbol{X}_m\|^{1+\alpha}}>e^{\theta(2^{k-1}R)^{1+\alpha}})\le\frac{\mathbb{E}[e^{\theta\|\boldsymbol{X}_m\|^{1+\alpha}}]}{e^{\theta(2^{k-1}R)^{1+\alpha}}}=\mathbb{E}\Big[e^{\theta\|\boldsymbol{X}_m\|^{1+\alpha}}\Big]e^{-\theta(2^{k-1}R)^{1+\alpha}}\text{ under Assumption A5-B}. \tag{S77-B}$$

Assumption A5 ensures that each RHS of (S77) remains finite.

Next, we examine the term $\sup_{\boldsymbol{x}\in A_k}\ln\|p_0(\cdot|\boldsymbol{x})\|_\infty$. First, utilizing the expansion (S41), for $\boldsymbol{X}_m\in A_k$, we can express the density as $p_0(\boldsymbol{z}|\boldsymbol{x})=q_0(\boldsymbol{z}|\boldsymbol{x})(1+c_{\mathrm{d}t}(\boldsymbol{z}_c;\boldsymbol{x})\,\mathrm{d}t)+\mathrm{Rem}_{\mathrm{d}t}(\boldsymbol{z}_c;\boldsymbol{x},R_k)$. Noting that $q_0(\boldsymbol{z}|\boldsymbol{x})=\phi(\boldsymbol{z}_c)$, we obtain

$$\sup_{\boldsymbol{x}\in A_k}\|p_0(\cdot|\boldsymbol{x})\|_\infty\le\sup_{\boldsymbol{x}\in A_k}\|\phi(\cdot)\|_\infty+\sup_{\boldsymbol{x}\in A_k}\|\phi(\cdot)\,c_{\mathrm{d}t}(\cdot;\boldsymbol{x})\|_\infty\mathrm{d}t+\sup_{\boldsymbol{x}\in A_k}\|\mathrm{Rem}_{\mathrm{d}t}(\cdot;\boldsymbol{x},R_k)\|_\infty. \tag{S78}$$

Here, the first term on the RHS of (S78) is the supremum of the Gaussian density $\phi(\boldsymbol{z}_c)$, which is bounded by 1. The second term involves $c_{\mathrm{d}t}(\boldsymbol{z}_c;\boldsymbol{x})$, which is a polynomial in $\boldsymbol{z}_c$ with coefficients growing at most polynomially in $\boldsymbol{x}$. Since this polynomial is multiplied by the Gaussian decay, the term $\sup_{\boldsymbol{z}_c}\|\phi(\cdot)\,c_{\mathrm{d}t}(\cdot;\boldsymbol{x})\|_\infty$ can be bounded by $C(1+\|\boldsymbol{x}\|^r)$ for some finite global constant C. Given that $\sup_{\boldsymbol{x}\in A_k}\|\boldsymbol{x}\|=2^kR$, the second term is bounded by $C(1+(2^kR)^r)\mathrm{d}t$. Thus, (S78) can be rewritten as

$$\sup_{\boldsymbol{x}\in A_k}\|p_0(\cdot|\boldsymbol{x})\|_\infty\le 1+C\big(1+(2^kR)^r\big)\mathrm{d}t+\sup_{\boldsymbol{x}\in A_k}\|\mathrm{Rem}_{\mathrm{d}t}(\cdot;\boldsymbol{x},R_k)\|_\infty. \tag{S79}$$

The remaining term $\sup_{A_k}\|\mathrm{Rem}_{\mathrm{d}t}\|_\infty$ is bounded using (S42). Using the fact that $\|\mathrm{poly}(\boldsymbol{z}_c)\exp(-\gamma\|\boldsymbol{z}_c\|^2)\|_\infty<\infty$ for $\gamma>0$ and noting that the truncation radius applied here is $R_k=2^kR$ (note that we cannot use (S44) uniformly in k since $R_k=2^kR(\mathrm{d}t)$ can be arbitrary large for fixed $\mathrm{d}t$), there exists a finite uniform constant C_{Rem} such that

$$\|\mathrm{Rem}_{\mathrm{d}t}\|_\infty \le C_{\mathrm{Rem}}\,\mathrm{d}t^{3/2}\big(1+(2^kR)^{a_1}\big) \quad \text{under Assumption A5-A,} \tag{S80-A}$$

$$\|\mathrm{Rem}_{\mathrm{d}t}\|_\infty \le C_{\mathrm{Rem}}\Big(\mathrm{d}t^{3/2}+\big(1+(2^kR)^{b_2}\big)\Big)\exp\Big(c_B^{(2)}\big(1+(2^kR)^{b_3}\big)\mathrm{d}t\Big)\big(1+(2^kR)^{b_4}\big) \quad \text{under A5-B,} \tag{S80-B}$$

where the truncation term of the form $\exp(-(\cdot)_+^2/\mathrm{d}t)\le 1$ is omitted from the product for simplicity.

For convenience, we choose $\mathrm{d}t<1$ sufficiently small such that $R(\mathrm{d}t)>1$ hereafter. Combining the results above into (S79), there exist a uniform constant C_∞ and exponents $r_A,r_B>0$ such that

$$\sup_{\boldsymbol{x}\in A_k}\|p_0(\cdot|\boldsymbol{x})\|_\infty \le C_\infty\,2^{r_Ak}R^{r_A} \quad \text{under Assumption A5-A,} \tag{S81-A}$$

$$\sup_{\boldsymbol{x}\in A_k}\|p_0(\cdot|\boldsymbol{x})\|_\infty \le C_\infty\exp\big(2^{r_Bk}R^{r_B}\big) \quad \text{under Assumption A5-B,} \tag{S81-B}$$

where we replace $\mathrm{d}t$ by 1 using $\mathrm{d}t\le 1$ to simplify the upper bounds.

(S81) yield the bound of each S_k with uniform constant $C_A,C_B>0$,

$$S_k=\mathbb{P}(\boldsymbol{X}_m\in A_k)\sup_{\boldsymbol{x}\in A_k}\ln\|p_0(\cdot|\boldsymbol{x})\|_\infty \le C_A\,2^{-n(k-1)}\mathrm{R}^{-n}(r_Ak\ln 2+r_A\ln R) \quad \text{under Assumption A5-A,} \tag{S82-A}$$

$$S_k=\mathbb{P}(\boldsymbol{X}_m\in A_k)\sup_{\boldsymbol{x}\in A_k}\ln\|p_0(\cdot|\boldsymbol{x})\|_\infty \le C_B\,e^{-\theta(2^{k-1}R)^{1+\alpha}}2^{r_Bk}R^{r_B} \quad \text{under Assumption A5-B.} \tag{S82-B}$$

We now proceed to evaluate the infinite sum of these terms for a fixed $R(\mathrm{d}t)$. First, in the case of (S82-A), the exponential decay of P_k with respect to k completely dominates the polynomial growth of log-supnorm term $\sup_{A_k}\|\ln p_0(\cdot|\boldsymbol{x})\|_\infty$. Indeed, letting $r:=2^{-n}<1$ and applying the standard geometric series formulas $\sum_{j=1}^\infty r^{j-1}=\frac{1}{1-r}$ and $\sum_{j=1}^\infty j\,r^{j-1}=\frac{1}{(1-r)^2}$, we can simplify the upper bound of the infinite series. Substituting the choice $R(\mathrm{d}t)=\mathrm{d}t^{-\kappa}$ from (S43-A) yields

$$\sum_{k=1}^\infty S_k \le C_Ar_AR^{-n}\left(\frac{\ln R}{1-2^{-n}}+\frac{\ln 2}{(1-2^{-n})^2}\right)\le 4C_Ar_AR^{-n}\ln(2R)=4C_Ar_A\mathrm{d}t^{\kappa n}\ln\frac{2}{\mathrm{d}t^\kappa} \quad \text{under A5-A.} \tag{S83}$$

By choosing an integer $n\in\mathbb{N}$ such that $\kappa n>1$ (this is always possible since all moments are finite under A5-A), we ensure that $\sum_{k=1}^\infty S_k=o(\mathrm{d}t)$.

For the case of (S82-B), although the log-supnorm term exhibits exponential growth with respect to k, this growth is dominated by the double-exponential decay of P_k. To demonstrate this rigorously, let T_k denote the upper bound of S_k given in the RHS of (S82-B). Clearly, the inequality $\sum_k S_k\le\sum_k T_k$ holds. To bound the sum $\sum_k T_k$, we evaluate the ratio of consecutive terms $\rho(k):=T_{k+1}/T_k$:

$$\rho(k)=\frac{T_{k+1}}{T_k}=\frac{C_B\,e^{-\theta(2^kR)^{1+\alpha}}2^{r_B(k+1)}R^{r_B}}{C_B\,e^{-\theta(2^{k-1}R)^{1+\alpha}}2^{r_Bk}R^{r_B}}=2^{r_B}\exp\Big(-\theta(2^{1+\alpha}-1)R^{1+\alpha}2^{(k-1)(1+\alpha)}\Big). \tag{S84}$$

Since $\theta,a,R>0$, the ratio $\rho(k)$ is strictly decreasing in k, implying $\rho(k)\le\rho(1)$ for all $k\ge 1$. Furthermore, observing that $\rho(1)\to 0$ as $R\to\infty$ (which corresponds to $\mathrm{d}t\to 0$), there exists a global threshold $\mathrm{d}t_0>0$ such that $\rho(k)\le\rho(1)\le 1/2$ holds for all k whenever $\mathrm{d}t<\mathrm{d}t_0$. Under this condition, the sum is bounded by a geometric series:

$$\sum_{k=1}^\infty T_k=T_1+\sum_{k=2}^\infty\left(\prod_{j=1}^{k-1}\rho(j)\right)T_1\le T_1+\sum_{k=1}^\infty\rho(1)^kT_1\le T_1+\sum_{k=1}^\infty\left(\frac{1}{2}\right)^kT_1=2T_1.$$

Substituting $R(\mathrm{d}t)=(\ln(1/\mathrm{d}t))^\beta$ with $\beta(1+\alpha)>1$ from (S43-B) into the expression for T_1, we obtain

$$\sum_{k=1}^\infty S_k\le\sum_{k=1}^\infty T_k\le 2T_1=2^{r_B+1}C_B\left(\ln\frac{1}{\mathrm{d}t}\right)^{\beta r_B}\exp\left(-\theta\left(\ln\frac{1}{\mathrm{d}t}\right)^{\beta(1+\alpha)}\right)=o(\mathrm{d}t)\quad\forall\mathrm{d}t<\mathrm{d}t_0 \text{ under A5-B,} \tag{S85}$$

where the limit holds because the ratio $2T_1/\mathrm{d}t = 2T_1 \exp(\ln(1/\mathrm{d}t))$ vanishes as $\mathrm{d}t \to 0$. Specifically, considering the exponent in terms of $x = \ln(1/\mathrm{d}t)$, we have $x + \beta r_B \ln x - \theta x^{\beta(1+\alpha)} \to -\infty$ as $x \to \infty$, provided that $\beta(1+\alpha) > 1$. Consequently, combining (S83) and (S85), we establish the target conclusion for all cases,

$$\sum_{k=1}^{\infty} S_k = o(\mathrm{d}t). \tag{S86}$$

Having established via (S56), (S62), (S71), and (S76), together with the result (S86) just derived. Thus every term on the RHS of the decomposition (S51) is $o(\mathrm{d}t)$, we have successfully proven the main objective of this subsection:

$$I(\boldsymbol{\eta}_t; \boldsymbol{X}_m) - I(\boldsymbol{\eta}_t^G; \boldsymbol{X}_m) = o(\mathrm{d}t) \quad \text{as } \mathrm{d}t \to 0. \tag{S87}$$

Therefore the Gaussian proxy is asymptotically exact at order $\mathrm{d}t$ for mutual information.

G. The main identity

We now determine the exact leading-order term of $I(\boldsymbol{\eta}_t; \boldsymbol{X}_m)$ in powers of $\mathrm{d}t$. Here, the previously derived estimate (S87) plays a pivotal role. By Eq. (S45), the proxy variable $\boldsymbol{\eta}_t^G$ can be viewed as the output of an additive Gaussian channel with a small signal-to-noise ratio (SNR) of order $\mathrm{d}t$. In this regime, the exact leading term of the mutual information can be calculated using the I-MMSE relation established by Guo et al. [27].

Specifically, Guo et al. showed that for a Gaussian channel with input $\boldsymbol{S}$ and output $\boldsymbol{Y} := \sqrt{\mathrm{snr}}\boldsymbol{S} + \boldsymbol{N}$, the derivative of the mutual information with respect to snr equals half of the minimum mean-squared error (MMSE) [27]. Since $\mathrm{MMSE}(0) = \mathrm{Tr}\,\mathrm{Cov}(\boldsymbol{S})$ at $\mathrm{snr} = 0$, integrating this over small snr yields $I(\boldsymbol{Y}; \boldsymbol{S}) = \frac{\mathrm{snr}}{2}\mathrm{Tr}\,\mathrm{Cov}(\boldsymbol{S}) + o(\mathrm{snr})$. Applying this to our model (S45), we obtain the exact leading term:

$$I(\boldsymbol{\eta}_t^G; \boldsymbol{X}_m) = \frac{\mathrm{d}t}{2}\mathrm{Tr}\,\mathrm{Cov}\left[\boldsymbol{s}(\boldsymbol{X}_m, t_m)\right] + o(\mathrm{d}t). \tag{S88}$$

By the bound (S87), the actual information $I(\boldsymbol{\eta}_t; \boldsymbol{X}_m)$ shares this identical leading term. Here, $\boldsymbol{s} = \mathbf{B}_m^{-1}\boldsymbol{v}$. Using the continuity assumptions for $\boldsymbol{F}_I$ and $\mathbf{B}$ (A2) and for the density (A3), we have $\boldsymbol{v}_m = \boldsymbol{v}(\boldsymbol{X}_t, t) + o(1)$ and $\mathbf{B}_m^{-1} = \mathbf{B}(\boldsymbol{X}_t, t)^{-1} + o(1)$. Furthermore, utilizing the definition of the diffusion tensor $\mathbf{D} = \frac{1}{2}\mathbf{B}\mathbf{B}^{\intercal}$, we expand the covariance term as follows:

$$\begin{aligned} I(\boldsymbol{\eta}_t; \boldsymbol{X}_m) &= \frac{\mathrm{d}t}{2}\mathrm{Tr}\,\mathrm{Cov}\left[\mathbf{B}(\boldsymbol{X}_t, t)^{-1}\boldsymbol{v}(\boldsymbol{X}_t, t)\right] + o(\mathrm{d}t) \\ &= \frac{\mathrm{d}t}{2}\left(\frac{1}{2}\mathbb{E}\left[\boldsymbol{v}_t^{\intercal}\mathbf{D}_t^{-1}\boldsymbol{v}_t\right] - \frac{1}{2}\left\|\mathbb{E}[\mathbf{D}_t^{-1/2}\boldsymbol{v}_t]\right\|^2\right) + o(\mathrm{d}t) \\ &= \frac{\mathrm{d}t}{4}\sigma_t - \frac{\mathrm{d}t}{4}\left\|\mathbb{E}[\mathbf{D}_t^{-1/2}\boldsymbol{v}_t]\right\|^2 + o(\mathrm{d}t). \end{aligned} \tag{S89}$$

where $(\cdot)_t := (\cdot)(\boldsymbol{X}_t, t)$. The second equality uses the identity $\mathrm{Tr}\,\mathrm{Cov}(\boldsymbol{w}) = \mathbb{E}[\|\boldsymbol{w}\|^2] - \|\mathbb{E}[\boldsymbol{w}]\|^2$ for a random vector $\boldsymbol{w}$ and the relation $\|\mathbf{B}^{-1}\boldsymbol{v}\|^2 = \boldsymbol{v}^{\intercal}(\mathbf{B}\mathbf{B}^{\intercal})^{-1}\boldsymbol{v} = \frac{1}{2}\boldsymbol{v}^{\intercal}\mathbf{D}^{-1}\boldsymbol{v}$. The final equality substitutes the definition of the entropy production rate (S5).

Dividing by $\mathrm{d}t$ and taking the limit $\mathrm{d}t \to 0$, we establish the generalized main identity for multiplicative noise, extending Eq. (2) of the main text:

$$\sigma_t = 4 \lim_{\mathrm{d}t \to 0} \frac{I(\boldsymbol{\eta}_t; \boldsymbol{X}_m)}{\mathrm{d}t} + \|\mathbb{E}[\mathbf{D}_t^{-1/2}\boldsymbol{v}_t]\|^2. \tag{S90}$$

If $\mathbf{D}_t$ is spatially constant (i.e., additive noise), then $\mathbb{E}[\mathbf{D}_t^{-1/2}\boldsymbol{v}_t] = \mathbf{D}_t^{-1/2}\mathbb{E}[\boldsymbol{v}_t]$, and also $I(\boldsymbol{\eta}_t; \boldsymbol{X}_m) = I(\mathrm{d}\boldsymbol{x}; \boldsymbol{X}_m)$ since $\boldsymbol{\eta}_t = \mathbf{B}_m^{-1}\mathrm{d}\boldsymbol{x}/\sqrt{\mathrm{d}t}$ becomes linear transformation with coordinate-independent matrix $\frac{1}{\sqrt{\mathrm{d}t}}\mathbf{B}(t_m)^{-1}$. Hence Eq. (S90) reduces exactly to the main identity in the main text, Eq. (2).

The rest of the SM only propagates this result to projections, subsystem decompositions, and the RBC applications.

II. PROOFS OF THE REMAINING RESULTS

In this section, we prove the remaining main-text results—Eqs. (5) and (6b)—by first establishing how the identity behaves under linear transformations and then applying that result to bipartite subsystem projections.

A. Extension of the Main identity to Linear Transformations

While (S89) provides a powerful result for the full system, it is necessary to extend this framework to accommodate subsystems or their linear combinations. To this end, let $T_{\boldsymbol{\eta}}$ and $T_{\boldsymbol{X}}$ be deterministic linear transformations acting independently on $\boldsymbol{\eta}_t$ and $\boldsymbol{X}_m$, respectively. These transformations need not be identical nor invertible; in particular, they can represent projection maps onto subspaces. In this setting, we first aim to establish the following asymptotic equivalence:

$$\Delta I_T \coloneqq I\big(T_{\boldsymbol{\eta}}(\boldsymbol{\eta}_t); T_{\boldsymbol{X}}(\boldsymbol{X}_m)\big) - I\big(T_{\boldsymbol{\eta}}(\boldsymbol{\eta}_t^G); T_{\boldsymbol{X}}(\boldsymbol{X}_m)\big) = o(\mathrm{d}t). \tag{S91}$$

This generalizes (S87). The proof follows a structure almost identical to that presented in Section I.F. As in Section I.F, we introduce the same compact ball $B(0,R)$ of radius $R = R(\mathrm{d}t)$ defined in (S43), along with the same indicator variable S such that $S = 1$ if $\boldsymbol{X}_m \in B(0,R)$ and $S = 0$ otherwise. Now we introduce the shorthand variables

$$\boldsymbol{U} \coloneqq T_{\boldsymbol{\eta}}(\boldsymbol{\eta}_t), \quad \boldsymbol{U}^G \coloneqq T_{\boldsymbol{\eta}}(\boldsymbol{\eta}_t^G), \quad \boldsymbol{W} \coloneqq T_{\boldsymbol{X}}(\boldsymbol{X}_m),$$

for simplicity. Using the standard information-theoretic identity $I(A;\boldsymbol{W}) = I(A;S) - I(A;S|\boldsymbol{W}) + I(A;\boldsymbol{W}|S)$ for $A = \boldsymbol{U}, \boldsymbol{U}^G$, we can rearrange ΔI_T to obtain a decomposition analogous to (S47),

$$|\Delta I_T| \le \big|I(\boldsymbol{U};S) - I(\boldsymbol{U}^G;S) - I(\boldsymbol{U};S|\boldsymbol{W}) + I(\boldsymbol{U}^G;S|\boldsymbol{W})\big| + \mathbb{P}(S=1)|\Delta I_{T1}| + \mathbb{P}(S=0)|\Delta I_{T0}|, \tag{S92}$$

where $\Delta I_{Tk} \coloneqq I(\boldsymbol{U};\boldsymbol{W}|S=k) - I(\boldsymbol{U}^G;\boldsymbol{W}|S=k)$. Furthermore, since S is a binary random variable, the universal bound $I(X;S) \le H(S)$ and $I(X;S|Y) \le H(S|Y) \le H(S)$ hold for any arbitrary random variables X and Y. Consequently, the magnitude of the terms in the first four terms on RHS of (S92) is strictly bounded by $4H(S)$, hence $o(\mathrm{d}t)$ because of (S56) in step 2 of Section I.F.

Moreover, we have $|\Delta I_{T0}| \le I\big(\boldsymbol{U};\boldsymbol{W}|S=0\big) + I\big(\boldsymbol{U}^G;\boldsymbol{W}|S=0\big)$. By the data processing inequality (DPI), this sum is bounded above by $I(\boldsymbol{\eta}_t;\boldsymbol{X}_m|S=0) + I(\boldsymbol{\eta}_t^G;\boldsymbol{X}_m|S=0)$. Since $\mathbb{P}(S=0)\big[I(\boldsymbol{\eta}_t;\boldsymbol{X}_m|S=0) + I(\boldsymbol{\eta}_t^G;\boldsymbol{X}_m|S=0)\big] = o(\mathrm{d}t)$ due to (S50), (S76), and (S86) in step 5-6 of Section I.F, the last term in (S92) is also $o(\mathrm{d}t)$. For the remaining term $|\Delta I_{T1}|$, following the exact same decomposition logic as in (S48), we can bound it as

$$|\Delta I_{T1}| \le \underbrace{D_{\mathrm{KL}}\big(p_1^T(\boldsymbol{u},\boldsymbol{w}) \| q_1^T(\boldsymbol{u},\boldsymbol{w})\big)}_{A_T} + \underbrace{\left|\mathbb{E}_{\boldsymbol{X}_m \in B(0,R)}\left[\int_{\mathbb{R}^{\dim(\boldsymbol{u})}} \mathrm{d}\boldsymbol{u}\, \big(p_1^T(\boldsymbol{u}|\boldsymbol{w}) - q_1^T(\boldsymbol{u}|\boldsymbol{w})\big) \ln \frac{q_1^T(\boldsymbol{u}|\boldsymbol{w})}{q_1^T(\boldsymbol{u})}\right]\right|}_{B_T}. \tag{S93}$$

Here, $p_1^T(\boldsymbol{u},\boldsymbol{w})$ and $p_1^T(\boldsymbol{u}|\boldsymbol{w})$ serve as shorthand notations for the joint and conditional probability densities of the transformed actual dynamics, evaluated at $T_{\boldsymbol{\eta}}(\boldsymbol{\eta}_t) = \boldsymbol{u}$ and $T_{\boldsymbol{X}}(\boldsymbol{X}_m) = \boldsymbol{w}$ given $S = 1$. The symbol q_1^T denotes the analogous probability densities for the transformed Gaussian proxy $T_{\boldsymbol{\eta}}(\boldsymbol{\eta}_t^G)$. Because the marginal laws of $\boldsymbol{X}_m$ (and thus $T_{\boldsymbol{X}}(\boldsymbol{X}_m)$) are identical for both processes, the term A in (S48) can be rewritten as the joint relative entropy $D_{\mathrm{KL}}\big(p_1(\boldsymbol{z},\boldsymbol{x}) \| q_1(\boldsymbol{z},\boldsymbol{x})\big)$. Applying the DPI to this divergence directly implies $A_T \le A$. Given that $A = o(\mathrm{d}t)$ was established in (S62), it straightforwardly follows that $A_T = o(\mathrm{d}t)$. Consolidating these results into (S92), we obtain

$$|\Delta I_T| \le B_T + o(\mathrm{d}t).$$

Therefore what we need is bounding $B_T = o(\mathrm{d}t)$. To this end, let us focus on the region $S = 1$. The key point is that, even after applying the transformation T, the same argument as in Step 4 of Section I.F still applies.

If $T_{\boldsymbol{\eta}} = 0$, then $B_T = 0$ trivially. Otherwise we restrict the codomain of $T_{\boldsymbol{\eta}}$ to its image space $\mathrm{Im}(T_{\boldsymbol{\eta}})$, so that the transformed Gaussian noise has covariance $\boldsymbol{\Sigma} \coloneqq T_{\boldsymbol{\eta}} T_{\boldsymbol{\eta}}^\intercal$ which is nondegenerate on this space. Then we obtain $\left\|\boldsymbol{\Sigma}^{-\frac{1}{2}} T_{\boldsymbol{\eta}} \boldsymbol{y}\right\| \le \|\boldsymbol{y}\|$ for all $\boldsymbol{y} \in \mathbb{R}^d$ since $\|\boldsymbol{\Sigma}^{-1/2} T_{\boldsymbol{\eta}} \boldsymbol{y}\|^2 = \boldsymbol{y}^\intercal T_{\boldsymbol{\eta}}^\intercal (T_{\boldsymbol{\eta}} T_{\boldsymbol{\eta}}^\intercal)^{-1} T_{\boldsymbol{\eta}} \boldsymbol{y}$, and the matrix $T_{\boldsymbol{\eta}}^\intercal (T_{\boldsymbol{\eta}} T_{\boldsymbol{\eta}}^\intercal)^{-1} T_{\boldsymbol{\eta}}$ is an orthogonal projection.

Now let ϕ_Σ denote the corresponding Gaussian density. The transformed Gaussian proxy then satisfies

$$\boldsymbol{U}^G = \sqrt{\mathrm{d}t}\, T_{\boldsymbol{\eta}}\, \boldsymbol{s}(\boldsymbol{X}_m) + \boldsymbol{\xi}, \quad \boldsymbol{\xi} \sim \mathcal{N}(0, \boldsymbol{\Sigma}). \tag{S94}$$

Consequently,

$$q_1^T(\boldsymbol{u}|\boldsymbol{w}) = \mathbb{E}\Big[\phi_\Sigma\Big(\boldsymbol{u} - \sqrt{\mathrm{d}t}\, T_{\boldsymbol{\eta}}\, \boldsymbol{s}(\boldsymbol{X}_m)\Big)\Big|\boldsymbol{W} = \boldsymbol{w}, S = 1\Big],$$
$$q_1^T(\boldsymbol{u}) = \mathbb{E}\Big[\phi_\Sigma\Big(\boldsymbol{u} - \sqrt{\mathrm{d}t}\, T_{\boldsymbol{\eta}}\, \boldsymbol{s}(\boldsymbol{X}_m)\Big)\Big|S = 1\Big].$$

Using the bound $\left\|\sqrt{\mathrm{d}t}\,\boldsymbol{s}(\boldsymbol{x})\right\| \le C_\epsilon \mathrm{d}t^{\frac{1}{2}-\epsilon}$ on $B(0,R)$ from (S65) with $\|\boldsymbol{\Sigma}^{-1/2}T_{\boldsymbol{\eta}}\boldsymbol{y}\| \le \|\boldsymbol{y}\|$ for both $\boldsymbol{y}=\boldsymbol{s}(\boldsymbol{x})$ and $\boldsymbol{y}=\boldsymbol{z}$, the same argument as in (S66)–(S68) in Step 4 of Section I.F yields

$$\left|\ln \frac{q_1^T(\boldsymbol{u}|\boldsymbol{w})}{q_1^T(\boldsymbol{u})}\right| \le C_\epsilon \mathrm{d}t^{\frac{1}{2}-\epsilon}\left\|\boldsymbol{\Sigma}^{-\frac{1}{2}}\boldsymbol{u}\right\| + C_\epsilon \mathrm{d}t^{1-2\epsilon} \tag{S95}$$

Next, since $\boldsymbol{W}=T_{\boldsymbol{X}}(\boldsymbol{X}_m)$ is a deterministic function of $\boldsymbol{X}_m$,

$$p_1^T(\boldsymbol{u}|\boldsymbol{w}) - q_1^T(\boldsymbol{u}|\boldsymbol{w}) = \mathbb{E}\left[\int \mathrm{d}\boldsymbol{z}\,\delta\big(\boldsymbol{u}-T_{\boldsymbol{\eta}}\boldsymbol{z}\big)\big(p_1(\boldsymbol{z}|\boldsymbol{x})-q_1(\boldsymbol{z}|\boldsymbol{x})\big)\middle|\boldsymbol{W}=\boldsymbol{w}\right].$$

Hence, by Jensen's inequality, for any nonnegative function g,

$$\int \mathrm{d}\boldsymbol{u}\big|p_1^T(\boldsymbol{u}|\boldsymbol{w}) - q_1^T(\boldsymbol{u}|\boldsymbol{w})\big|g(\boldsymbol{u}) \le \mathbb{E}\left[\int \mathrm{d}\boldsymbol{z}\big|p_1(\boldsymbol{z}|\boldsymbol{x})-q_1(\boldsymbol{z}|\boldsymbol{x})\big|g(T_{\boldsymbol{\eta}}\boldsymbol{z})\middle|\boldsymbol{W}=\boldsymbol{w}\right]. \tag{S96}$$

Now one can choose $g(\boldsymbol{u}) = 1+\|\boldsymbol{\Sigma}^{-1/2}\boldsymbol{u}\|$. Since $1+\|\boldsymbol{\Sigma}^{-1/2}\boldsymbol{u}\| = 1+\|\boldsymbol{\Sigma}^{-1/2}T_{\boldsymbol{\eta}}\boldsymbol{z}\| \le 1+\|\boldsymbol{z}\|$, substituting this into the bound (S96) and taking expectation over $S=1$ yields

$$B_T \le C_\epsilon \mathrm{d}t^{\frac{1}{2}-\epsilon}\mathbb{E}\left[\int \mathrm{d}\boldsymbol{z}|p_1(\boldsymbol{z}|\boldsymbol{x})-q_1(\boldsymbol{z}|\boldsymbol{x})|\|\boldsymbol{z}\|\right] + C_\epsilon \mathrm{d}t^{1-2\epsilon}\mathbb{E}\left[\int \mathrm{d}\boldsymbol{z}|p_1(\boldsymbol{z}|\boldsymbol{x})-q_1(\boldsymbol{z}|\boldsymbol{x})|\right]. \tag{S97}$$

Finally, from the density expansion (S41) and the remainder bound (S44), and by the same argument as in (S69)–(S71) in Step 4 of section 1.F, the integrals on the RHS of (S97) are all $O(\mathrm{d}t)$. The technical details are identical and therefore omitted. Consequently $B_T = o(\mathrm{d}t)$, which proves (S91).

Next, let us evaluate the term $I(\boldsymbol{U}^G;\boldsymbol{W})$. Normalizing the proxy defined in (S94) yields

$$\hat{\boldsymbol{U}}^G = \sqrt{\mathrm{d}t}\,\hat{\boldsymbol{s}}(\boldsymbol{X}_m) + \boldsymbol{N}, \quad \boldsymbol{N}\sim\mathcal{N}(0,\mathbf{I}),$$

where $\hat{\boldsymbol{U}}^G := \boldsymbol{\Sigma}^{-1/2}\boldsymbol{U}^G$ and $\hat{\boldsymbol{s}} := \boldsymbol{\Sigma}^{-1/2}T_{\boldsymbol{\eta}}\,\boldsymbol{s}$.

Since the linear transformation $\boldsymbol{U}^G \mapsto \hat{\boldsymbol{U}}^G$ is strictly invertible on the image space $\mathrm{Im}(T_{\boldsymbol{\eta}})$, it preserves mutual information, implying $I(\boldsymbol{U}^G;\boldsymbol{W}) = I(\hat{\boldsymbol{U}}^G;\boldsymbol{W})$. Furthermore, because the channel signal is entirely determined by $\hat{\boldsymbol{s}}$, the variables form a Markov chain $\boldsymbol{W}\to\hat{\boldsymbol{s}}\to\hat{\boldsymbol{U}}^G$. This Markov property ensures that the conditional mutual information vanishes ($I(\hat{\boldsymbol{U}}^G;\boldsymbol{W}|\hat{\boldsymbol{s}})=0$), which allows us to decompose the mutual information as follows:

$$I(\hat{\boldsymbol{U}}^G;\boldsymbol{W}) = I(\hat{\boldsymbol{U}}^G;\hat{\boldsymbol{s}}) - I(\hat{\boldsymbol{U}}^G;\hat{\boldsymbol{s}}|\boldsymbol{W}).$$

Then, by the same small-SNR I-MMSE expansion used in (S88) of Section I.G,

$$I(\hat{\boldsymbol{U}}^G;\hat{\boldsymbol{s}}) = \frac{\mathrm{d}t}{2}\mathrm{Tr}\,\mathrm{Cov}(\hat{\boldsymbol{s}}) + o(\mathrm{d}t),$$

and, similarly, conditioning on W and averaging,

$$I(\hat{\boldsymbol{U}}^G;\hat{\boldsymbol{s}}|\boldsymbol{W}) = \frac{\mathrm{d}t}{2}\mathrm{Tr}\,\mathbb{E}[\mathrm{Cov}(\hat{\boldsymbol{s}}|\boldsymbol{W})] + o(\mathrm{d}t).$$

Combining these results with the law of total covariance $\mathrm{Cov}(X) = \mathbb{E}[\mathrm{Cov}(X|Y)] + \mathrm{Cov}(\mathbb{E}[X|Y])$, we obtain

$$I(\hat{\boldsymbol{U}}^G;\boldsymbol{W}) = \frac{\mathrm{d}t}{2}\mathrm{Tr}\,\mathrm{Cov}(\mathbb{E}[\hat{\boldsymbol{s}}|\boldsymbol{W}]) + o(\mathrm{d}t) = \frac{\mathrm{d}t}{2}\mathrm{Tr}\,\boldsymbol{\Sigma}^{-1}\mathrm{Cov}(\mathbb{E}[T_{\boldsymbol{\eta}}\boldsymbol{s}(\boldsymbol{X}_m,t_m)|T_{\boldsymbol{X}}(\boldsymbol{X}_m)]) + o(\mathrm{d}t). \tag{S98}$$

Now, to justify replacing the midpoint $(\boldsymbol{X}_m,t_m)$ on RHS of (S98) by the initial point $(\boldsymbol{X}_t,t)$, let $\boldsymbol{W}_u := T_{\boldsymbol{X}}(\boldsymbol{X}_u)$ and $\boldsymbol{m}_u(\boldsymbol{w}) := \mathbb{E}[T_{\boldsymbol{\eta}}\,\boldsymbol{s}(\boldsymbol{X}_u,u)\mid \boldsymbol{W}_u=\boldsymbol{w}]$ for $u\in[t,t+\mathrm{d}t]$. Since $T_{\boldsymbol{X}}$ is linear, after a fixed linear change of coordinates adapted to $T_{\boldsymbol{X}}$, $\boldsymbol{m}_u(\boldsymbol{w})$ can be written as the ratio of fiber integrals of $T_{\boldsymbol{\eta}}\,\boldsymbol{s}(\boldsymbol{x},u)p(\boldsymbol{x},u)$ and $p(\boldsymbol{x},u)$; hence, by the same continuity/moment assumptions used in Section I.G, $\boldsymbol{m}_u$ is continuous in $(\boldsymbol{w},u)$ with at most polynomial growth. Since $\boldsymbol{W}_m = \boldsymbol{W}_t + o_{L^2}(1)$, we obtain $\boldsymbol{m}_{t_m}(\boldsymbol{W}_m) = \boldsymbol{m}_t(\boldsymbol{W}_t) + o_{L^2}(1)$ and therefore $\mathrm{Cov}\big(\boldsymbol{m}_{t_m}(\boldsymbol{W}_m)\big) = \mathrm{Cov}\big(\boldsymbol{m}_t(\boldsymbol{W}_t)\big) + o(1)$. Combining all these results yields the exact analytic form of the information rate,

$$\lim_{\mathrm{d}t\to 0}\frac{I\big(T_{\boldsymbol{\eta}}(\boldsymbol{\eta}_t);T_{\boldsymbol{X}}(\boldsymbol{X}_m)\big)}{\mathrm{d}t} = \frac{1}{2}\mathrm{Tr}\big[(T_{\boldsymbol{\eta}}T_{\boldsymbol{\eta}}^\intercal)^{-1}\mathrm{Cov}(\mathbb{E}[\,T_{\boldsymbol{\eta}}\mathbf{B}(\boldsymbol{X}_t,t)^{-1}\boldsymbol{v}(\boldsymbol{X}_t,t)\mid T_{\boldsymbol{X}}(\boldsymbol{X}_t)])\big], \tag{S99}$$

where $(T_{\boldsymbol{\eta}}T_{\boldsymbol{\eta}}^\intercal)^{-1}$ is inverse on $\mathrm{Im}(T_{\boldsymbol{\eta}})$. The expression (S99) will serve as a crucial building block in subsequent proofs.

B. Proof of the local EP identity (Eq. 5)

We now derive the subsystem identity directly from (S99). Choosing $T_{\boldsymbol{\eta}} = P_A$, $T_{\boldsymbol{X}} = \mathbf{I}$, (S99) gives

$$\lim_{\mathrm{d}t\to 0} \frac{I(P_A\,\boldsymbol{\eta}_t; \boldsymbol{X}_m)}{\mathrm{d}t} = \frac{1}{2}\mathrm{Tr}\,\mathrm{Cov}(\mathbf{B}_A^{-1}\boldsymbol{v}_A).$$

In the additive-noise case,

$$P_A\,\boldsymbol{\eta}_t = \frac{1}{\sqrt{\mathrm{d}t}}\mathbf{B}_A^{-1}\mathrm{d}\boldsymbol{x}_A,$$

and since $\mathbf{B}_A$ is constant, this is an invertible linear transformation of $\mathrm{d}\boldsymbol{x}_A$. Hence mutual information is preserved: $I(P_A\,\boldsymbol{\eta}_t; \boldsymbol{X}_m) = I(\mathrm{d}\boldsymbol{x}_A; \boldsymbol{x}_m)$. Now using $\mathbf{B}_A\mathbf{B}_A^{\intercal} = 2\mathbf{D}_A$, we obtain

$$\mathrm{Tr}\,\mathrm{Cov}(\mathbf{B}_A^{-1}\boldsymbol{v}_A) = \frac{1}{2}\Big(\langle \boldsymbol{v}_A^{\intercal}\mathbf{D}_A^{-1}\boldsymbol{v}_A\rangle - \langle \boldsymbol{v}_A\rangle^{\intercal}\mathbf{D}_A^{-1}\langle \boldsymbol{v}_A\rangle\Big).$$

Therefore

$$\lim_{\mathrm{d}t\to 0} \frac{I(\mathrm{d}\boldsymbol{x}_A; \boldsymbol{x}_m)}{\mathrm{d}t} = \frac{1}{4}\Big(\langle \boldsymbol{v}_A^{\intercal}\mathbf{D}_A^{-1}\boldsymbol{v}_A\rangle - \langle \boldsymbol{v}_A\rangle^{\intercal}\mathbf{D}_A^{-1}\langle \boldsymbol{v}_A\rangle\Big).$$

Recalling $\sigma_A = \langle \boldsymbol{v}_A^{\intercal}\mathbf{D}_A^{-1}\boldsymbol{v}_A\rangle$ and $\sigma_A^{\mathrm{mf}} \coloneqq \langle \boldsymbol{v}_A\rangle^{\intercal}\mathbf{D}_A^{-1}\langle \boldsymbol{v}_A\rangle$, we immediately obtain

$$\sigma_A = 4\,\mathcal{I}(\mathrm{d}\boldsymbol{x}_A; \boldsymbol{x}_m) + \sigma_A^{\mathrm{mf}}, \tag{S100}$$

which is Eq. (5) in the main text.

C. Proof of current-space expression of the EP channels (Eq. 6b)

Self channel contribution is defined by $\sigma_A^{\mathrm{self}} \coloneqq 4\,\mathcal{I}(\mathrm{d}\boldsymbol{x}_A; \boldsymbol{x}_m^A) + \sigma_A^{\mathrm{mf}}$. To evaluate it, we again use (S99), now with $T_{\boldsymbol{\eta}} = T_{\boldsymbol{X}} = P_A$. Then

$$\lim_{\mathrm{d}t\to 0} \frac{I(P_A\,\boldsymbol{\eta}_t; P_A\boldsymbol{X}_m)}{\mathrm{d}t} = \frac{1}{2}\mathrm{Tr}\,\mathrm{Cov}\big(\mathbb{E}[\mathbf{B}_A^{-1}\boldsymbol{v}_A \mid \boldsymbol{x}_A]\big).$$

Let $\tilde{\boldsymbol{v}}_A \coloneqq \mathbb{E}[\boldsymbol{v}_A|\boldsymbol{x}_A]$. Since $\mathbf{B}_A$ is constant and $P_A\boldsymbol{X}_m = \boldsymbol{x}_m^A$, the previous relation becomes

$$\begin{aligned}\lim_{\mathrm{d}t\to 0} \frac{I(\mathrm{d}\boldsymbol{x}_A; \boldsymbol{x}_m^A)}{\mathrm{d}t} &= \frac{1}{2}\mathrm{Tr}\,\mathrm{Cov}\big(\mathbf{B}_A^{-1}\tilde{\boldsymbol{v}}_A\big)\\ &= \frac{1}{4}\Big(\langle \tilde{\boldsymbol{v}}_A^{\intercal}\mathbf{D}_A^{-1}\tilde{\boldsymbol{v}}_A\rangle - \langle \tilde{\boldsymbol{v}}_A\rangle^{\intercal}\mathbf{D}_A^{-1}\langle \tilde{\boldsymbol{v}}_A\rangle\Big).\end{aligned}$$

Because $\langle \tilde{\boldsymbol{v}}_A\rangle = \langle \boldsymbol{v}_A\rangle$, adding the mean-flow term gives $\sigma_A^{\mathrm{self}} = \langle \tilde{\boldsymbol{v}}_A^{\intercal}\mathbf{D}_A^{-1}\tilde{\boldsymbol{v}}_A\rangle$.

For the interaction channel contribution, the chain rule gives

$$I(\mathrm{d}\boldsymbol{x}_A; \boldsymbol{x}_m^{-A}|\boldsymbol{x}_m) = I(\mathrm{d}\boldsymbol{x}_A; \boldsymbol{x}_m) - I(\mathrm{d}\boldsymbol{x}_A; \boldsymbol{x}_m^A),$$

where $\boldsymbol{x}_m^{-A} \coloneqq (I - P_A)\boldsymbol{X}_m$. Applying (S99) to the first term with $T_{\boldsymbol{\eta}} = P_A$, $T_{\boldsymbol{X}} = I$, and to the second with $T_{\boldsymbol{\eta}} = T_{\boldsymbol{X}} = P_A$, yields

$$\lim_{\mathrm{d}t\to 0} \frac{I(\mathrm{d}\boldsymbol{x}_A; \boldsymbol{x}_m^{-A}|\boldsymbol{x}_m^A)}{\mathrm{d}t} = \frac{1}{4}\mathrm{Tr}\left[\mathbf{D}_A^{-1}\left\{\mathrm{Cov}(\boldsymbol{v}_A) - \mathrm{Cov}(\widetilde{\boldsymbol{v}}_A)\right\}\right].$$

By the law of total covariance and $\widetilde{\boldsymbol{v}}_A = \mathbb{E}[\boldsymbol{v}_A|\boldsymbol{x}_A]$,

$$\mathrm{Cov}(\boldsymbol{v}_A) - \mathrm{Cov}(\widetilde{\boldsymbol{v}}_A) = \mathbb{E}[\mathrm{Cov}(\boldsymbol{v}_A|\boldsymbol{x}_A)].$$

Hence

$$\sigma_{A|-A}^{\mathrm{int}} = \langle (\boldsymbol{v}_A - \widetilde{\boldsymbol{v}}_A)^{\intercal}\mathbf{D}_A^{-1}(\boldsymbol{v}_A - \widetilde{\boldsymbol{v}}_A)\rangle.$$

Together with the expression for σ_A^{self} above, this gives Eq. (6b) of the main text.

III. NUMERICAL ILLUSTRATION OF THE IDENTITY

In this section, we numerically illustrate the multiplicative-noise identity (S90) derived in Sec. I using a simple one-dimensional diffusion on a ring. This example is analytically transparent while retaining genuinely state-dependent noise, allowing us to compare the small-dt mutual information slope directly with the theoretical prediction. It also provides a useful contrast between midpoint and endpoint conditioning. In Sec. I.E, we showed that under midpoint conditioning the $\sqrt{\mathrm{d}t}$-order non-Gaussian corrections cancel, so that the conditional law is captured by a Gaussian channel up to $O(\mathrm{d}t)$ corrections; no analogous simplification is expected for endpoint conditioning. We therefore focus on the midpoint-conditioned quantity $I(\boldsymbol{\eta}_t; \boldsymbol{X}_m)$, and report $I(\boldsymbol{\eta}_t; \boldsymbol{X}_t)$ only as a control illustrating why the exact identity is formulated with the midpoint rather than with an endpoint.

A. Model and observables

We consider a one-dimensional Itô diffusion on a ring $x \in [0, 2\pi)$,

$$\begin{aligned} \mathrm{d}X_t &= F(X_t)\,\mathrm{d}t + \sqrt{2D(X_t)}\,\mathrm{d}W_t \qquad (\mathrm{mod}\ 2\pi), \\ \text{with} \quad D(x) &= 2 + \sin x, \quad F(x) = D'(x) + f = \cos x + f. \end{aligned} \tag{S101}$$

This choice yields a smooth uniformly elliptic diffusion coefficient, $1 \le D(x) \le 3$, on a compact state space. It therefore lies within the regular setting considered in Sec. I, where compact configuration spaces such as motion on a ring are natural examples covered by the assumption A5-A. Also the model is analytically convenient because its steady state can be calculated exactly. Writing the stationary current in one dimension as

$$j_{ss}(x) = F(x)p_{ss}(x) - \partial_x[D(x)p_{ss}(x)],$$

one finds that the choice $F = D' + f$ gives a uniform stationary density

$$p_{ss} = \frac{1}{2\pi},$$

together with a constant current

$$j_{ss} = \frac{f}{2\pi}.$$

Accordingly, the current velocity is spatially uniform,

$$v_{ss}(x) = \frac{j_{ss}(x)}{p_{ss}(x)} = f.$$

Thus $f = 0$ corresponds to equilibrium, whereas $f \neq 0$ generates a nonequilibrium steady state with a nonzero circulating current on the ring.

In the multiplicative-noise setting, the natural observable is not the raw increment itself but the midpoint-whitened increment

$$\eta_t \coloneqq \frac{1}{\sqrt{\mathrm{d}t}} B(X_m)^{-1}(X_{t+\mathrm{d}t} - X_t), \qquad B(x) = \sqrt{2D(x)} = \sqrt{4 + 2\sin x},$$

introduced in Sec. I. We therefore take the midpoint-conditioned mutual information $I(\eta_t; X_m)$ as the primary quantity of interest. For comparison, we also compute $I(\eta_t; X_t)$, which serves as a control rather than as the quantity appearing in the exact identity. This comparison is conceptually useful because Sec. I.E shows that midpoint conditioning removes the $\sqrt{\mathrm{d}t}$-order non-Gaussian correction, whereas endpoint conditioning does not share this cancellation.

In the present model, although the current velocity itself is spatially constant, $v_{ss}(x) = f$, the effective normalized signal entering the midpoint-conditioned channel is

$$B(x)^{-1} v_{ss}(x) = \frac{f}{\sqrt{2D(x)}} = \frac{f}{\sqrt{4 + 2\sin x}},$$

which remains position dependent through the diffusion amplitude. The midpoint therefore remains informative about η_t whenever $f \neq 0$. Specializing the general multiplicative-noise identity (S90) to this steady-state model yields

$$\sigma = \left\langle \frac{f^2}{D(X)} \right\rangle, \quad 4 \lim_{\mathrm{d}t \to 0} \frac{I(\eta_t; X_m)}{\mathrm{d}t} = \sigma - \left\langle \frac{f}{\sqrt{D(X)}} \right\rangle^2$$

B. Simulation and mutual information estimation

To generate short-time trajectory pairs, we sampled the initial point X_t directly from the exact steady state of the ring model. Since $p_{ss}(x) = 1/(2\pi)$, this was done by drawing X_t uniformly on $[0, 2\pi)$, thereby avoiding any burn-in or equilibration procedure. For each time step $\mathrm{d}t$, we set $h = \mathrm{d}t/2$ and constructed the pair (X_m, η_t) through two successive half-steps. The midpoint variable was defined as $X_m = X_{t+h}$, reduced modulo 2π, while the total increment was taken from the corresponding unwrapped trajectory so that short-time displacement remained consistent with the local SDE dynamics.

Because the present example involves state-dependent diffusion, we did not use a plain Euler-Maruyama update. Instead, each half-step was propagated with the one-dimensional Milstein scheme [24, 44]. Writing $\sigma(x) = \sqrt{2D(x)}$, with $D(x) = 2 + \sin x$ and $F(x) = D'(x) + f$, the first half-step was evolved as

$$X_m^{un} = X_t + F(X_t)h + \sigma(X_t)\Delta W_1 + \frac{1}{2}D'(X_t)\big((\Delta W_1)^2 - h\big),$$

followed by $X_m = X_m^{un} \mod 2\pi$. The second half-step was then evolved from X_m^{un} using the same Milstein correction, but with coefficients evaluated at the midpoint:

$$X_{t+\mathrm{d}t}^{un} = X_m^{un} + F(X_m)h + \sigma(X_m)\Delta W_2 + \frac{1}{2}D'(X_m)\big((\Delta W_2)^2 - h\big).$$

Here ΔW_1 and ΔW_2 are independent Gaussian increments with variance h. The normalized midpoint-whitened increment was finally constructed as

$$\eta_t = \frac{X_{t+\mathrm{d}t}^{un} - X_t}{\sqrt{\mathrm{d}t}\, B(X_m)}, \qquad B(x) = \sqrt{2D(x)} = \sqrt{4 + 2\sin x}.$$

This choice is consistent with the diffusion-metric normalization introduced in Sec. I and, for multiplicative noise, retains the $O(\mathrm{d}t)$ corrections relevant to mutual information estimation.

We estimated the mutual information using a Kraskov-Stögbauer-Grassberger (KSG) k-nearest-neighbor (knn) estimator with $k = 10$. Since X_m is an angular variable on the ring, the midpoint coordinate was treated as periodic with period 2π, while η_t was treated as nonperiodic. Operationally, we used the Chebyshev norm and implemented periodicity only in the X_m coordinate through a periodic cKDTree representation. To reduce the small positive bias that is typical when the true mutual information is close to zero, we subtracted a permutation baseline,

$$\hat{I}_{\mathrm{deb}}(X;Y) = \hat{I}_{\mathrm{raw}}(X;Y) - \frac{1}{n_{\mathrm{perm}}}\sum_{l=1}^{n_{\mathrm{perm}}} \hat{I}_{\mathrm{raw}}(X;\pi_l(Y)),$$

with $n_{\mathrm{perm}} = 3$. We applied the same estimator both to the midpoint-conditioned quantity $I(\eta_t; X_m)$, which is the object entering the exact identity, and to the initial point-conditioned quantity $I(\eta_t; X_t)$, which we report only as a comparison quantity.

For each driving strength f and each value of $\mathrm{d}t$, we used $N = 10^6$ independently generated samples and repeated the calculation over five random seeds. The reported error bars correspond to the standard deviation across seeds. To quantify the small-$\mathrm{d}t$ slope, we considered two regression forms. First, we used an unconstrained linear fit,

$$\hat{I}(\mathrm{d}t) \approx m_{\mathrm{lin}}\,\mathrm{d}t + c,$$

mainly as a visual guide and to expose the approximately f-independent baseline offset associated with finite-sample and estimator fluctuations. Second, we used a constrained quadratic fit,

$$\hat{I}(\mathrm{d}t) \approx m_{\mathrm{quad}}\,\mathrm{d}t + b\,\mathrm{d}t^2,$$

which enforces $I(0) = 0$ and was used to estimate the leading coefficient in the $\mathrm{d}t \to 0$ limit. In the figures below, the linear fit is shown only for comparison, whereas the coefficient m_{quad} is taken as the numerical estimate of the asymptotic mutual information rate.

C. Numerical results

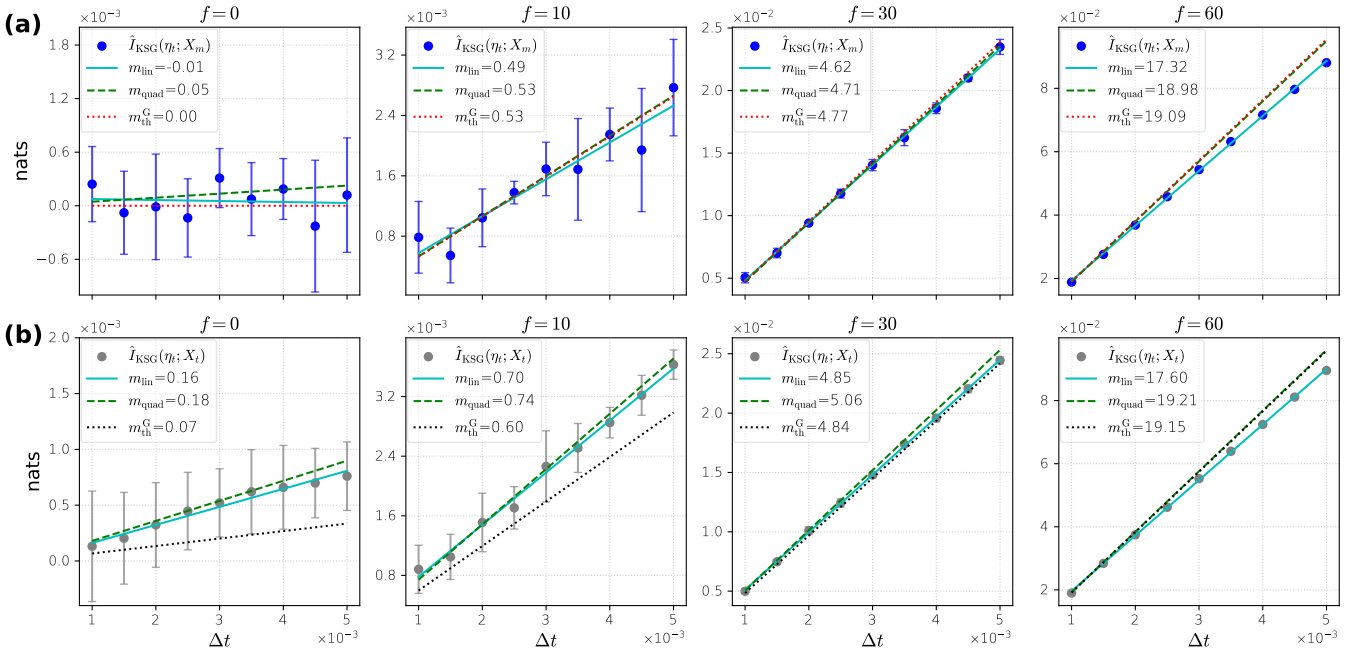

FIG. S1. Mutual information versus Δt for the multiplicative-noise model of Sec. III.A–B. **(a)** $I_{\mathrm{KSG}}(\eta_t; X_m)$ using the time midpoint X_m. **(b)** $I_{\mathrm{KSG}}(\eta_t; X_t)$ using the initial point X_t. Columns from left to right correspond to $f = 0, 10, 30, 60$. Error bars show the standard deviation over five independent runs. Cyan solid, green dashed, and red dotted lines denote the linear fit, the line from the first-order coefficient of the quadratic fit, and the theoretical slope from the Gaussian-channel approximation, respectively; the corresponding slope values are shown in each panel.

The numerical results are shown in Fig. S1. Three main features are worth emphasizing. First, midpoint and initial point conditioning lead to different Gaussian-channel predictions, and the numerical data clearly reflect this distinction. Most notably, at $f = 0$, where the ring diffusion is in equilibrium, the midpoint-conditioned theory predicts a vanishing slope and the measured $I_{\mathrm{KSG}}(\eta_t; X_m)$ is likewise consistent with zero within numerical uncertainty. By contrast, the initial point conditioned quantity I_{KSG} exhibits a positive slope already at $f = 0$, both in the Gaussian approximation and in the direct numerical estimates. Second, the initial point conditioned data show that the Gaussian approximation is not uniformly reliable, especially in the weak-drive regime. For $f = 0$ and $f = 10$, the measured small-dt slope deviates visibly from the Gaussian-channel prediction, whereas the midpoint conditioned results remain in close agreement with the theoretical value. This is consistent with the analysis of Sec. I.E: under midpoint conditioning, the effect of the spatial variation of $D(x)$ is canceled at the relevant order, while no analogous cancellation occurs for conditioning on the initial point. Third, in the larger-drive regime, where finite-dt effects become more pronounced, the coefficient m_{quad} obtained from the constrained fit $I(\mathrm{d}t) \approx m_{\mathrm{quad}}\mathrm{d}t + b\,\mathrm{d}t^2$ provides a more faithful estimate of the asymptotic slope than a simple linear fit, most clearly for the $f = 60$. This is consistent with the theoretical small-time form $I(\mathrm{d}t) = m\,\mathrm{d}t + o(\mathrm{d}t)$, for which the quadratic fit isolates the leading-order coefficient while partially absorbing subleading finite-time corrections.

IV. SIMULATION DETAILS FOR THE MAIN TEXT FIGS.2–3

A. Numerical procedure for Fig. 2

The caption of Fig. 2 lists the model equations and parameter values; here we record only the additional details needed for reproduction. Panels (a,b) were not generated from trajectory data, but from the closed-form steady state of linear additive-noise model

$$\dot{\boldsymbol{X}} = \mathbf{A}\boldsymbol{X} + \sqrt{2\mathbf{D}}\boldsymbol{\xi}, \quad \mathbf{A} = \begin{pmatrix} -k & s+a \\ s-a & -k \end{pmatrix}, \quad \mathbf{D} = \mathbf{I}. \tag{S102}$$

For each parameter set, the stationary covariance $\mathbf{C}$ was obtained from the Lyapunov equation

$$\mathbf{A}\mathbf{C} + \mathbf{C}\mathbf{A}^{\intercal} + 2\mathbf{D} = \mathbf{0}. \tag{S103}$$

The black arrows show the drift $\boldsymbol{F}(\boldsymbol{x}) = \mathbf{A}\boldsymbol{x}$, the yellow arrows show the local mean velocity

$$\boldsymbol{v}(\boldsymbol{x}) = \boldsymbol{F}(\boldsymbol{x}) - \mathbf{D}\nabla \ln p_{ss}(\boldsymbol{x}) = (\mathbf{A} + \mathbf{D}\mathbf{C}^{-1})\boldsymbol{x}, \tag{S104}$$

and the contours show the stationary Gaussian density $p_{ss}(\boldsymbol{x}) \propto \exp\left(-\frac{1}{2}\boldsymbol{x}^{\intercal}\mathbf{C}^{-1}\boldsymbol{x}\right)$. The crosses mark the midpoint locations used for the conditional displacement plots in panels (c,d).

For panels (c,d), conditional displacement clouds were generated from the same linear model (S102) using a two-half-step Euler-Maruyama update with $\Delta t = 10^{-3}$. In each case, 10^5 trajectories were initialized at the origin and propagated for 10^4 steps, and the first 10^3 steps were discarded as burn-in. The midpoint $\boldsymbol{X}_m$ was recorded after the first half-step, and the full displacement was defined by $\Delta\boldsymbol{X} = \boldsymbol{X}_{t+\Delta t} - \boldsymbol{X}_t$. Conditional samples were then selected by restricting $\boldsymbol{X}_m$ to two square windows centered at $(-0.6, 0.6)$ and $(0.6, -0.6)$, with half-width 0.02 in each coordinate. The panels show scatter plots of the resulting conditional displacements together with their one-dimensional marginals. Dashed crosshairs indicate the sample means, solid black lines indicate the origin, and the dashed circles indicate the 1.96σ scale of each conditional cloud.

For panels (e,f), the linear model (S102) and the nonlinear model

$$\begin{aligned}\dot{x} &= -x + y + \epsilon y^3 + \sqrt{2}\,\xi_x,\\ \dot{y} &= -y + \sqrt{2}\,\xi_y,\end{aligned}$$

with $\mathbf{D} = \mathrm{I}$, were used. The information rate was estimated from midpoint-conditioned displacements. In the additive-noise setting,

$$I(\Delta\boldsymbol{X}; \boldsymbol{X}_m) = \frac{\Delta t}{4}\mathrm{Tr}\big(\mathbf{D}^{-1}\mathrm{Cov}(\boldsymbol{v}(\boldsymbol{X}_m))\big) + o(\Delta t), \quad \boldsymbol{v}(\boldsymbol{x}_m) = \frac{\mathbb{E}[\Delta\boldsymbol{X}|\boldsymbol{X}_m = \boldsymbol{x}_m]}{\Delta t}.$$

For the main-text panels, we therefore used the regression-based estimator

$$\hat{I}_{\mathrm{MMSE}}(\Delta t) := \frac{\Delta t}{4}\mathrm{Tr}\left(\mathbf{D}^{-1}\hat{\mathrm{Cov}}(\hat{\boldsymbol{v}}(\boldsymbol{X}_m))\right), \tag{S105}$$

where $\hat{\boldsymbol{v}}(\boldsymbol{x}_m)$ was obtained by regressing $\Delta\boldsymbol{X}$ on $\boldsymbol{X}_m$. The subscript MMSE refers to the leading-order small-SNR Gaussian/I-MMSE expression underlying this estimator [27]. In practice, polynomial features of the midpoint coordinates were used together with sample splitting (cross-fitting) to reduce overfitting bias. For each parameter value, $\hat{I}_{\mathrm{MMSE}}(\Delta t)$ was evaluated over a range of Δt and fitted both by

$$\hat{I}(\Delta t) \approx m_{\mathrm{lin}}\Delta t + c \tag{S106}$$

and by the constrained quadratic form

$$\hat{I}(\Delta t) \approx m_{\mathrm{quad}}\Delta t + b\Delta t^2, \quad \hat{I}(0) = 0. \tag{S107}$$

The main text reports m_{quad} as the representative estimate of the $\Delta t \to 0$ slope.

As an independent cross-check, we also computed direct mutual information estimates with a debiased KSG k-nearest-neighbor estimator for $k = 10$, denoted by $\hat{I}_{\mathrm{KSG}}$. These KSG calculations were not used for the main-text panels, but only to verify that the regression-based estimator $\hat{I}_{\mathrm{MMSE}}$ gives consistent small-Δt slopes. At fixed sample size, $\hat{I}_{\mathrm{KSG}}$ exhibits substantially larger fluctuations than $\hat{I}_{\mathrm{MMSE}}$, especially in the nonlinear case, while remaining consistent with the same $\sigma/4$ trend. The corresponding summary for the linear model and comparison plots for the nonlinear model are shown in Fig. S2–S4.

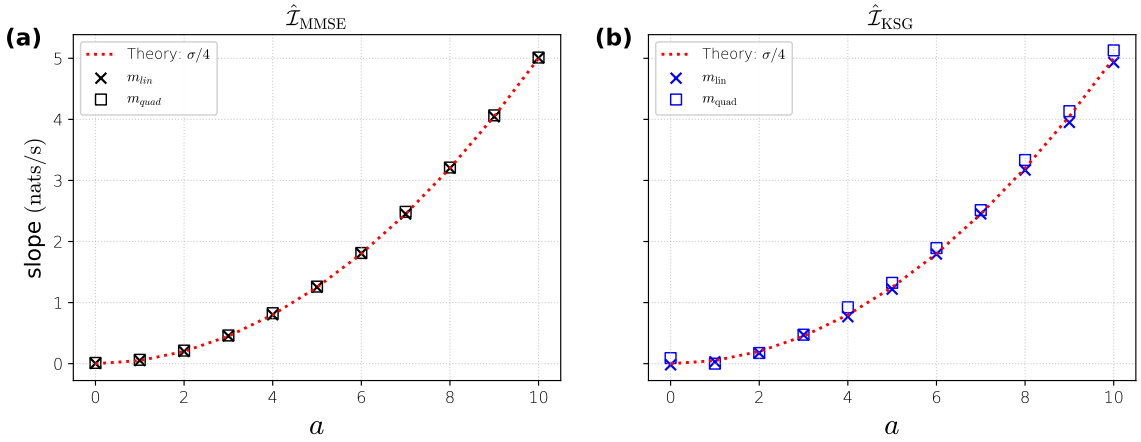

FIG. S2. Summary of slope estimates for the linear model used in Fig. 2e of the main text. Panel **(a)** shows $\hat{\mathcal{I}}_{\mathrm{MMSE}}$, and panel **(b)** shows $\hat{\mathcal{I}}_{\mathrm{KSG}}$, as a function of the nonequilibrium parameter a. Crosses denote the slope m_{lin} obtained from a linear fit in $\mathrm{d}t$, while squares denote the first-order coefficient m_{quad} extracted from a quadratic fit. The red dotted line indicates the theoretical prediction, $\sigma/4$, where σ is the EP rate.

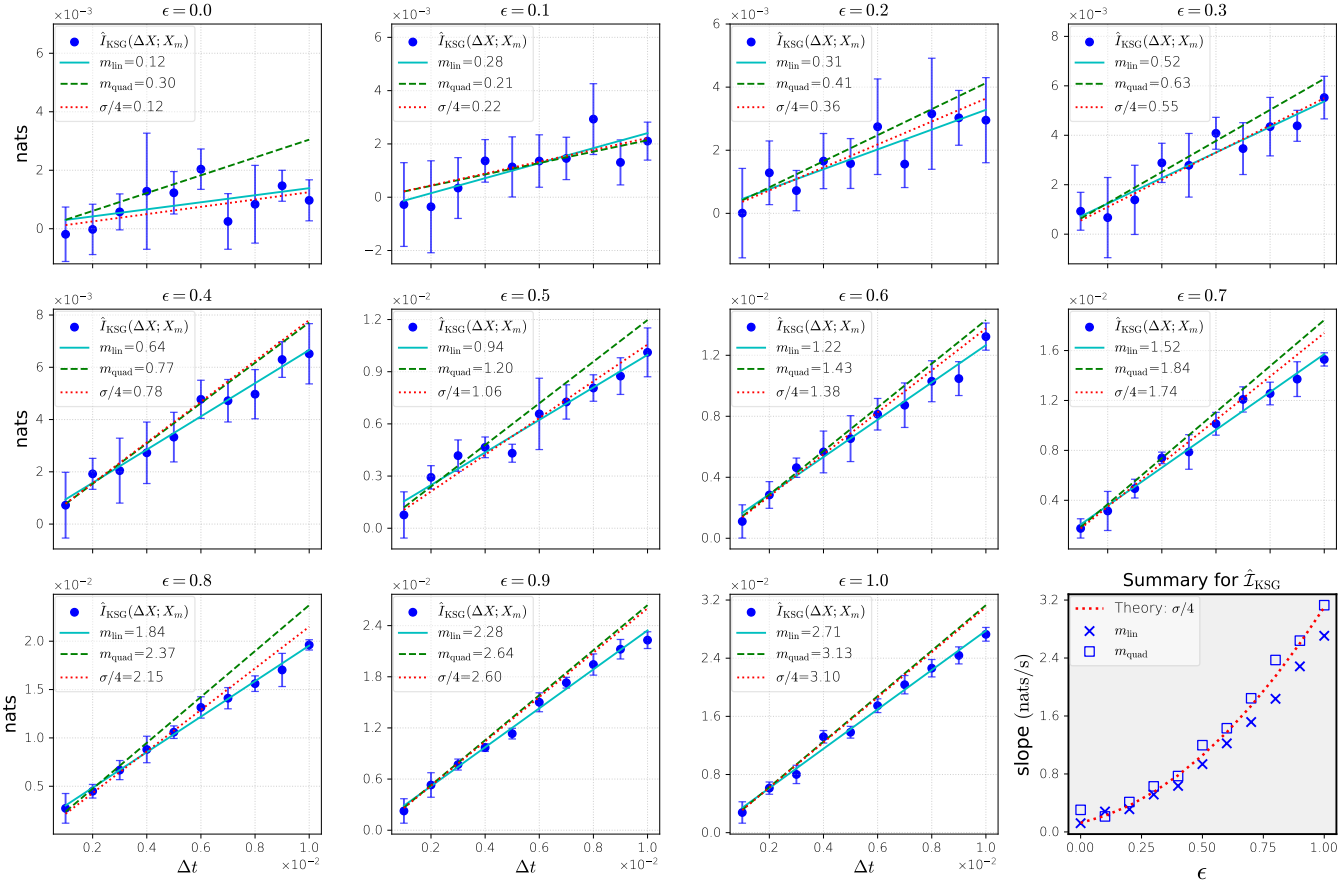

FIG. S3. Numerical results for the nonlinear model used in Fig. 2f of the main text, shown for $\hat{I}_{\mathrm{KSG}}(\Delta \boldsymbol{X}; \boldsymbol{X}_m)$. The first 11 panels display the mutual information as a function of $\mathrm{d}t$ for $\epsilon = 0.0, 0.1, \cdots, 1.0$ and the last panel summarizes the extracted slopes over the full range of ϵ. Blue circles show the mean $\hat{I}_{\mathrm{KSG}}$, and error bars denote the standard deviation over independent 5 runs. The cyan solid line is the linear fit, the green dashed line is the line determined by the first-order coefficient extracted from a quadratic fit in $\mathrm{d}t$, and the red dotted line indicates the simulated EP rate divided by four, $\sigma/4$. The corresponding slope values are listed in each panel.

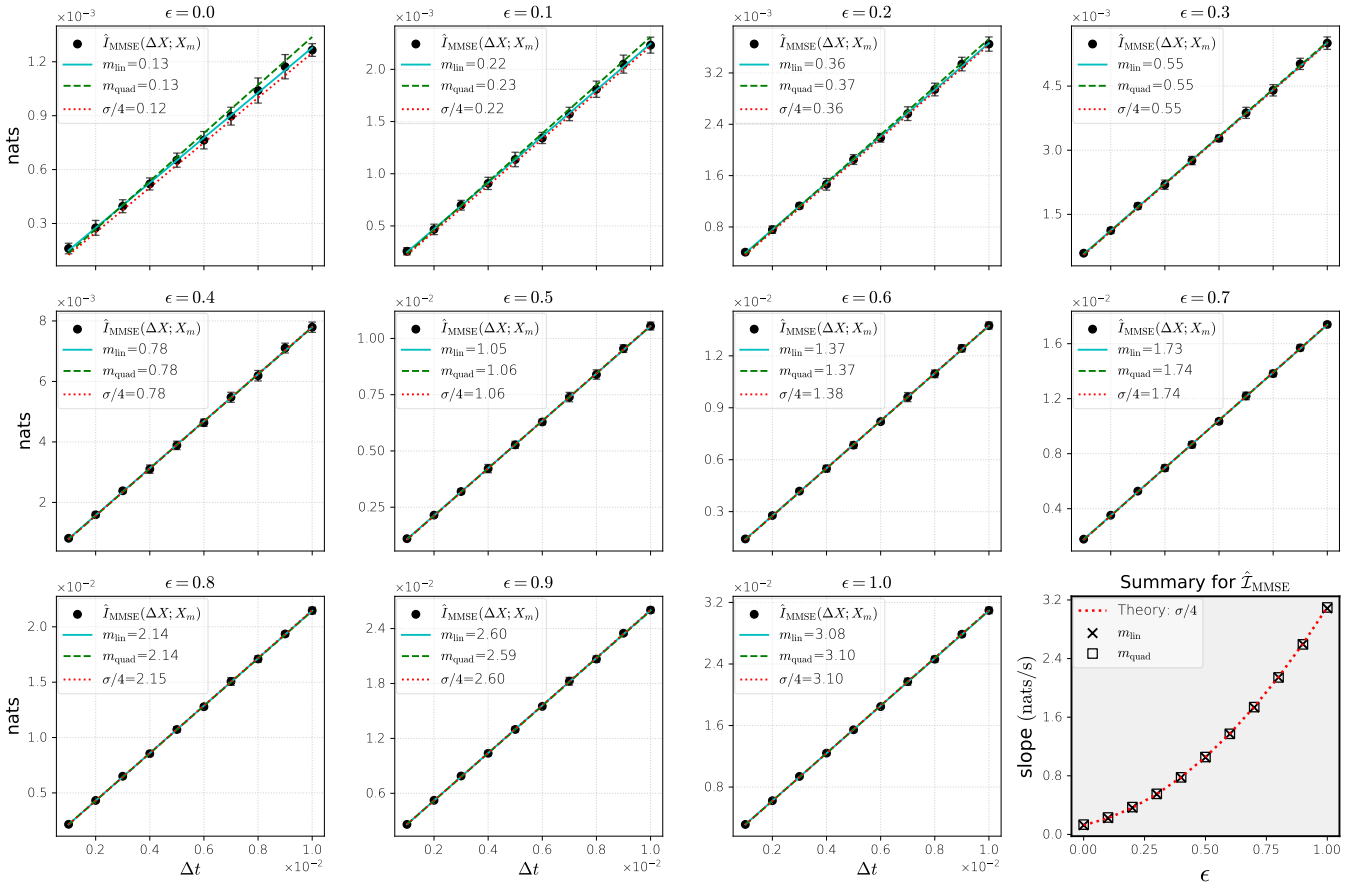

FIG. S4. Same as Fig. S3, but for $\hat{I}_{\mathrm{MMSE}}$. Black circles show the numerical estimates, the cyan solid line the linear fit, the green dashed line the first-order coefficient extracted from a quadratic fit in dt, and the red dotted line the simulated value $\sigma/4$.

For panel (e), the EP rate was evaluated analytically for the linear Ornstein-Uhlenbeck process. With stationary covariance $\mathbf{C}$ and local mean velocity $\boldsymbol{v}$ in (S103) and (S104), the total EP rate is given by

$$\sigma = \langle \boldsymbol{v}^{\intercal}\mathbf{D}^{-1}\boldsymbol{v}\rangle = \mathrm{Tr}\left[\mathbf{D}^{-1}(\mathbf{A}+\mathbf{D}\mathbf{C}^{-1})\mathbf{C}(\mathbf{A}+\mathbf{D}\mathbf{C}^{-1})^{\intercal}\right].$$

Using the Lyapunov equation (S103), this is equivalently written as

$$\sigma = \mathrm{Tr}(\mathbf{D}^{-1}\mathbf{A}\mathbf{C}\mathbf{A}^{\intercal}) + \mathrm{Tr}(\mathbf{A}). \tag{S108}$$

For panel(f), the EP rate was estimated independently from the midpoint discretization of the Stratonovich expression

$$\sigma = \left\langle \boldsymbol{F}(\boldsymbol{X}_t)^{\intercal}\mathbf{D}^{-1} \circ \dot{\boldsymbol{X}}_t \right\rangle,$$

namely

$$\hat{\sigma} = \frac{1}{\Delta t}\left\langle \boldsymbol{F}\left(\frac{\boldsymbol{X}_t + \boldsymbol{X}_{t+\Delta t}}{2}\right)^{\intercal}\mathbf{D}^{-1}\left(\boldsymbol{X}_{t+\Delta t} - \boldsymbol{X}_t\right)\right\rangle.$$

For each value of ϵ, 10^4 trajectories of length 10^4 steps were simulated with $\Delta t = 10^{-3}$, starting from the origin, and the initial 5×10^3 steps were discarded as burn-in. This estimator follows from the steady-state identity

$$\sigma = \int \frac{\boldsymbol{j}_{ss}^{\intercal}\mathbf{D}^{-1}\boldsymbol{j}_{ss}}{p_{ss}}\mathrm{d}\boldsymbol{x} = \int \boldsymbol{F}^{\intercal}\mathbf{D}^{-1}\boldsymbol{j}_{ss}\mathrm{d}\boldsymbol{x},$$

where the second equality uses $\boldsymbol{j}_{ss} = \boldsymbol{F}p_{ss} - \mathbf{D}\nabla p_{ss}$ together with stationarity, so that the term involving $\nabla \ln p_{ss}$ integrates to zero under the adopted boundary conditions.

B. Numerical procedure for Fig. 3

For Fig. 3, we first collect the general steady-state expressions for the self and interaction EP rates in linear Langevin systems, together with the quantities entering the learning-rate bound, and then summarize the panel-specific calculations and results. Panel (a) is purely schematic and is therefore omitted.

- **B-1. Steady-state formulas for $\sigma_A^{\mathrm{self}}, \sigma_{A|B}^{\mathrm{int}}, \dot{I}_A$ and $\mathcal{R}_A$**

We consider the linear Langevin dynamics in steady-state,

$$\dot{\boldsymbol{x}}(t) = \mathbf{A}\boldsymbol{x}(t) + \mathbf{B}\,\boldsymbol{\xi}(t).$$

Let $\mathbf{C}$ denote the steady-state covariance matrix of the system, which satisfies the Lyapunov equation (S103). To define the time midpoint variable $\boldsymbol{x}_m \coloneqq \boldsymbol{x}(t+\mathrm{d}t/2)$, let $\boldsymbol{y}_1$ denote the Brownian increment $\mathbf{B}\,\mathrm{d}\boldsymbol{W}_1 \sim \mathcal{N}(0, \mathbf{B}\mathbf{B}^{\intercal}\mathrm{d}t/2) = \mathcal{N}(0, \mathbf{D}\mathrm{d}t)$ over the first half-step $[t, t+\mathrm{d}t/2]$, and let $\boldsymbol{y}_2$ denote the increment over the second half-step $[t+\mathrm{d}t/2, t+\mathrm{d}t]$. Then, the relevant variables are given by

$$\mathrm{d}\boldsymbol{x} = \mathbf{A}\boldsymbol{x}\,\mathrm{d}t + \boldsymbol{y}_1 + \boldsymbol{y}_2, \quad \boldsymbol{x}_m = \boldsymbol{x} + \mathbf{A}\boldsymbol{x}\frac{\mathrm{d}t}{2} + \boldsymbol{y}_1.$$

Under the partition $\{A, B\}$, we write

$$\boldsymbol{x} = \begin{pmatrix} \boldsymbol{x}_A \\ \boldsymbol{x}_B \end{pmatrix}, \quad \mathrm{d}\boldsymbol{x} = \begin{pmatrix} \mathrm{d}\boldsymbol{x}_A \\ \mathrm{d}\boldsymbol{x}_B \end{pmatrix}, \quad \boldsymbol{x}_m = \begin{pmatrix} \boldsymbol{x}_m^A \\ \boldsymbol{x}_m^B \end{pmatrix},$$
$$\mathbf{A} = \begin{pmatrix} \mathbf{A}_{AA} & \mathbf{A}_{AB} \\ \mathbf{A}_{BA} & \mathbf{A}_{BB} \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} \mathbf{C}_{AA} & \mathbf{C}_{AB} \\ \mathbf{C}_{BA} & \mathbf{C}_{BB} \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} \mathbf{D}_{AA} & \mathbf{D}_{AB} \\ \mathbf{D}_{BA} & \mathbf{D}_{BB} \end{pmatrix}.$$

For Gaussian random variables $\boldsymbol{X}$ and $\boldsymbol{Y}$, the mutual information $I(\boldsymbol{X}; \boldsymbol{Y})$ is given by

$$I(\boldsymbol{X}; \boldsymbol{Y}) = -\frac{1}{2}\log\det\left(I - \mathrm{Var}(\boldsymbol{X})^{-1}\mathrm{Cov}(\boldsymbol{X}, \boldsymbol{Y})\mathrm{Var}(\boldsymbol{Y})^{-1}\mathrm{Cov}(\boldsymbol{Y}, \boldsymbol{X})\right).$$

Hence $I(\mathrm{d}\boldsymbol{x}_A; \boldsymbol{x}_m^A)$ can be evaluated by taking $\boldsymbol{X} = \mathrm{d}\boldsymbol{x}_A$ and $\boldsymbol{Y} = \boldsymbol{x}_m^A$,

$$\begin{aligned} I(\mathrm{d}\boldsymbol{x}_A; \boldsymbol{x}_m^A) &= -\frac{1}{2}\log\det\left(I - \mathrm{Var}(\mathrm{d}\boldsymbol{x}_A)^{-1}\mathrm{Cov}(\mathrm{d}\boldsymbol{x}_A, \boldsymbol{x}_m^A)\mathrm{Var}(\boldsymbol{x}_m^A)^{-1}\mathrm{Cov}(\boldsymbol{x}_m^A, \mathrm{d}\boldsymbol{x}_A)\right) \\ &= -\frac{1}{2}\log\det\left(I - (2\mathbf{D}_{AA}\mathrm{d}t + O(\mathrm{d}t^2))^{-1}(\mathbf{K}_{AA}\mathrm{d}t + O(\mathrm{d}t^2))(\mathbf{C}_{AA} + O(\mathrm{d}t^2))^{-1}(\mathbf{K}_{AA}\mathrm{d}t + O(\mathrm{d}t^2))^{\intercal}\right) \\ &= -\frac{1}{2}\log\det\left(I - ((2\mathbf{D}_{AA})^{-1}\mathbf{K}_{AA}\mathbf{C}_{AA}^{-1}\mathbf{K}_{AA}^{\intercal})\mathrm{d}t + O(\mathrm{d}t^2)\right) \\ &= \frac{1}{4}\mathrm{Tr}\left[\mathbf{D}_{AA}^{-1}\mathbf{K}_{AA}\mathbf{C}_{AA}^{-1}\mathbf{K}_{AA}^{\intercal}\right]\mathrm{d}t + O(\mathrm{d}t^2), \end{aligned}$$

where $\mathbf{K} \coloneqq \mathbf{A}\mathbf{C} + \mathbf{D}$. The last equality follows from $\log\det(I - \epsilon\mathbf{M}) = -\epsilon\mathrm{Tr}[\mathbf{M}] + O(\epsilon^2)$. Note that we used the Lyapunov equation (S103), which cancels the $O(\mathrm{d}t)$ term in $\mathrm{Var}(\boldsymbol{x}_m^A)$.

Likewise, the conditional mutual information $I(\boldsymbol{X}; \boldsymbol{Y} \mid \boldsymbol{Z})$ is given by

$$I(\boldsymbol{X}; \boldsymbol{Y} \mid \boldsymbol{Z}) = -\frac{1}{2}\log\det\left(I - \mathrm{Var}(\boldsymbol{X}|\boldsymbol{Z})^{-1}\mathrm{Cov}(\boldsymbol{X}, \boldsymbol{Y}|\boldsymbol{Z})\mathrm{Var}(\boldsymbol{Y}|\boldsymbol{Z})^{-1}\mathrm{Cov}(\boldsymbol{Y}, \boldsymbol{X}|\boldsymbol{Z})\right).$$

Here, for jointly Gaussian variables $\boldsymbol{X}, \boldsymbol{Y}$ and $\boldsymbol{Z}$,

$$\mathrm{Cov}(\boldsymbol{X}, \boldsymbol{Y} \mid \boldsymbol{Z}) = \mathrm{Cov}(\boldsymbol{X}, \boldsymbol{Y}) - \mathrm{Cov}(\boldsymbol{X}, \boldsymbol{Z})\,\mathrm{Var}(\boldsymbol{Z})^{-1}\,\mathrm{Cov}(\boldsymbol{Z}, \boldsymbol{Y}),$$

and

$$\mathrm{Var}(\boldsymbol{X} \mid \boldsymbol{Z}) = \mathrm{Var}(\boldsymbol{X}) - \mathrm{Cov}(\boldsymbol{X}, \boldsymbol{Z})\mathrm{Var}(\boldsymbol{Z})^{-1}\mathrm{Cov}(\boldsymbol{Z}, \boldsymbol{X}).$$

Using these results, we obtain

$$\begin{aligned} I(\mathrm{d}\boldsymbol{x}_A; \boldsymbol{x}_m^B \,|\, \boldsymbol{x}_m^A) &= -\frac{1}{2}\log\det\left(I - \mathrm{Var}(\mathrm{d}\boldsymbol{x}_A|\boldsymbol{x}_m^A)^{-1}\mathrm{Cov}(\mathrm{d}\boldsymbol{x}_A, \boldsymbol{x}_m^B|\boldsymbol{x}_m^A)\mathrm{Var}(\boldsymbol{x}_m^B|\boldsymbol{x}_m^A)^{-1}\mathrm{Cov}(\boldsymbol{x}_m^B, \mathrm{d}\boldsymbol{x}_A|\boldsymbol{x}_m^A)\right) \\ &= -\frac{1}{2}\log\det\left(I - (2\mathbf{D}_{AA}\mathrm{d}t + O(\mathrm{d}t^2))^{-1}(\mathbf{L}_{AB|A}\mathrm{d}t + O(\mathrm{d}t^2))(\mathbf{C}_{B|A} + O(\mathrm{d}t^2))^{-1}(\mathbf{L}_{AB|A}\mathrm{d}t + O(\mathrm{d}t^2))^{\intercal}\right) \\ &= -\frac{1}{2}\log\det\left(I - ((2\mathbf{D}_{AA})^{-1}\mathbf{L}_{AB|A}\mathbf{C}_{B|A}^{-1}\mathbf{L}_{AB|A}^{\intercal})\mathrm{d}t + O(\mathrm{d}t^2)\right) \\ &= \frac{1}{4}\mathrm{Tr}\left[\mathbf{D}_{AA}^{-1}\mathbf{L}_{AB|A}\mathbf{C}_{B|A}^{-1}\mathbf{L}_{AB|A}^{\intercal}\right]\mathrm{d}t + O(\mathrm{d}t^2), \end{aligned}$$

where $\mathbf{C}_{B|A} \coloneqq \mathbf{C}_{BB} - \mathbf{C}_{BA}\mathbf{C}_{AA}^{-1}\mathbf{C}_{AB}$ and $\mathbf{L}_{AB|A} \coloneqq \mathbf{K}_{AB} - \mathbf{K}_{AA}\mathbf{C}_{AA}^{-1}\mathbf{C}_{AB}$.

Therefore, the analytic forms of the self and interaction EP rates are given by,

$$\sigma_A^{\mathrm{self}} \coloneqq \lim_{\mathrm{d}t\to 0} \frac{4I(\mathrm{d}\boldsymbol{x}_A; \boldsymbol{x}_m^A)}{\mathrm{d}t} = \mathrm{Tr}\left[\mathbf{D}_{AA}^{-1}\mathbf{K}_{AA}\mathbf{C}_{AA}^{-1}\mathbf{K}_{AA}^{\intercal}\right], \tag{S109}$$

$$\sigma_{A|B}^{\mathrm{int}} \coloneqq \lim_{\mathrm{d}t\to 0} \frac{4I(\mathrm{d}\boldsymbol{x}_A; \boldsymbol{x}_m^B \mid \boldsymbol{x}_m^A)}{\mathrm{d}t} = \mathrm{Tr}\left[\mathbf{D}_{AA}^{-1}\mathbf{L}_{AB|A}\mathbf{C}_{B|A}^{-1}\mathbf{L}_{AB|A}^{\intercal}\right]. \tag{S110}$$

The learning-rate in Matsumoto et al. (2025) [20], or equivalently, the information flow in Horowitz (2015) [45] is defined as

$$\dot{I}_A(\boldsymbol{x}_A; \boldsymbol{x}_B) = \int \boldsymbol{J}_A(\boldsymbol{x}) \cdot \nabla_{\boldsymbol{x}_A} \ln p(\boldsymbol{x}_B|\boldsymbol{x}_A)\, \mathrm{d}\boldsymbol{x}.$$

Since $\boldsymbol{x}_B|\boldsymbol{x}_A \sim \mathcal{N}(\mathbf{C}_{BA}\mathbf{C}_{AA}^{-1}\boldsymbol{x}_A, \mathbf{C}_{B|A})$, we obtain

$$\nabla_{\boldsymbol{x}_A} \ln p(\boldsymbol{x}_B|\boldsymbol{x}_A) = \mathbf{C}_{AA}^{-1}\mathbf{C}_{AB}\mathbf{C}_{B|A}^{-1} \underbrace{\left(\boldsymbol{x}_B - \mathbf{C}_{BA}\mathbf{C}_{AA}^{-1}\boldsymbol{x}_A\right)}_{=:\boldsymbol{r}_B}$$

while $\boldsymbol{J}_A = p(\boldsymbol{x})\boldsymbol{v}_A$ where

$$\begin{aligned}\boldsymbol{v}_A &= \left[(\mathbf{A} + \mathbf{D}\mathbf{C}^{-1})\boldsymbol{x}\right]_A = (\mathbf{K}\mathbf{C}^{-1}\boldsymbol{x})_A = (\mathbf{K}\mathbf{C}^{-1})_{AA}\boldsymbol{x}_A + (\mathbf{K}\mathbf{C}^{-1})_{AB}\boldsymbol{x}_B \\ &= \left[(\mathbf{K}\mathbf{C}^{-1})_{AA} + (\mathbf{K}\mathbf{C}^{-1})_{AB}\mathbf{C}_{BA}\mathbf{C}_{AA}^{-1}\right]\boldsymbol{x}_A + (\mathbf{K}\mathbf{C}^{-1})_{AB}\boldsymbol{r}_B.\end{aligned}$$

Note that $\mathrm{Cov}(\boldsymbol{x}_A, \boldsymbol{r}_B) = 0$ and $\mathrm{Var}(\boldsymbol{r}_B) = \mathbf{C}_{B|A}$. Combining these results yields

$$\dot{I}_A = \mathrm{Tr}\left[\mathbf{C}_{AA}^{-1}\mathbf{C}_{AB}\left((\mathbf{K}\mathbf{C}^{-1})_{AB}\right)^{\intercal}\right] = \mathrm{Tr}\left[\mathbf{C}_{AA}^{-1}\mathbf{C}_{AB}\mathbf{C}_{B|A}^{-1}\mathbf{L}_{AB|A}^{\intercal}\right]. \tag{S111}$$

We emphasize that although (S111) is formally well-defined for any Gaussian diffusion, its interpretation as the information-flow term entering a subsystem-resolved second-law balance in the sense of Horowitz (2015) requires the partition A, B to be bipartite, i.e. $\mathbf{D}_{AB} = \mathbf{D}_{BA} = \mathbf{0}$ [45].

Lastly, to write $\mathcal{R}_A$ explicitly in the steady-state linear Gaussian case, we first calculate the conditional Fisher information matrix

$$\mathbf{F}_A^{|A} \coloneqq \left\langle \left(\nabla_{\boldsymbol{x}_A} \ln p(\boldsymbol{x}_B|\boldsymbol{x}_A)\right)\left(\nabla_{\boldsymbol{x}_A} \ln p(\boldsymbol{x}_B|\boldsymbol{x}_A)\right)^{\intercal} \right\rangle = \mathbf{C}_{AA}^{-1}\mathbf{C}_{AB}\mathbf{C}_{B|A}^{-1}\mathbf{C}_{BA}\mathbf{C}_{AA}^{-1},$$

and hence

$$\mathcal{R}_A \coloneqq \mathrm{Tr}\left[\mathbf{D}_{AA}\mathbf{F}_A^{|A}\right]^{-1} = \mathrm{Tr}\left[\mathbf{D}_{AA}\mathbf{C}_{AA}^{-1}\mathbf{C}_{AB}\mathbf{C}_{B|A}^{-1}\mathbf{C}_{BA}\mathbf{C}_{AA}^{-1}\right]^{-1}. \tag{S112}$$

• B-2. Details for models and calculations of Fig. 3

For panels (b,c), the curves were not generated from trajectory data, but were evaluated analytically from the steady-state Gaussian statistics of three-dimensional linear cascade model shown in Fig. 3(a), with partition $A = \{x, y\}$ and $B = \{z\}$. For each parameter value, the total EP rate σ_{AB}, self/interaction EP rate $\sigma_A^{\mathrm{self}}, \sigma_{A|B}^{\mathrm{int}}$, learning rate $\dot{I}_A$, and information resistance $\mathcal{R}_A$ were computed from (S108)–(S112), while the local EP rate σ_A were obtained as $\sigma_A = \sigma_A^{\mathrm{self}} + \sigma_{A|B}^{\mathrm{int}}$.

For panel (b), we fixed $\beta = 1$ and varied the internal coupling α in the drift matrix

$$\mathbf{A} = \begin{pmatrix} -2 & \alpha & 0 \\ 0 & -2 & 1 \\ 0 & 0 & -2 \end{pmatrix}, \quad \mathbf{D} = \mathrm{I}.$$

The analytic form of σ_A^{self} and $\sigma_{A|B}^{\mathrm{int}}$ is given by

$$\begin{aligned}\sigma_A^{\mathrm{self}}(\alpha) &= \frac{289\alpha^2(19\alpha^2 + 272)}{64(323\alpha^2 + 4608)} = \frac{4913}{18432}\alpha^2 + O(\alpha^4) \approx 0.267\alpha^2, \\ \sigma_{A|B}^{\mathrm{int}}(\alpha) &= \frac{23409\alpha^4 + 706048\alpha^2 + 5308416}{2(305\alpha^2 + 4352)(323\alpha^2 + 4608)} = \frac{9}{68} - \frac{5057}{5326848}\alpha^2 + O(\alpha^4) \approx 0.132 - 0.001\alpha^2.\end{aligned}$$

Therefore $\sigma_A^{\rm self}$ increases markedly with α, whereas $\sigma_{A|B}^{\rm int}$ remains nearly unchanged.

For panel (c), we fixed $\alpha = 1$ and varied the upstream coupling β in

$$\mathbf{A} = \begin{pmatrix} -2 & 1 & 0 \\ 0 & -2 & \beta \\ 0 & 0 & -2 \end{pmatrix}, \quad \mathbf{D} = \mathrm{I}.$$

In this case,

$$\sigma_A^{\rm self}(\beta) = \frac{(\beta^2+16)^2(19\beta^2+272)}{64(3\beta^4+576\beta^2+4352)} = \frac{1}{4} + \frac{\beta^2}{64} + O(\beta^4) \approx 0.25 + 0.0156\beta^2$$
$$\sigma_{A|B}^{\rm int}(\beta) = \frac{\beta^2(\beta^8+560\beta^6+83776\beta^4+1218560\beta^2+4734976)}{2(\beta^4+304\beta^2+4352)(3\beta^4+576\beta^2+4352)} = \frac{\beta^2}{8} + O(\beta^4) \approx 0.125\beta^2.$$

This shows that $\sigma_{A|B}^{\rm int}$ is nearly an order of magnitude more sensitive to β than $\sigma_A^{\rm self}$. Because $\alpha = 1$, changes in β also alter the steady-state distribution of partition $A = \{x, y\}$, and hence weakly affect $\sigma_A^{\rm self}$ as well. However, this effect is much smaller.

For panel (d), we used the same α sweep as in panel (b), with $\beta = 1$. At $\alpha = 1$, we obtain

$$\sigma_A^{\rm self} = 0.26649, \quad \sigma_{A|B}^{\rm int} = 0.13147, \quad \sigma_A = 0.39795,$$

and

$$\dot{I}_A = 0.11767, \quad \mathcal{R}_A = 9.4871, \quad |\dot{I}_A|^2\mathcal{R}_A = 0.13137.$$

Thus, $\sigma_{A|B}^{\rm int}/\sigma_A = 0.33036$, while $|\dot{I}_A|^2\mathcal{R}_A/\sigma_{A|B}^{\rm int} = 0.99924$. As α increase, the growing self channel contribution therefore makes the previous bound looser, whereas the sharpened interaction channel bound remains nearly saturated in this model.

V. SHARPENED LEARNING-RATE BOUND

Beyond the single toy model considered in the main text, we examine the sharpened learning-rate bound for a broader set of randomly generated linear Ornstein-Uhlenbeck (OU) systems and for linear OU models fitted to red-blood-cell (RBC) data.

A. Randomly generated Linear OU systems

For each dimension $d = 2, ..., 7$ we retained 10^4 accepted samples from a proposal ensemble with full drift matrix $\mathbf{A}$ and diagonal diffusion matrix $\mathbf{D}$. Before rejection, the magnitudes of the entries of $\mathbf{A}$ and the diagonal entries of $\mathbf{D}$ were drawn independently from log-uniform distributions on $[1, 10^3]$. Off-diagonal signs of $\mathbf{A}$ were chosen at random, while each diagonal entry was taken negative with probability 0.9. We then retained only physical samples for which $\mathbf{A}$ was Hurwitz, the stationary covariance $\mathbf{C}$ solving the Lyapunov equation (S103) was positive definite, and the total steady-state EP was positive. For each accepted sample, we randomly partitioned the d variables into two nonempty subsets A and B, with $A \cup B = \{1, \cdots, d\}$. The values are evaluated the steady-state Gaussian quantities $\sigma_{A|B}^{\rm int}, \mathcal{R}_A$, and $|\dot{I}_A|$ using the analytic formulas derived above. We then plotted the dimensionless ratio

$$\rho_{\rm int} := \frac{\left|\dot{I}_A\right|^2 \mathcal{R}_A}{\sigma_{A|B}^{\rm int}}$$

against $\sigma_{A|B}^{\rm int}$ on a logarithmic horizontal axis, with colors indicating the dimension d in the left panel of Fig. S5. The dashed horizontal line at unity marks saturation of Eq. (8). As expected, the sampled points remain below this line over a broad range of interaction EP rate values. The right panel additionally shows $\sigma_{A|B}^{\rm int}/\sigma_A$, the fraction of the previous local EP upper bound retained by the sharpened bound; smaller values indicate greater tightening.

It is worth noting that all $d = 2$ samples lie exactly on the saturation line. This is not a numerical artifact. In a two-dimensional bipartite system each subsystem is one-dimensional, and at steady-state the self part vanishes identically. Indeed, with $\mathbf{K} = \mathbf{AC} + \mathbf{D}$, the Lyapunov equation (S103) implies $\mathbf{K} + \mathbf{K}^{\intercal} = \mathbf{0}$, so any 1×1 diagonal block must satisfy $\mathbf{K}_{AA} = 0$; hence $\sigma_A^{\rm self} = 0$, and similarly $\sigma_B^{\rm self} = 0$. More generally, any one-dimensional subsystem has zero self EP rate at steady-state. In the scalar-scalar case one can find $|\dot{I}_A|^2\mathcal{R}_A = \sigma_{A|B}^{\rm int}$.

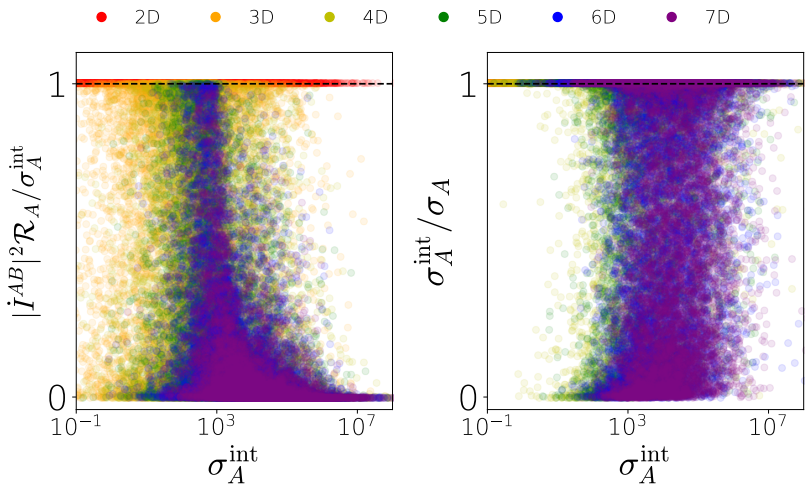

FIG. S5. Numerical assessment of the sharpened learning-rate bound in randomly generated stationary linear OU systems. For each dimension $d = 2, \cdots, 7$, 10^4 accepted samples are shown, with colors indicating d. (Left) The saturation ratio $|\dot{I}_A|^2\mathcal{R}_A/\sigma^{\rm int}_{A|B}$. (Right) The ratio $\sigma^{\rm int}_{A|B}/\sigma_A$ of the sharpened to the previous upper bound. Both are plotted against the interaction EP rate $\sigma^{\rm int}_{A|B}$. The dashed lines mark unity. In the scalar-scalar case ($d = 2$), the sharpened bound is saturated and the self channel vanishes, so both ratios equal unity.

B. Fitted RBC models

We use the fitted RBC parameter sets reported in Terlizzi et al. (2024) [33]. The parameter values for the OT-sensing and OM datasets are listed in Tables I and II, respectively. These correspond to distinct measurement modalities: OT-sensing tracks probe-bead fluctuations of the membrane, whereas OM analyzes free-standing contour fluctuations. The OT-sensing table contains two passivated cells and five active cells, whereas the OM table contains one passivated cell and six active cells.

For each parameter set we construct the three-dimensional linear Langevin model,

$$\dot{\boldsymbol{X}} = \mathbf{A}\boldsymbol{X} + \sqrt{2\mathbf{D}}\,\boldsymbol{\xi}, \qquad \boldsymbol{X} = (x, y, \eta)^{\intercal},$$

with

$$\mathbf{A} = \begin{pmatrix} -\mu_x k_x & \mu_x k_{\rm int} & 0 \\ \mu_y k_{\rm int} & -\mu_y k_y & \mu_y \\ 0 & 0 & -1/\tau \end{pmatrix}, \qquad \mathbf{D} = \begin{pmatrix} \mu_x k_B T & 0 & 0 \\ 0 & \mu_y k_B T & 0 \\ 0 & 0 & \epsilon^2/\tau \end{pmatrix}.$$

Here x is the measured outer-membrane coordinate, y the hidden membrane-cortex attachment, and η the active force. The steady-state covariance matrix $\mathbf{C}$ is obtained from the Lyapunov equation (S103). Throughout this section, we use the tabulated fitted parameter sets directly and without additional fitting or trajectory simulation. Entropies are measured in units of k_B, so the learning-rates and EP rates have units of s^{-1}.

Parameters	**P1**	**P2**	**A1**	**A2**	**A3**	**A4**	**A5**
k_x [10^{-3} pN/nm]	6.0	8.1	6.5	15.1	3.7	7.17	9.5
μ_x [10^4 nm/(pN s)]	2.6	2.1	2.8	3.6	2.6	1.6	1.7
k_y [10^{-2} pN/nm]	1.7	29.0	1.6	1.6	2.9	1.3	1.78
μ_y [10^4 nm/(pN s)]	140	31.0	2.9	2.3	0.57	1.96	1.94
$k_{\rm int}$ [10^{-3} pN/nm]	1.8	6.0	4.5	4.2	2.3	2.88	2.84
ϵ [pN]	0.7	2.1	2.6	2.4	4.7	3.86	4.41
τ [10^{-2} s]	200	400	8.0	1.1	2.5	17.0	7.7

TABLE I. Parameters for OT-sensing experiments.

Parameters	**P1**	**A1**	**A2**	**A3**	**A4**	**A5**	**A6**
k_x [10^{-2} pN/nm]	59.0	5.37	5.07	5.23	4.61	5.39	8.3
μ_x [10^4 nm/(pN s)]	0.8	1.2	1.3	1.2	1.4	1.2	0.9
k_y [10^{-2} pN/nm]	5.6	1.93	1.89	1.92	1.95	2.10	1.35
μ_y [10^4 nm/(pN s)]	0.082	4.85	4.89	4.79	4.84	4.95	4.65
$k_{\rm int}$ [10^{-3} pN/nm]	91.0	3.8	3.57	3.8	3.50	3.94	4.16
ϵ [pN]	0.2	4.42	4.37	4.43	4.40	4.45	4.52
τ [10^{-2} s]	750	6.0	4.1	6.0	3.9	3.3	7.0

TABLE II. Parameters for OM experiments.

Common note. Parameter sets are taken from Ref. [33]. We use $k_B T = 4.142\,\mathrm{pN}\cdot\mathrm{nm}$.

C. Observable-hidden partition and EP decomposition of the RBC

For the RBC model, we partition the variables into the observable coordinate $\{x\}$ and the hidden block $\{y, \eta\}$. The total steady-state EP rate is evaluated from Eq. (S108), while the self and interaction channels of the hidden block are evaluated from Eqs. (S109) and (S110), respectively. This gives

$$\sigma_{\rm tot} = \sigma_x + \sigma_{y\eta}, \qquad \sigma_{y\eta} = \sigma^{\rm self}_{y\eta} + \sigma^{\rm int}_{y\eta|x}.$$

In this setting, because the observable x sector is one-dimensional, its steady-state self channel contribution vanishes identically, and $\sigma_x = \sigma^{\rm int}_{x|y\eta}$.

For each fitted cell, we therefore compute the three normalized quantities

$$\frac{\sigma_x}{\sigma_{\rm tot}}, \qquad \frac{\sigma^{\rm int}_{y\eta|x}}{\sigma_{\rm tot}}, \qquad \frac{\sigma^{\rm self}_{y\eta}}{\sigma_{\rm tot}},$$

which sum to unity up to numerical precision. The bars in Fig. S6 display these fractions for each active fitted cell separately.

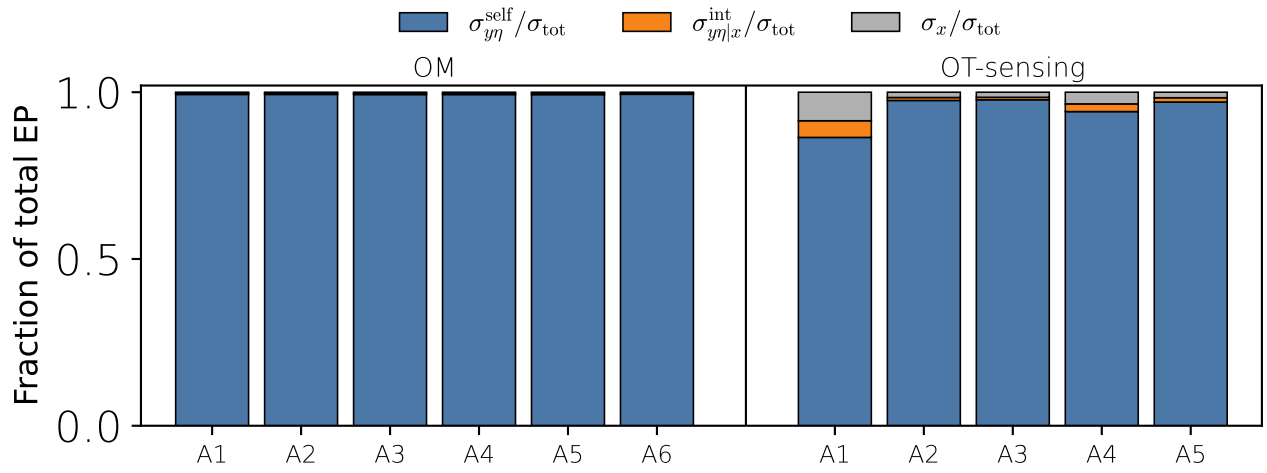

FIG. S6. Cell-resolved EP decomposition for the observable-hidden partition $\{x\}|\{y, \eta\}$ in the fitted RBC models of Ref. [33]. Each bar corresponds to one active-cell parameter set from Tables I and II, ordered as in the tables, and shows the fractions $\sigma_x/\sigma_{\rm tot}$ (gray), $\sigma^{\rm int}_{y\eta|x}/\sigma_{\rm tot}$ (orange), and $\sigma^{\rm self}_{y\eta}/\sigma_{\rm tot}$ (blue). The left and right blocks show the OM ($n = 6$) and OT-sensing ($n = 5$) fits, respectively. All quantities are evaluated analytically from the stationary Gaussian model without additional fitting. For every active fitted cell, self EP within the hidden block is the dominant contribution. Passivated-cell results are discussed in the text.

For the active OM fits, the hidden-block self channel contribution accounts for $0.9928 - 0.9938$ of the total EP, with an arithmetic mean of 0.9932. The active OT-sensing fits show broader cell-to-cell variation, but the same hierarchy persists: the self fraction ranges from 0.8645 to 0.9773, with a mean of 0.9459. The mean hidden-block interaction fractions are only 0.00358 and 0.02044 for OM and OT-sensing, respectively. Thus, for this partition, most dissipation occurs within the hidden block rather than in its interaction channel with the observable coordinate. The cell-resolved presentation confirms that this hierarchy is not an artifact of averaging over heterogeneous fits.

D. Sharpening of the learning-rate bound for RBC

We apply the learning-rate bound to the hidden block, i.e., $A = \{y, \eta\}$ and $B = \{x\}$ in Eq. (8). Define

$$L_{y\eta} \coloneqq \left|\dot{I}_{y\eta}\right|^2 \mathcal{R}_{y\eta}.$$

For the centered stationary OU models, the previous local-EP bound is $L_{y\eta} \le \sigma_{y\eta}$, whereas the sharpened bound is $L_{y\eta} \le \sigma^{\rm int}_{y\eta|x}$. Their saturation ratios and the tightening factor are

$$\rho_{\rm old} = \frac{L_{y\eta}}{\sigma_{y\eta}}, \quad \rho_{\rm int} = \frac{L_{y\eta}}{\sigma^{\rm int}_{y\eta|x}}, \quad \mathcal{T} = \frac{\sigma_{y\eta}}{\sigma^{\rm int}_{y\eta|x}}.$$

The improvement is precisely the removal of $\sigma_{y\eta}^{\mathrm{self}}$ from the upper bound. In contrast, applying the bound to the scalar observable sector gives no such improvement because $\sigma_x^{\mathrm{self}} = 0$.

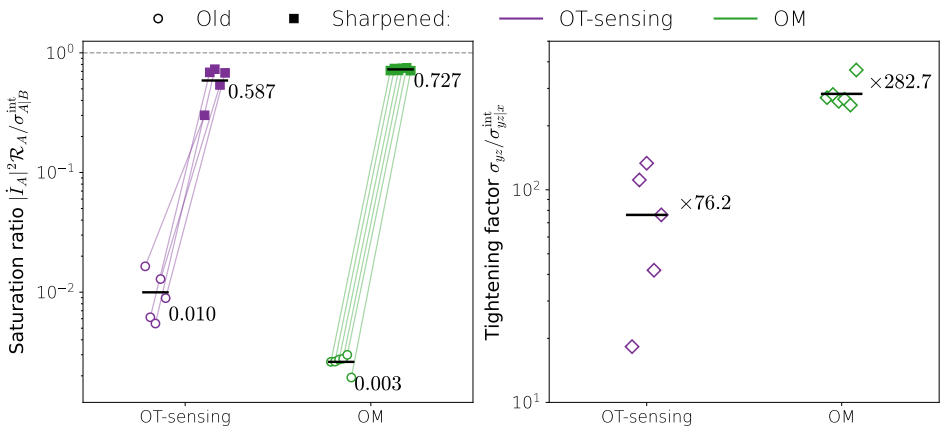

FIG. S7. Sharpening of the learning-rate bound for the hidden block $\{y, \eta\}$ relative to the observable coordinate $\{x\}$. (Left) saturation ratios for the previous bound (open circles) and the sharpened bound (filled squares); lines connect the two values for the same active-cell fit. The dashed horizontal line marks saturation. (Right) the tightening factor $\mathcal{T}$ for each active-cell fit. Purple and Green denote OT-sensing ($n = 5$) and OM ($n = 6$), respectively. Short black horizontal lines and their numerical labels indicate arithmetic means of the per-cell quantities, not ratios of ensemble-averaged rates. Passivated-cell results, whose saturation ratios involve very small numerators and denominators, are reported in the text only.

As shown in Fig. S7, the mean saturation ratio increases from 0.00998 to 0.587 for OT-sensing and from 0.00262 to 0.727 for OM. The mean per-cell tightening factors are 76.2 and 282.7, respectively. Individual factors range from 18.3 to 133.3 for OT-sensing and from 250.2 to 366.2 for OM. These model-based results illustrate that removing self dissipation can substantially sharpen the learning-rate bound while preserving a nontrivial degree of saturation.

For the passivated fits OT P1, OT P2, and OM P1, $L_{y\eta}$ is 1.67×10^{-5}, 1.47×10^{-7}, and $4.69\times10^{-10}\ s^{-1}$, respectively, much smaller than those of the active fits (10.0–$40.5\,\mathrm{s}^{-1}$). Their sharpened saturation ratios remain well defined and satisfy the bound, taking values of 0.896, 0.806, and 0.975. However, these relatively high ratios arise from dividing two very small rates and should not be interpreted as indicating appreciable absolute information flow. To avoid overstating their biophysical significance, we report the passivated-cell results in the text and retain their parameters in Tables I and II, while restricting the figures to the active fits.

VI. HYDRODYNAMIC COARSE-GRAINING REVERSAL: ANALYTIC CALCULATION AND FINITE-RESOLUTION RECOVERY

A. Model and parameters

We consider the three-dimensional stationary OU model introduced for two hydrodynamically coupled particles driven through a stochastic trap center [21]. We write $\boldsymbol{z} := (x_1, x_2, \lambda)^{\intercal}$ and use the convention

$$\mathrm{d}\boldsymbol{z} = -\mathbf{F}\boldsymbol{z}\,\mathrm{d}t + \sqrt{2\mathbf{D}}\,\mathrm{d}W, \tag{S113}$$

where

$$\mathbf{F} = \begin{pmatrix} 1/\tau_1 & \epsilon/\tau_2 & -1/\tau_1 \\ \epsilon/\tau_1 & 1/\tau_2 & -\epsilon/\tau_1 \\ 0 & 0 & 1/\tau_\lambda \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} D_0 & \epsilon D_0 & 0 \\ \epsilon D_0 & D_0 & 0 \\ 0 & 0 & \theta D_0 \end{pmatrix}.$$

The parameters are $\tau_1 = 1.5\,ms$, $\tau_2 = 2.1\,ms$, $\tau_\lambda = 4.0\,ms$, $D_0 = 0.16\,\mu m^2 s^{-1}$, and $\theta = 0.56$. The two conditions used in Fig. 4d are $\epsilon = 0.29$ and $\epsilon = 0.49$. ϵ and θ are dimensionless. All logarithms are natural, so rates in this section are reported in $k_B s^{-1}$ after applying the factor four in the midpoint identity.

B. Analytic evaluation of the ϵ derivatives in Fig. 4c

We use the closed-form expressions for linear systems in Eqs. (S109)–(S110), together with the Lyapunov equation (S103). For the dynamics considered here, the drift matrix is $\mathbf{A} = -\mathbf{F}$. We define

$$\mathbf{F}_\epsilon \equiv \partial_\epsilon \mathbf{F}, \qquad \mathbf{D}_\epsilon \equiv \partial_\epsilon \mathbf{D}, \qquad \mathbf{C}_\epsilon \equiv \partial_\epsilon \mathbf{C}. \tag{S114}$$

The derivatives $\mathbf{F}_\epsilon$ and $\mathbf{D}_\epsilon$ follow directly from the model matrices. Differentiating the Lyapunov equation $\mathbf{FC}+\mathbf{CF}^\mathsf{T} = 2\mathbf{D}$ with respect to ϵ gives a second continuous Lyapunov equation for the covariance derivative,

$$\mathbf{FC}_\epsilon + \mathbf{C}_\epsilon \mathbf{F}^\mathsf{T} = 2\mathbf{D}_\epsilon - \mathbf{F}_\epsilon \mathbf{C} - \mathbf{C}\mathbf{F}_\epsilon^\mathsf{T}. \tag{S115}$$

After solving Eq. (S115) for $\mathbf{C}_\epsilon$, we differentiated Eqs. (S109)–(S110) with respect to ϵ and substituted $\mathbf{F}_\epsilon$, $\mathbf{D}_\epsilon$, and $\mathbf{C}_\epsilon$. This yields the ϵ derivatives of the individual channel rates shown in Fig. 4(c).

The differentiation itself was performed analytically; however, we did not expand the resulting expressions into fully symbolic closed forms. Instead, at each value of ϵ, Eq. (S115) was solved numerically using the continuous Lyapunov solver in SciPy. Thus, the curves in Fig. 4(c) do not rely on finite-difference approximations with respect to ϵ.

C. Synthesis of the trajectories used for Fig. 4d

Synthetic trajectories were generated with the exact discrete transition of the stationary OU process. For observation interval δt and stationary covariance $\mathbf{C}$, the initial state is drawn from $\mathcal{N}(\mathbf{0}, \mathbf{C})$, and subsequent states satisfy

$$\boldsymbol{z}_{k+1} = \mathbf{M}\boldsymbol{z}_k + \boldsymbol{\eta}_k, \quad \mathbf{M} = \exp(-\mathbf{F}\delta t), \quad \boldsymbol{\eta}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{C} - \mathbf{MCM}^\mathsf{T}). \tag{S116}$$

The Gaussian innovations are independent across steps and independent of the initial state. This construction initializes the process in stationarity and avoids time-integration bias. At each ϵ, we generated 40 independent trajectories of duration $50\,s$ sampled at $\delta t = 10^{-4}\,s$, giving 500001 recorded states per trajectory using (S116).

D. Midpoint mutual information and zero-time extrapolation

For this linear Gaussian model, the increment and midpoint are jointly Gaussian. We therefore estimate their (conditional) information directly from empirical covariances, rather than $\hat{I}_{MMSE}$ or $\hat{I}_{KSG}$ from Sec. IV.A. The Gaussian (conditional) information formula is exact at the population level for this model. For an integer half-lag n, we form

$$\tau_n = 2n\,\delta t, \quad d\boldsymbol{z}_{k,n} = \boldsymbol{z}_{k+2n} - \boldsymbol{z}_k, \quad \boldsymbol{z}_{\boldsymbol{m}}^{(k,n)} = \boldsymbol{z}_{k+n},$$

where τ_n is the total lag, $d\boldsymbol{z}_{k,n}$ is the increment, and $\boldsymbol{z}_{\boldsymbol{m}}^{(k,n)}$ is the recorded temporal midpoint (no interpolation is used). For each lag, all available overlapping windows within a trajectory enter the empirical covariance of $\left(d\boldsymbol{z}_{k,n}, \boldsymbol{z}_m^{(k,n)}\right)$.

For Gaussian blocks U, V, W,

$$I(U;V|W) = \frac{1}{2}\log\left(\frac{\det \boldsymbol{\Sigma}_{U|W}\det \boldsymbol{\Sigma}_{V|W}}{\det \boldsymbol{\Sigma}_{UV|W}}\right),$$

where for arbitrary T,

$$\boldsymbol{\Sigma}_{T|W} = \boldsymbol{\Sigma}_{TT} - \boldsymbol{\Sigma}_{TW}\boldsymbol{\Sigma}_{WW}^{-1}\boldsymbol{\Sigma}_{WT}.$$

An empty conditioning block gives ordinary MI. Before the covariance is formed, each coordinate of the increment and midpoint blocks is divided by its sample standard deviation. This rescaling leaves Gaussian MI and CMI unchanged and improves numerical conditioning.

For each channel we fit the five finite-lag values $n = 1, \cdots, 5$ to a polynomial constrained to pass through the origin,

$$I_i(\tau_n) = a_{i1}(\tau_n/\delta t) + a_{i2}(\tau_n/\delta t)^2 + a_{i3}(\tau_n/\delta t)^3, \quad \hat{\sigma}_i = 4a_{i1}/\delta t. \tag{S117}$$

We call this the W5-cubic protocol. Figure S8 displays the raw information values and the mean inferred slopes. For example, at $\epsilon = 0.29$, the W5-linear, quadratic, and cubic total-rate estimates have means 200.665, 259.069, and $266.663\,k_\mathrm{B}s^{-1}$, respectively, compared with the exact value $267.261\,k_\mathrm{B}s^{-1}$.

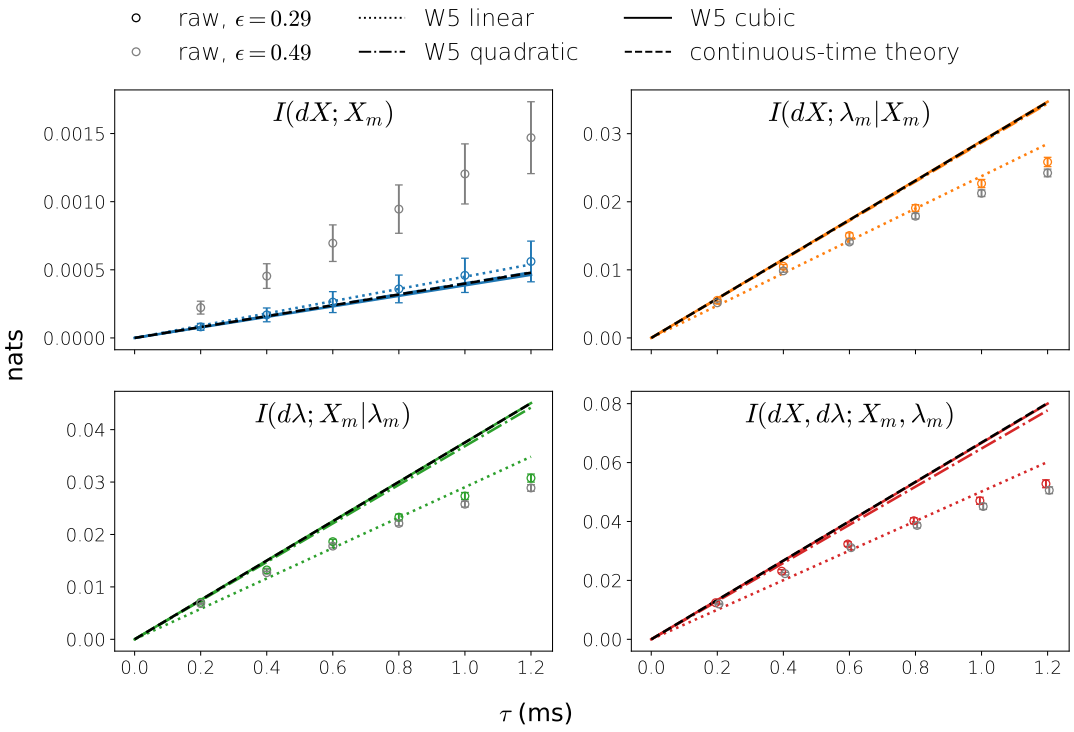

FIG. S8. Midpoint information and inferred zero-time slopes. The panels show $I(dX; X_m)$ (upper left, blue), $I(dX; \lambda_m | X_m)$ (upper right, orange), $I(d\lambda; X_m | \lambda_m)$ (lower left, green) and $I(d\boldsymbol{z}; \boldsymbol{z}_m)$ (lower right, red). Open symbols and error bars show the mean and SD across 40 trajectories at each lag. For each panel, channel colors denote $\epsilon = 0.29$ and gray denotes $\epsilon = 0.49$. For $\epsilon = 0.29$, dotted, dash-dotted, and solid lines are τ times the mean slope from the W5-linear, quadratic, and cubic fits, respectively. Black dashed lines are $\tau\sigma_i/4$. These lines display the inferred zero-time slopes. The sixth lag is displayed for context but excluded from the W5 fits.

E. Protocol sensitivity and finite-resolution recovery

We assess finite-resolution recovery using the same 40 trajectories at each ϵ. For channel i, with exact rate σ_i, the empirical bias and root-mean-square error (RMSE) are

$$\mathrm{bias_i} = \frac{1}{40}\sum_{\mathrm{r}=1}^{40} \hat{\sigma}_{\mathrm{i,r}} - \sigma_{\mathrm{i}}, \quad \mathrm{RMSE_i} = \left[\frac{1}{40}\sum_{\mathrm{r}=1}^{40} (\hat{\sigma}_{\mathrm{i,r}} - \sigma_{\mathrm{i}})^2\right]^{\frac{1}{2}}.$$

Figure S9 compares quadratic and cubic fits over W4, W5, and W6, where Wm uses $n = 1, \cdots, m$. We also evaluate the closure residual within each trajectory,

$$c_r = \hat{\sigma}^{\mathrm{self}}_{X,r} + \hat{\sigma}^{\mathrm{int}}_{X|\lambda,\,r} + \hat{\sigma}^{\mathrm{int}}_{\lambda|X,\,r} - \hat{\sigma}_{\mathrm{tot,\,r}}.$$

Quadratic fits increasingly underestimate the interaction and total rates as the window widens, while cubic fits suppress most of this bias over W4-W6. At W5, the total-rate biases are -0.598 and $-0.541\,k_{\mathrm{B}}s^{-1}$ for the cubic fits, compared with -8.192 and $-8.101\,k_{\mathrm{B}}s^{-1}$ for the quadratic fit. The corresponding cubic RMSEs are 6.294 and $5.233\,k_{\mathrm{B}}s^{-1}$, or 2.35% and 2.03% of the exact total rates.

W5-cubic provides a common protocol for all channels without channel-specific tuning. It uses five lag values for three coefficients, leaving two residual degrees of freedom, compared with one for W4-cubic. These lag values are correlated, so this count describes the fit rather than independent statistical replication. The small X self channel retains larger relative uncertainty: its relative RMSEs are 32.2% and 22.6%. We therefore use W5-cubic as a practical choice for this model and sampling resolution, without claiming that it is optimal for every channel or other systems.

The W5-cubic closure residual is small but detectable: its means are 0.709 and $0.697\,k_{\mathrm{B}}s^{-1}$, with mean relative residuals 0.266% and 0.271%. Thus, finite-data extrapolation does not enforce perfectly exact closure at either condition.

Figure S10 resolves the W5-cubic results into individual trajectories. For each condition, shaded regions show the 95% Student-t confidence interval for the ensemble mean. The theoretical rate, denoted by a black star, lies within

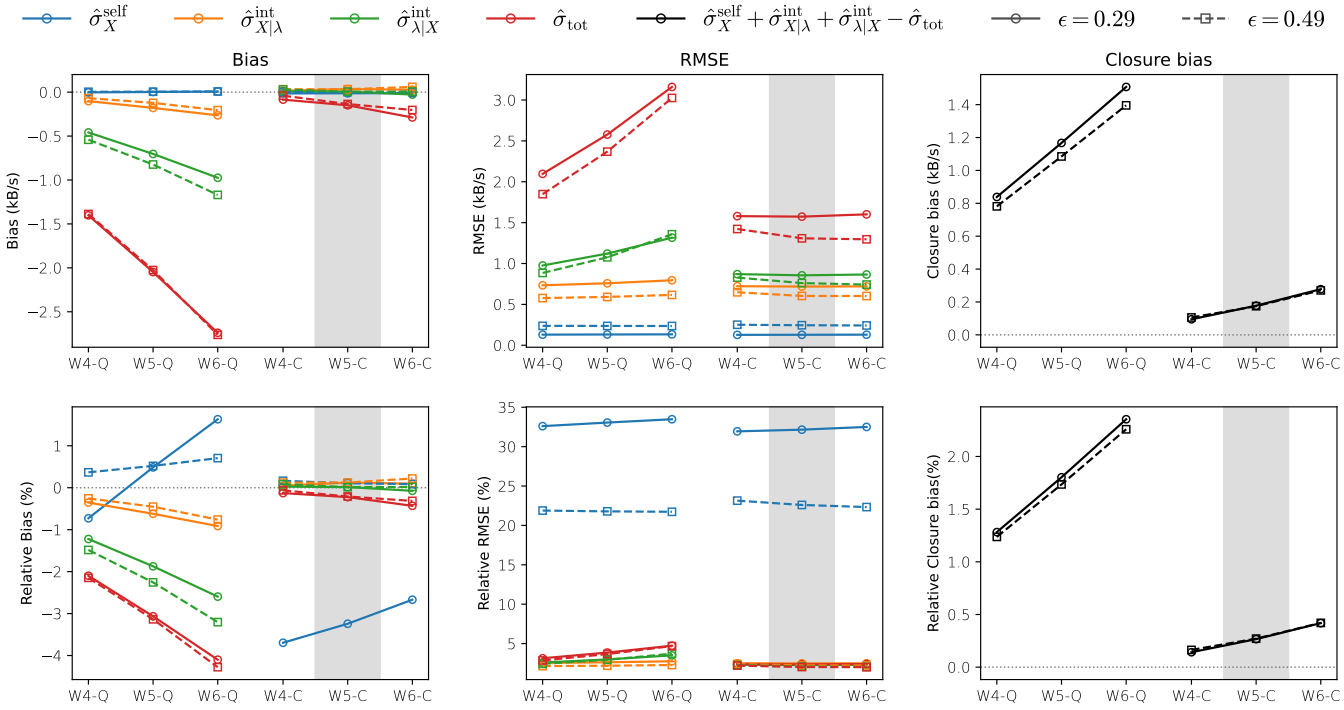

FIG. S9. Sensitivity to the lag window and polynomial order. Columns show channel bias, channel RMSE, and mean closure residual. The upper row reports dimensional quantities in $k_{\mathrm{B}}s^{-1}$. The lower row reports percentages. Relative channel bias and RMSE are normalized by the corresponding exact rate. Relative closure bias is $100\langle c_r/\hat{\sigma}_{\mathrm{tot,r}}\rangle$. Colors identify the four channels, with black for closure. Solid lines with circles and dashed lines with squares denote $\epsilon = 0.29$ and $\epsilon = 0.49$, respectively. Q and C indicate quadratic and cubic fits. The shaded W5-cubic protocol is used in Fig. 4d in the main text.

each of the eight displayed intervals. The spread of the individual points is visibly larger than the uncertainty in the ensemble mean, particularly for the X self channel. Overlapping windows within a trajectory are not treated as independent realizations for these intervals.

The changes in Fig. 4d are differences of the two ensemble means, $\Delta\bar{\hat{\sigma}}_i = \bar{\hat{\sigma}}_i(0.49) - \bar{\hat{\sigma}}_i(0.29)$. Their 95% confidence intervals are computed using the Welch formula for two independent ensembles. The interaction losses exceed the self-channel gain, recovering the coarse-graining reversal as in the main text.

The change obtained from the three-channel sum also agrees closely with the directly estimated total change: their difference is $-0.012\,k_{\mathrm{B}}s^{-1}$, with 95% CI $[-0.146, 0.123]\,k_{\mathrm{B}}s^{-1}$. This interval is computed by first forming c_r within each trajectory, preserving the channel covariances, and then taking the difference between the two independent ensemble means of c_r. This agreement of changes is distinct from the small nonzero closure residual at each individual condition.

F. Cross-check against short-time TUR inference

We cross-check the total EP obtained from midpoint information against the short-time thermodynamic-uncertainty-relation (TUR) inference of Manikandan, Gupta, and Krishnamurthy [46], using the linear-current basis employed in subsequent applications [47, 48] and the hydrodynamic studies [21, 22]. Both methods are evaluated on the same trajectories, so their agreement tests consistency between two inference routes for this application.

Let m be the lag in recorded sampling steps, $h_m = m\delta t$, and let $\tilde{\boldsymbol{z}}_k$ denote the trajectory after subtracting its empirical mean. The nine current features are

$$\boldsymbol{\psi}_{ab}^{(k,m)} = (\tilde{\boldsymbol{z}}_{k+m} - \tilde{\boldsymbol{z}}_k)_a \left(\frac{\tilde{\boldsymbol{z}}_{k+m} + \tilde{\boldsymbol{z}}_k}{2}\right)_b, \qquad a, b \in \{1, 2, 3\}.$$

We form the empirical feature mean $\boldsymbol{\mu}_m$ and covariance $\boldsymbol{\Xi}_m$, both using the normalization $1/N_m$, where N_m is the number of available windows. The optimized quantity is

$$Q_m = 2\boldsymbol{\mu}_m^{\intercal}\boldsymbol{\Xi}_m^{-1}\boldsymbol{\mu}_m. \tag{S118}$$

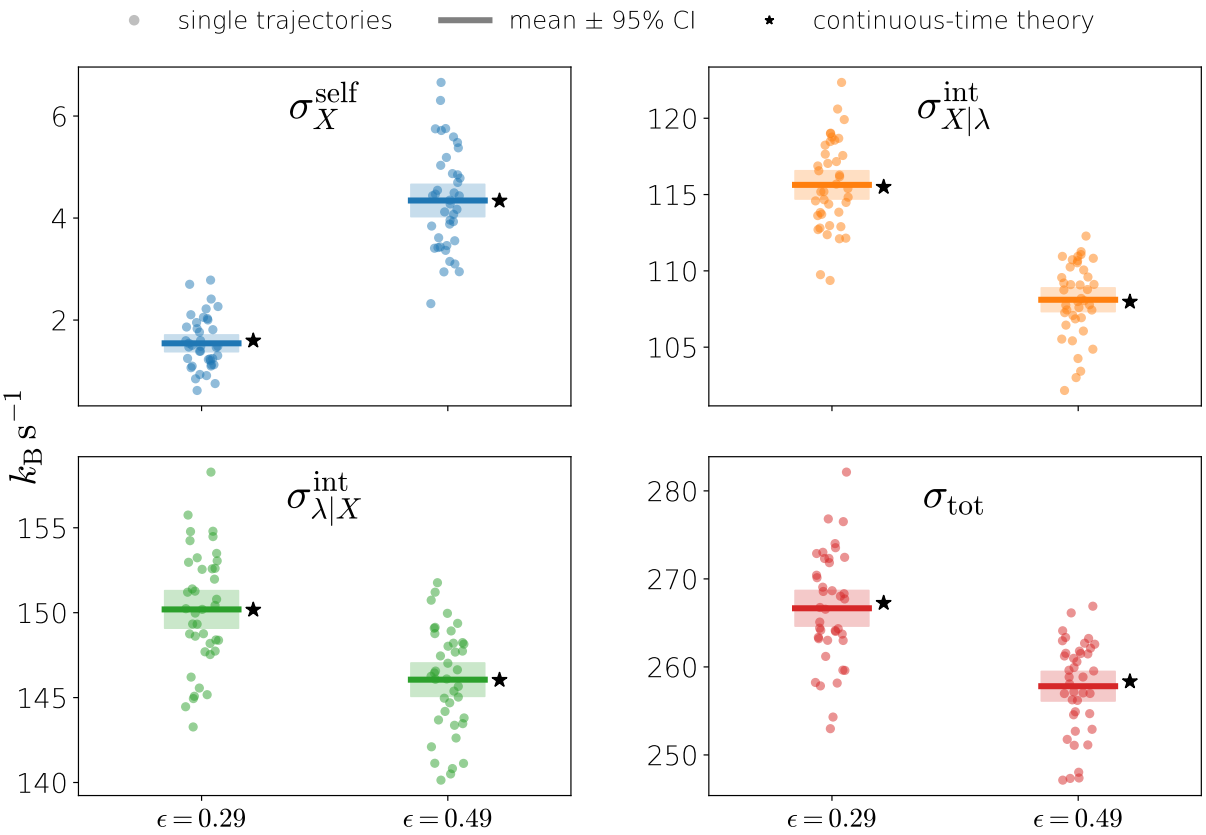

FIG. S10. W5-cubic channel rates at the two conditions used in Fig. 4d, with the same panel order as Fig. S8. Colored points show 40 independent trajectory estimates at $\epsilon = 0.29$ (left) and 0.49 (right) in each panel. Black stars show the exact continuous-time rates. Colored horizontal bars mark the ensemble means and their 95% Student-t confidence intervals. These intervals describe uncertainty in the mean, while the point clouds show trajectory-to-trajectory variation. All eight theoretical rates lie within their respective intervals.

The short-time EP estimate is Q_m/h_m. The arithmetic mean of the two endpoints is part of this current construction; unlike the midpoint-MI identity, it does not require a recorded state at the temporal midpoint.

Finite-lag bias remains appreciable if this quantity is used without extrapolation. At the shortest lag $h_1 = \delta t$, the mean estimates are 250.593 and 242.346 $k_{\mathrm{B}}s^{-1}$, giving bias -16.668 and $-16.003\,k_{\mathrm{B}}s^{-1}$, respectively (about -6.2%). At $2\delta t$, the biases are -32.101 and $-31.090\,k_{\mathrm{B}}s^{-1}$. We therefore also extrapolate the TUR quantity to the continuous-time target as in (S117), using

$$Q_m \simeq b_1(h_m/\delta t) + b_2(h_m/\delta t)^2 + b_3(h_m/\delta t)^3, \quad \hat{\sigma}_{\mathrm{TUR}} = b_1/\delta t.$$

This is a zero-intercept cubic fit of Q_m; it carries no additional factor four.

We use two lag conventions in Fig. S11. The matched-physical-lag check uses $m = 2, 4, 6, 8, 10$ for both methods, corresponding to 0.2–1.0 ms. The native-lag check retains this midpoint-MI window but uses $m = 1, 2, 3, 4, 5$ for TUR, or 0.1–0.5 ms. The latter gives each method its five shortest accessible lags; it does not involve an optimization over lag choices. Both fits are performed separately on every trajectory.

At $\epsilon = 0.29$, the midpoint-MI, matched-lag TUR, and native-lag TUR means are 266.663, 264.197, and 266.857 $k_{\mathrm{B}}s^{-1}$; at $\epsilon = 0.49$, they are 257.808, 255.768, and 258.004 $k_{\mathrm{B}}s^{-1}$. The exact values are 267.261 and 258.349 $k_{\mathrm{B}}s^{-1}$. For the native-lag check, the trajectory-level Pearson correlations are 0.983 and 0.980, and the RMS differences between the two methods are 1.179 and 1.066 $k_{\mathrm{B}}s^{-1}$. At matched physical lags, the corresponding correlations are 0.944 and 0.886, with RMS differences 3.311 and 3.201 $k_{\mathrm{B}}s^{-1}$ and mean TUR-minus-MI offsets -2.466 and $-2.401\,k_{\mathrm{B}}s^{-1}$. Thus, the two routes give closely agreeing trajectory-level EP values, with small protocol-dependent offsets. This cross-check supports the consistency of the finite-resolution midpoint-information analysis for the present model and sampling conditions; the result is not a claim of general estimator superiority.

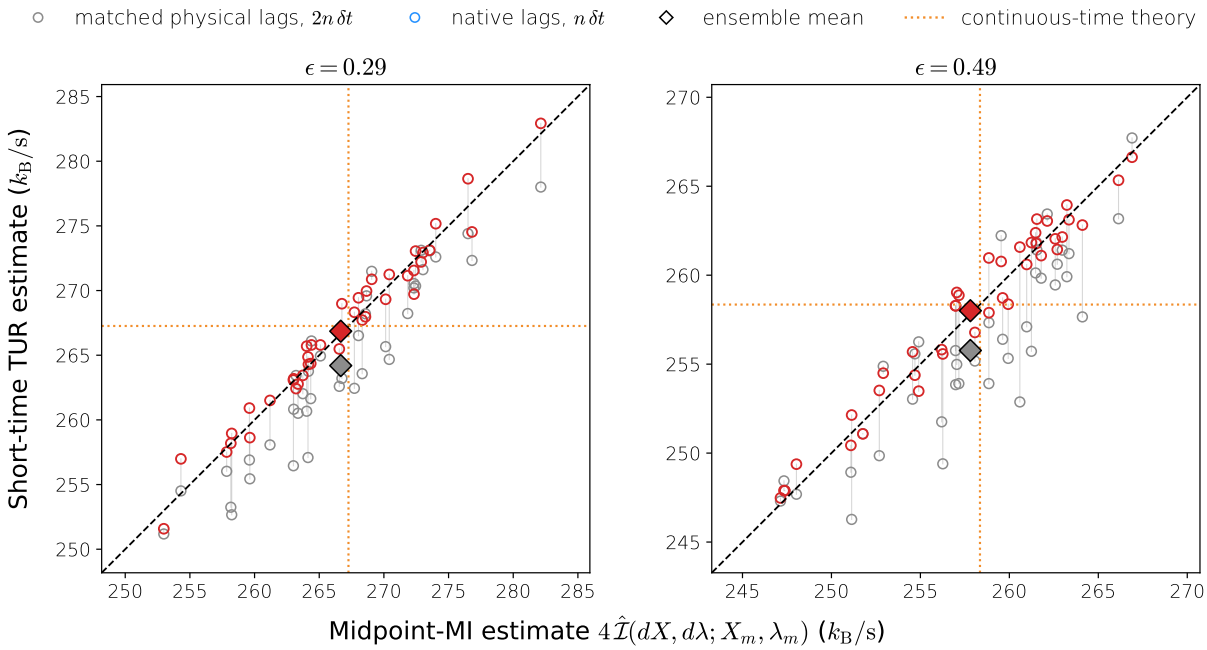

FIG. S11. Cross-check of the inferred total EP against short-time TUR inference at $\epsilon = 0.29$ (left) and 0.49 (right). Each open symbol corresponds to the same trajectory on both axes. The horizontal coordinate is the midpoint-MI W5-cubic estimate. Gray symbols use a TUR W5-cubic fit at matched physical lags $2n\delta t$; red symbols use the five native TUR lags $n\delta t$, with $n = 1, 2, 3, 4, 5$. Faint vertical connectors join the two TUR results for each trajectory. Diamonds mark pairs of ensemble means. The black dashed diagonal denotes equality, and dotted orange horizontal and vertical lines mark the exact total EP. The annotated RMSEs are those of the two TUR protocols relative to the exact rate.